OBE Curriculum

Department of Forestry and Environmental Science

Bachelor of Science (Honours)

Session: 2025-26

Shahjalal University of Science and Technology

Sylhet, Bangladesh

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OVERVIEW OF THE UNIVERSITY AND DEPARTMENT

Name of the University	**Shahjalal University of Science and Technology, Sylhet**
Establishment of the University	**25 August, 1986**
Founder Vice Chancellor of the University	**Professor Dr. Sadruddin Ahmed Chawdhury**
Current Vice Chancellor of the University	**Professor Dr. A. M. Sarwaruddin Chowdhury**
First Academic Session of the University	**1990-1991**
Website of the University	**www.sust.edu**
E-mail of the University	**registrar@sust.edu**
Name of the Department	**Forestry and Environmental Science (FES)**
First Academic Session of the Department	**1998-1999**
Website of the Department	**www.sust.edu/d/fes**
E-mail of the Department	**fes@sust.edu**
PABX Extension of the Department	**2273**
Founder Head of the Department	**Professor Dr. Narayan Saha**
Current Head of the Department	**Professor Dr. Mohammad Redowan**
Programs Offering	**B.Sc. (Honours), M.Sc. and Ph.D.**

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List of Faculty Members

SN	Designation	Full Name	PABX	Cell Phone
1	**Professors**	Dr. Narayan Saha	2273, 2604	01716072440
2		Dr A. Z. M. Manzoor Rashid	2273,2606	01711302555
3		Dr. Md. Qumruzzaman Chowdhury	2273, 2294	01734161326
4		Dr. Mohammad Belal Uddin	2273, 2605	01727767431
5		Dr. Romel Ahmed	-	01874934514
6		Dr. Mohammad Redowan	2273	01818465634
7		Dr. Mizanur Rahman	-	01711153231
8		Dr. Swapan Kumar Sarker	-	01791069892
9		Dr. Mohammed Abu Sayed Arfin Khan	-	01917174537
10		Dr. Kazi Mohammad Masum	-	01816052842
11		Dr. Farzana Raihan	-	01712207882
12		Dr. Mahmuda Islam	-	01712172658
13		Dr. Mahmuda Sharmin	-	01671635120
14		Dr. Mohammad Golam Kibria	-	01716627144
15	**Associate Professors**	Ms. Fahmida Sultana	-	01911720167
16		Dr. Mohammed Masum Ul Haque	-	01748714848
17		Ms. Nusrat Islam	-	01717113566
18	**Assistant Professors**	Mr. Anup Datta	-	01786010404
19		Mr. Md. Saifuzzaman Bhuiyan	-	01628598829
20		Mr. Abdur Rakib Bhuiyan	-	01916937563
21		Ms. Rabeya Sultana	-	01672229303
22		Ms. Rahela Khatun	-	01794738298
23	**Lecturers**	Mr. Md. Shamim Reza Saimun	-	01521236118
24		Ms. Farhana Bintay Hye	-	01949824158
25		Ms. Nilima Ferdous	-	01680254406

PART - A

1. Title of the Academic Program: Bachelor of Science (Honours)

2. Name of the University: Shahjalal University of Science and Technology (SUST)

3. Vision of the University:

To be a leading university of excellence in Science and Technology with a strong national commitment and significant international impact.

4. Mission of the University:

M01 To advance learning and knowledge through teaching and research in science and technology.

M02 To serve as a center for knowledge creation, technological innovation and transfer among academia, industry, and society.

M03 To assist in transferring Bangladesh a country with sustainable economic growth and equitable social development.

5. Name of the Program Offering Entity: Department of Forestry and Environmental Science

6. Vision of the Program Offering Entity:

The department of Forestry and Environmental Science (FES) will be the learning- and research-based department of choice for the meritorious and top-ranked students of Bangladesh.

7. **Mission of the Program Offering Entity**:

M1. Enhancing/enriching Forestry graduates' knowledge-based capacity to identify, clarify and provide best possible solutions to emerging issues and challenges related to individual, workplace, society, and the country.

M2. Developing leadership and managerial capacities of Forestry graduates for the relevant organizations.

M3. Imparting humanistic attitudes towards community, forest and nature conservation.

8. Objectives of the Program Offering Entity:

The objectives of the Program Offering Entity are to offer the courses and conduct researches under the following dimensions:

- Ø Knowledge and Understanding
- Ø Skills and Processes
- Ø Values and Attitudes

9. Name of the Degree: B.Sc. (Honours) in Forestry

10. Description of the Program:

The B.Sc. (Honours) in Forestry program will consist of eight semesters. There will be two semesters in an academic year. The duration of the first semester will span from 1st January to 30th June and the second semester

from 1st July to 31st December. Duration of each semester will be of 20 weeks, of which 14 weeks classes, 2 weeks preparatory leave and 4 weeks for semester final examinations. These 20 weeks may not be contiguous to accommodate various holidays. Recess before the semester final examinations may coincide with holidays. The final grading will be completed before the beginning of each semester. Total credit for B.Sc. (Honours) in Forestry program will be 160.

11. Graduate Attributes:

GA1: In-depth discipline-specific knowledge and professional skills.

GA2: Beyond-discipline knowledge, insight and skills.

GA3: Capacity for critical, creative, and evidence-based thinking to solve complex problems.

GA4: Awareness of and sensitivity to ethics and ethical standard on interpersonal, social, research, and professional levels.

GA5: Valuing integrity, environmental sustainability, and civic engagement.

GA6: Understanding of social and civic responsibilities and the rights of individuals and groups.

GA7: Willingness to life-long learning through directed/guided and self-directed/independent study.

GA8: Capacity for self-reflection, self-discovery, and personal development.

GA9: Capable of communicating effectively in a range of contexts.

GA10: Digital capabilities: well prepared for living, learning, and working in a digital society.

GA11: Professionalism and leadership readiness: engage in professional behavior and have the potential to take leadership roles in their chosen careers and communities.

12. Program Educational Objectives (PEOs):

PEO1. Help the students to conceptualize the basic concepts, theories, principles, processes, and procedures of the core areas of Forestry.

PEO2. Enables the students to demonstrate high levels of analytical and critical thinking skills to solve conceptual and quantitative problems in forestry and environment related fields through the application of fundamental principles of Forestry and environmental science.

PEO3. Develop skills to demonstrate safe and acceptable skills in laboratory procedures and experimental designs.

PEO4. Help the students to acquire the basic tools needed to carry out Forestry and environmental research independently or within a team.

PEO5. To enhance the effective communication skills in the form of written, oral and in interactive presentation mode.

PEO6. Help students to develop curiosity and interest in making scientific investigation, personal integrity and to realize the roles of forestry graduates on entrepreneurship as well as industrial, environmental, social and economic aspects nationally and globally.

PEO7. Acquaint students with moral and ethical values and their practices to handle the research findings, maintaining secrecy of Intellectual Property (IP) and development of patriotism and community services.

13. Program Learning Outcomes (POs):

A. Fundamental Skill	
PO1	Basic concepts, theories, principles, processes and laboratory procedures, and experimental designs of the core areas of Forestry and related areas.
PO2	Speaking and writing abilities
PO3	Mathematical, Statistical and Computer skills
B. Social Skill	
PO4	Knowledge of social sciences
PO5	Knowledge of behavioural sciences
C. Thinking Skill	
PO6	Research abilities
PO7	Problem solving abilities
D. Personal Skill	
PO8	Moral and ethical values
PO9	Leadership and team works qualities
PO10	Cultural and patriotic values

14. Mapping mission of the university with PEOs

PEOs	Mission 01	Mission 02	Mission 03
PEO 1	X		
PEO 2	X		
PEO 3	X		
PEO 4		X	
PEO 5		X	X
PEO 6		X	X
PEO 7		X	X

15. Mapping POs with the PEOs

POs	PEO1	PEO2	PEO3	PEO4	PEO5	PEO6	PEO7
PO 1	X	X					
PO 2					X		
PO 3		X					
PO 4						X	X
PO 5						X	X
PO 6			X	X			
PO 7			X	X			
PO 8							X
PO 9					X	X	X
PO 10						X	X

16. Mapping courses with the POs

Courses	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
FES 08211121	X			X				X		X
FES 08211124	X			X			X		X	
FES 05211121	X	X				X	X			
FES 05211122	X									
AMS 05111101	X	X								
AMS 05111102	X									
FES 08211122	X						X		X	
CHE 05311101	X		X							
CHE 05311102	X		X							
MAT 05411101			X				X			
ENG 02311101		X							X	
FES 08211221	X									
FES 08211222	X					X				
FES 08211223	X					X	X			
FES 08211224	X					X	X		X	
FES 08211225	X	X					X			
AMS 05221201	X	X			X	X	X	X	X	
AMS 05221202	X	X	X			X	X	X	X	
AMS 07311201	X									
AMS 07311202	X		X						X	
FES 08211226	X						X		X	
SOC 03141201	X			X	X					
MAT 05411201			X				X		X	
FES 05322121	X	X								
FES 05322122	X	X								
FES 05212121	X	X	X			X	X	X		
FES 05212122	X		X			X	X			
FES 05112121	X		X			X	X			
FES 05112122	X		X			X	X			
FES 08212122	X						X		X	
PHY 05332101	X		X							
ECO 03112101	X			X	X					
STA 05422101	X		X							
FES 07222221	X					X				
FES 07222222	X					X				
FES 05112221	X		X			X	X	X	X	
FES 05112222	X		X			X	X			

FES 08212221	X									
FES 08212222	X						X		X	
BUS 04112201	X			X	X					
GEE 05322201	X	X				X	X			
CSE 06122216	X		X							
AMS 05222201	X	X	X		X	X	X	X	X	
AMS 05222202	X	X	X		X	X	X	X	X	
FES 08213121	X		X							
FES 08213122	X		X							
FES 08213123	X						X			
FES 08213124	X						X			
FES 08113121	X									
FES 08113122	X									
FES 08213126	X	X					X		X	
FES 08113123	X		X	X		X	X	X	X	X
FES 08113125	X			X	X					
FES 07313131	X									
FES 05323221	X			X			X		X	X
FES 05323222	X	X	X				X		X	X
FES 07223221	X						X			
FES 07223222	X						X			
FES 05223223	X			X		X		X	X	X
FES 07323221	X	X	X				X			
FES 08213221	X					X				
FES 08213222	X	X					X		X	
FES 08213223	X									
FES 08213224	X			X			X		X	
AMS 05423202	X	X	X			X				
FES 08214121	X					X				
FES 08214122	X		X			X			X	
FES 05114121	X		X			X				
FES 05114122	X		X			X				
FES 07314121	X			X			X		X	
AMS 05424101	X		X			X				
AMS 05424102	X		X			X				
FES 05214121	X			X		X		X	X	X
FES 00314122	X	X							X	
FES 07224121	X					X				
FES 08214221					X					

FES 08214222	X	X	X	X	X	X	X	X	X	X
FES 05214221	X					X			X	
FES 08214224	X	X	X	X	X	X	X	X	X	X
FES 05224221	X					X				
FES 05224223	X		X		X		X		X	
FES 05214225	X			X			X			

PART - B

17. Structure of the Curriculum

a) Duration of the programs:

B.Sc. (Hons.) in Forestry: Years: 4 Semesters: 8

b) Admission Requirements:

B.Sc. (Hons.) in Forestry program: The admission committee of the university will conduct the admission process for Bachelor's degree as per the rules. The student will be admitted in the first semester of an academic year in the individual discipline of different schools. Students are eligible for apply in the admission test for the program, who passed S.S.C. and H.S.C. examinations from Science group and got minimum GPA 3.0 in each examination and must be 7.00 in both examinations or minimum Grade B in case of A Level students. For admission in the program, in H.S.C. Science group needs minimum GPA 3.5 or minimum Grade B in case A Level students in both Biology and Mathematics. However the admission of foreign students will be subjected to the verification of successful academic records of O Level and A Level and minimum SAT score minimum 800.

Every student has to maintain her/his student status by enrolling, paying tuition fees, and doing the registration for required credits every successive semester following the requirements: For admissions to higher semester (2nd to 8th) and course registration following requisites and steps have to be strictly maintained:

(1) Completion of 100 level courses is mandatory for student's admission in semester of 300level courses.

(2) Completion of 200 level courses is required for admission in semester of 400 level courses.

(3) A student having incomplete 100 level courses shall be allowed for admission in her/his next available semester of 100-200 level courses until s/he completes all of 100 level courses.

(4) A student having incomplete 200 level courses shall be allowed for admission in her/his next available semester of 200-300 level courses until s/he completes all of 200 level courses.

(5) Once a student reaches to 8th semester of program s/he will be kept at this level, if necessary, till the specified last semester of the undergraduate program for completion of credit requirement of graduation.

Every admission/course registration of a student will be counted and adjusted from the total number of semesters of the program to determine her/his remaining period of study. Student advisors of all disciplines will advise every student about her/his courses for registration and monitor her/his performances. Accordingly, a student has to register for her/his courses and pay necessary dues within the first 4 (Four) weeks of every semester (2 more weeks for late registration). A student will not be allowed to appear in the examination if her/his semester and examination fee is not fully paid off. Foreign student must have valid Visa/residential permit to appear in the examination and that has to be checked by the student advisor and the head of the discipline/institute.

A student will be eligible for re-admission in the first year first semester of the subsequent session if s/he was present in at least 25% of the classes of her/his core and elective courses of the same semester. A student has to take re-admission if her/his student status is not maintained or one or more semesters have annulled because of disciplinary action taken against him/her. In the case of semester annulment, the student has to re-admit the

same level of semester. The level (Year and Semester) of re-admitter will be determined by his/her completed credits. The re-admitted students will always be assigned the original Registration number.

c) Graduating credits: B.Sc. (Hons.): 160

d) Total class weeks in a semester: 14

e) Minimum CGPA requirements for graduation: B.Sc. (Hons.): 2

f) Maximum academic years of completion: B.Sc. (Hons.): 6

g) Category of Courses			
Course Category	Course Type	Course Title	Credits
General Education (GED) Courses	Theory	Botany	3
		General Chemistry	3
		Calculus and Differential Equations	2
		English Language	2
		Wildlife Biology	2
		Survey and Settlement	2
		Sociology for Forestry	3
		Matrix and Vector Analysis	2
		General Physics	3
		Principles of Economics	3
		Statistics	2
		Cost and Management Accounting	2
		Physical Geography	2
		Wildlife Ecology and Management	2
		Research Methodology	3
	Sessional	Botany Laboratory	1
		Chemistry Laboratory	2
		Wildlife Biology Laboratory	1
		Survey and Settlement Laboratory	1
		Database Management and Programming Laboratory	3
		Wildlife Ecology and Management Laboratory	1
		Research Methodology Laboratory	1
Core/ Compulsory Courses	Theory	Introduction to Forestry	2
		Introduction to Environmental Science	2
		Dendrology and Species Silviculture	3
		Forest Tree Physiology	3
		Non-Wood Forest Products	2

		Geology and Soil Science	3
		Forest Ecology	2
		Forest Entomology	2
		Wood Structure and Properties	3
		Mycology, Forest Pathology and Forest Protection	3
		Principles and Practices of Silviculture	3
		Forest Mensuration and Inventory	3
		Plantation Silviculture	2
		Production and Management of Tea, Coffee and Rubber	2
		Agroforestry and Upland Farming Systems	3
		Forest and Environmental Economics	3
		Forest Based Industries and Community Development	3
		Remote Sensing and Geographic Information System	3
		Wood Seasoning and Preservation	2
		Water Resources, Watershed and Wetland Management	2
		Forest Harvesting and Transportation Engineering	3
		Mangrove Forestry	3
		Social Forestry and Forestry Extension	3
		Forest Development Planning	2
		Forest Management	3
		Genetics and Tree Improvement	3
		Urban Forestry and Land Use Planning	3
		Environmental Pollution and Climate Change	3
		Forest and Environmental Policy, Law and Administration	3
		Environmental Impact Assessment	2
		Biodiversity Conservation	2
		Protected Area Management	2
	Sessional	Introduction to Forestry Laboratory	1
		Introduction to Environmental Science Laboratory	1
		Fieldwork in Forests (Swamp Forest Ecosystems)	1
		Dendrology and Species Silviculture Laboratory	1
		Forest Tree Physiology Laboratory	1
		Fieldwork in Forests (Forest-Protected Areas)	1
		Geology and Soil Science Laboratory	1
		Forest Ecology Laboratory	1
		Forest Entomology Laboratory	1
		Fieldwork in Forests (Wetland Ecosystem)	1

		Wood Structure and Properties Laboratory	1
		Mycology, Forest Pathology and Forest Protection Laboratory	1
		Fieldwork in Forests (Tropical Hill Forest)	1
		Forest Mensuration and Inventory Laboratory	1
		Plantation Silviculture Laboratory	1
		Production and Management of Tea, Coffee and Rubber Laboratory	1
		Fieldwork in Forests (Deciduous Sal Forest)	1
		Remote Sensing and Geographic Information System Laboratory	1
		Wood Seasoning and Preservation Laboratory	1
		Fieldwork in Forests (Mangrove Forest)	1
		Forest Management Plan	3
		Genetics and Tree Improvement Laboratory	1
Optional/ Elective Courses	Theory	Dendrochronology	2
		Environmental Governance and Diplomacy	3
	Sessional	Biostatistics Laboratory	2
Capstone Courses	Sessional	Seminar	1
		Research Project	4
		Internship	1
Total		**79+3 (Optional) Courses**	**159 + 7 (Optional)**

18. Year/Semester wise distribution of courses for B.Sc. (Hons.) in Forestry Program

a) First Year / Semester courses:

First Year First Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 08211121	Introduction to Forestry	Core	2	0	2.0	
FES 08211124	Introduction to Forestry Laboratory	Core	0	1.5	1.0	
FES 05211121	Introduction to Environmental Science	Core	2	0	2.0	
FES 05211122	Introduction to Environmental Science Laboratory	Core	0	1.5	1.0	
AMS 05111101	Botany	General Education	3	0	3.0	
AMS 05111102	Botany Laboratory	General Education	0	1.5	1.0	
FES 08211122	Fieldwork in Forests (Swamp Forest Ecosystems)	Core	0	2	1.0	
CHE 05311101	General Chemistry	General Education	3	0	3.0	
CHE 05311102	Chemistry Laboratory	General Education	0	3	2.0	
MAT 05411101	Calculus and Differential Equations	General Education	2	0	2.0	
ENG 02311101	English Language	General Education	2	0	2.0	
Total			**14**	**9.5**	**20**	

First Year Second Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 08211221	Dendrology and Species Silviculture	Core	3	0	3	
FES 08211222	Dendrology and Species Silviculture Laboratory	Core	0	1.5	1	
FES 08211223	Forest Tree Physiology	Core	3	0	3	
FES 08211224	Forest Tree Physiology Laboratory	Core	0	1.5	1	
FES 08211225	Non-Wood Forest Products	Core	2	0	2	
AMS 05221201	Wildlife Biology	General Education	2	0	2	
AMS 05221202	Wildlife Biology Laboratory	General Education	0	1.5	1	
AMS 07311201	Survey and Settlement	General Education	2	0	2	
AMS 07311202	Survey and Settlement Laboratory	General Education	0	1.5	1	
FES 08211226	Fieldwork in Forests (Forest-Protected Areas)	Core	0	2	1	
SOC 03141201	Sociology for Forestry	General Education	3	0	3	
MAT 05411201	Matrix and Vector Analysis	General Education	2	0	2	
Total			**17**	**8**	**22**	

b) Second Year/ Semester courses:

Second Year First Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 05322121	Geology and Soil Science	Core	3	0	3	
FES 05322122	Geology and Soil Science Laboratory	Core	0	1.5	1	
FES 05212121	Forest Ecology	Core	3	0	3	
FES 05212122	Forest Ecology Laboratory	Core	0	1.5	1	
FES 05112121	Forest Entomology	Core	2	0	2	
FES 05112122	Forest Entomology Laboratory	Core	0	1.5	1	
FES 08212122	Fieldwork in Forests (Wetland Ecosystem)	Core	0	2	1	
PHY 05332101	General Physics	General Education	3	0	3	
ECO 03112101	Principles of Economics	General Education	3	0	3	
STA 05422101	Statistics	General Education	2	0	2	
Total			**16**	**6.5**	**20**	

Second Year Second Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 07222221	Wood Structure and Properties	Core	3	0	3	
FES 07222222	Wood Structure and Properties Laboratory	Core	0	1.5	1	
FES 05112221	Mycology, Forest Pathology and Forest Protection	Core	3	0	3	
FES 05112222	Mycology, Forest Pathology and Forest Protection Laboratory	Core	0	1.5	1	
FES 08212221	Principles and Practice of Silviculture	Core	3	0	3	
FES 08212222	Fieldwork in Forests (Tropical Hill Forest)	Core	0	2	1	
BUS 04112201	Cost and Management Accounting	General Education	2	0	2	
GEE 05322201	Physical Geography	General Education	2	0	2	
CSE 06122216	Database Management and Programming Laboratoty	General Education	0	4.5	3	
AMS 05222201	Wildlife Ecology and Management	General Education	2	0	2	
AMS 05222202	Wildlife Ecology and Management Laboratory	General Education	0	1.5	1	
Total			**15**	**11**	**22**	

c)Third Year/ Semester courses:

Third Year First Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 08213121	Forest Mensuration and Inventory	Core	3	0	3	
FES 08213122	Forest Mensuration and Inventory Laboratory	Core	0	1.5	1	
FES 08213123	Plantation Silviculture	Core	2	0	2	
FES 08213124	Plantation Silviculture Laboratory	Core	0	1.5	1	
FES 08113121	Production and Management of Tea, Coffee and Rubber	Core	2	0	2	
FES 08113122	Production and Management of Tea, Coffee and Rubber Laboratory	Core	0	1.5	1	
FES 08213126	Fieldwork in Forests (Deciduous Sal Forest)	Core	0	2	1	
FES 08113123	Agroforestry and Upland Farming Systems	Core	3	0	3	
FES 08113125	Forest and Environmental Economics	Core	3	0	3	
FES 07313131	Forest Based Industries and Community Development	Core	3	0	3	
Total			**16**	**6.5**	**20**	

Third Year Second Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 05323221	Remote Sensing and Geographic Information System	Core	3	0	3	
FES 05323222	Remote Sensing and Geographic Information System Laboratory	Core	0	1.5	1	
FES 07223221	Wood Seasoning and Preservation	Core	2	0	2	
FES 07223222	Wood Seasoning and Preservation Laboratory	Core	0	1.5	1	
FES 05223223	Water Resources, Watershed and Wetland Management	Core	2	0	2	
FES 07323221	Forest Harvesting and Transportation Engineering	Core	3	0	3	
FES 08213221	Mangrove Forestry	Core	3	0	3	
FES 08213222	Fieldwork in Forests (Mangrove Forest)	Core	0	2	1	
FES 08213223	Social Forestry and Forestry Extension	Core	3	0	3	
FES 08213224	Forest Development Planning	Core	2	0	2	
AMS 05423202	Biostatistics Laboratory	Optional	0	3	2	
Total			**18**	**5+3**	**21+2**	

d) Fourth Year/ Semester courses:

Fourth Year First Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 08214121	Forest Management	Core	3	0	3	
FES 08214122	Forest Management Plan	Core	0	6	3	
FES 05114121	Genetics and Tree Improvement	Core	3	0	3	
FES 05114122	Genetics and Tree Improvement Laboratory	Core	0	1.5	1	
FES 07314121	Urban Forestry and Land Use Planning	Core	3	0	3	
AMS 05424101	Research Methodology	General Education	3	0	3	
AMS 05424102	Research Methodology Laboratory	General Education	0	1.5	1	
FES 05214121	Environmental Pollution and Climate Change	Core	3	0	3	
FES 00314122	Seminar	Capstone	0	2	1	
FES 07224121	Dendrochronology	Optional	2	0	2	
Total			**15+2**	**11**	**21+2**	

Fourth Year Second Semester

Course Code	Course Title	Course Category	Hours/Week		Credits	Prerequisites
			Theory	Lab		
FES 08214221	Forest and Environmental Policy, Law and Administration	Core	3	0	3	
FES 08214222	Research Project	Capstone	0	8	4	
FES 05214221	Environmental Impact Assessment	Core	2	0	2	
FES 08214224	Internship	Capstone	0	2	1	
FES 05224221	Biodiversity Conservation	Core	2	0	2	
FES 05224223	Protected Area Management	Core	2	0	2	
FES 05214225	Environmental Governance and Diplomacy	Optional	3	0	3	
Total			**9 +3**	**10**	**14 +3**	

Note: Credit for optional courses will be added to the CGPA

PART C

19. Description of all courses of the program including the following information for each course:

Course No: FES 08211121	Credit: 2.0	Year: First	Semester: First
Introduction to Forestry			**Course Status: Theory**

Course Rationale:
This course is designed with a view to meet the needs of the contemporary world on national, regional and global forest issues, concepts and their significance. Various forest types, their composition, contribution of forest and environment sector in national and global development are vital thus imperative to offer basic and introductory ideas to the students to achieve detailed learning content and objectives. Pollution, impact of climate change, factors of biodiversity loss and degradation of forest are crucial to know as a beginner student.

Course Objectives:
The objectives of the course are-

- To introduce basic ideas about fundamentals concepts, definitions and features of forest, forest types, and species composition.
- To facilitate necessary knowledge about multiple benefits of forest and forestry practices of Bangladesh.
- To provide the basic knowledge on major issues in forestry.
- To familiarize students with forest-related institutions in Bangladesh.

Course Content:

1. **Introduction:** Definitions and meaning of important terms used in forestry; scopes, objectives and importance of forestry; recent trends in Forestry (e.g., from timber production management system to sustainable forestry, decentralization and devolution of power in natural resource management).
2. **Stratification Within a Forest**: canopy class; growth stage of a tree; forest strata in tropical ecosystem
3. **Forest Types:** Major forest ecosystems of the world; Location, distribution, and species composition of the different forest types of Bangladesh; Legal classification of the forests of Bangladesh.
4. **Major Forestry Issues:** Responsible factors of forestry problems (deforestation, degradation, encroachment; habitat fragmentation; Human-, animal conflict etc.) and their probable solutions.
5. **Tangible and Intangible Benefits of Forest:** Ecological, economic, environmental, social and other benefits of forest.
6. **Forest-related Institutions in Bangladesh:** Functional role and administrative set-up and their linkage with forest management.
7. **Forestry in a Global Context**

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, the students will be able to

CO1 Formulate the fundamental concepts, definition and features of forestry.
CO2 Describe various forest types of Bangladesh, their significant characteristics and management principles.
CO3 Analyze various factors of forest degradation and the means of overcoming those problems.
CO4 Evaluate the structure and functions of various institutions of Bangladesh dealing and governing forest management, education and research**.**
CO5 Explain the global trends and development having regards to forest**.**

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10

CO1	3			1						
CO2	3			2		1				
CO3	3			2						
CO4	3					3	2		1	
CO5	3			2		3		3		2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01,CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01,CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL05	CA02, CA03,CA04, CA05, SA01
CO5	TL01, TL05	CA02,CA03, CA04, CA05, SA01

Recommended References:

1. Grebner, D. L., Bettinger, P., Siry, J. P., & Boston, K. (2021). *Introduction to forestry and natural resources*. Academic Press.
2. Rahman, L. M. (2016). Forest resources. In *Bangladesh National Conservation Strategy* (pp. 2–13). Government of the People's Republic of Bangladesh.
3. Reza, A. A., &Hasan, M. K. (2019). Forest biodiversity and deforestation in Bangladesh: The latest update. Forest *degradation around the world*. IntechOpen. https://doi.org/10.5772/intechopen.85986 *(add DOI or URL if available)*
4. Haque, S. M. S. (2013). *Introduction to forestry in Bangladesh* (331 p.). Institute of Forestry and Environmental Science, University of Chittagong, and USDA.
5. Dey, T. K. (2018). *Useful plants of Bangladesh* (387 p.).
6. Helms, J. A. (1998). *The dictionary of forestry* (2nd ed., 210 p.). Society of American Foresters.

Course No: FES 08211124	**Credit: 1.0**	**Year: First**	**Semester: First**
Introduction to Forestry Laboratory			**Course Status: Practical**

Course Rationale:
This laboratory course is designed to provide hands-on experience to enrich the understanding and skills regarding basic forestry concepts, types, and their significance from national and global perspectives. Key attributes like forest measurement, nursery techniques, and site-specific tree species selection criteria will be the salient part of the lab course.

Course Objectives:
The objectives of the course are to-

- Enrich students to recognize common tree species ,their basic characteristics and significance
- Enable students to perform simple tree measurements and basic nursery techniques
- Understand site-species matching, multiple use trees
- Develop knowledge base on key forestry issues of national and global significance

Course Content:

1. Learning major tree species of the SUST Campus
2. Presentation on national, regional and global forestry context
3. Basic understanding of the forest measurements
4. Key aspects of forest nursery

5. Site-specific plant types, multipurpose tree species (MPTs) and Medicinal Plants(MPs) of economic , economic and significance

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, the students will be able to

CO1 Recognise major forest tree species according to forest and ecosystem type of Bangladesh.
CO2 Perform basic exercise to measure trees.
CO3 Present observations and communication of basic forestry concepts and their significance

CO4 Develop knowledge base on key forestry issues of national and global significance

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3			1			1		1	
CO2	3						1		1	
CO3	3			1			1		1	
CO4				1						

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL02, TL03, TL04, TL05	CA02, CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL02, TL03, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL01, TL02, TL04,TL05	CA03, CA04, CA05, SA03

Recommended References:

Published documents referred and supplied by the Course teacher
Official website and links of the respective institutions
Real time field visit.

Course No: FES 05211121	**Credit: 2.0**	**Year: First**	**Semester: First**
Introduction to Environmental Science			**Course Status: Theory**

Course Rationale:
This course is designed to introduce students to the fundamental concepts of environmental science, including its historical development, major environmental challenges, and interdisciplinary scope. It aims to build foundational knowledge on pollution, climate systems, energy use, and natural hazards, while highlighting the interconnection of environmental science with fields such as natural sciences, engineering, and policy. Through this course, students will also gain awareness of institutional responses to environmental issues at both national and international levels, preparing them to understand and critically engage with contemporary environmental challenges.

Course Objectives:

The objectives of the course are to-

- Understand the fundamental concepts, scope, and interdisciplinary nature of environmental science.
- Develop awareness of major environmental issues such as pollution, climate change, and energy use.
- Gain basic knowledge of weather systems, environmental hazards, and disaster types.
- Familiarize themselves with the roles of national and international institutions in addressing environmental challenges.

Course Content:

1. **Basic Concept of Environmental Science:** Definition of Environmental Science; Environmental History: key events and milestones (brief overview); Environmental Science vs. Environmentalism; Scope and goals of Environmental Science; Relation to other disciplines of science; State of environmental awareness in society.
2. **Pollution and Pollutants:** Definition and types of pollution; Classification of pollutants; Transfer of pollutants across different environmental media; Basic concepts of biotransformation and bioaccumulation.
3. **Climate and Atmospheric Issues:** Greenhouse gases; Global warming and climate change; Acid rain; Ozone layer depletion; El Niño and La Niña phenomena.
4. **Weather Systems and Meteorology:** Definition and scope of meteorology; Basic understanding of weather and forecasting; Key instruments used in meteorological observations (e.g., thermometer, barometer, anemometer, rain gauge) – overview
5. **Environmental Hazards and Disasters:** Concepts of risk, hazard, disaster and vulnerability; Types of hazards (overview).
6. **Environmental Governance and Institutions:** Functional role of selected national organizations (e.g., DOE, MOFE, BCAS, ICCCAD) and international organizations (e.g., UNEP, IPCC, UNFCCC) – basic overview.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, the students will be able to

CO1 Explain key concepts, historical developments in environmental science, including its relationship with other scientific disciplines, and describe major environmental issues such as global warming, acid rain, ozone depletion, and their observable impacts.

CO2 Identify and classify various types of pollution, pollutants, and their effects on ecosystems and human health.

CO3 Demonstrate understanding of basic meteorological concepts, weather measurement tools, and their relevance to environmental studies.

CO4 Recognize and evaluate common environmental hazards and understand the role of environmental institutions in managing environmental problems.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2				1				
CO2	3	2					1			
CO3	3	2	2							
CO4	3	2				2				

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Botkin, D. B., & Keller, E. A. (2014). *Environmental science: Earth as a living planet* (9th ed.). Wiley.
2. Cunningham, W. P., & Cunningham, M. A. (2016). *Principles of environmental science: Inquiry and applications* (8th ed.). McGraw-Hill Education.
3. Rajan, S. C. (2011). *Environmental science and engineering* (Indian ed.). Vikas Publishing House.
4. Miller, G. T., &Spoolman, S. E. (2018). *Environmental science* (15th ed.). Cengage Learning.
5. Sharma, P. D. (2020). *Ecology and environment* (Latest ed.). Rastogi Publications.

Course No: **FES 05211122**	**Credit: 1.0**	**Year: First**	**Semester: First**
Introduction to Environmental Science Laboratory			**Course Status: Practical**

Course Rationale:

This laboratory course is designed to provide hands-on experience in monitoring and analyzing key environmental parameters such as air and water quality and noise pollution. It equips students with fundamental skills in detecting environmental pollutants using field and laboratory techniques.

Course Objectives:

The objectives of the course are to:

- Enhance the skill of students in measuring basic air quality parameters (SPM).
- Apply the knowledge to determine key physical and chemical properties of water and wastewater.
- Developing the skill of students in analyzing environmental noise levels.

Course Content:

1. **Air Quality Analysis:** Measurement of suspended particulate matter (SPM) using basic methods.
2. **Water and Wastewater Testing:** Determination of major physical and chemical properties of water and wastewater- pH, EC, turbidity, TDS, TSS, total solids, total hardness, total alkalinity, DO, BOD.
3. **Noise Pollution Monitoring:** Analyze environmental noise using sound level meter.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, the students will be able to

CO1 Measure suspended particulate matter (SPM) in ambient air.
CO2 Determine basic physical and chemical properties of water and wastewater.
CO3 Analyze environmental noise levels.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	1	1			3	1		1	
CO2	3	1	1			3	1		1	
CO3	3	1	1			3	1		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1 Any documents supplied by the related course teacher(s).

Course No: AMS 05111101	**Credit: 3.0**	**Year: First**	**Semester: First**
Botany			**Course Status: Theory**

Course Rationale:

This course introduces students to fundamental concepts in Botany, focusing on its role in tree improvement, genetics, and forest management. The primary goal is to enhance crop productivity to meet growing demands while promoting environmental stability.

Course Objectives:

The objectives of the course are-

- To help students conceptualize the basic theories of general botany.
- To develop the ability to effectively address forest management challenges by integrating principles of silviculture, forest genetics, tree improvement, and other key aspects of forestry.

Course Content:

1. **Introduction:** Basic concepts *and related terms* used in Forest Botany, *scope of forest botany in forestry*
2. **Plant Morphology and Anatomy:** Plant organs and tissue systems; Vascular plant and their types; Anatomy and function of leaf, stem and root; Structure and function of flower, fruit and seed.
3. **Phytogeography:** Salient features of hydrophytes, xerophytes, halophytes and mesophytes, and their adaptations.
4. **Cytology:** Ultra-structure and function of each plant cell organelles; Molecular structure of DNA and RNA; Phases of Mitosis and Meiosis, *Cell division, differentiation, dedifferentiation, cell growth and cell and apoptosis*
5. **Plant Breeding:** Mendel's laws; Di- and Tri-hybridization; Goodness of Di- and Tri-hybridization; Polyploidies; Crossing over; Linkage; Transcription and Translation; Gene *expression,* Regulation and Mutation; and Genetic mapping.
6. **Reproductive biology of higher plants:** Pollination; Fertilization; Dispersal of fruits and seeds; asexual reproduction.
7. **Embryology:** Basic concepts of embryology. *Formation and development of plant embryo-microsporogenesis, megasporogenesis, embryogenesis and endosperm develop*
8. **Growth and development:** Basic concepts of growth and development.
9. **Economic Botany:** *Economically* Important plant species concerning economic importance commonly found in the forests and homesteads.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Understand the concept of forest botany and its economic importance, identify the morphological structures of plants, developmental aspects of male and female reproductive systems, pollination, fertilization process, structure, function of reproductive organs in plants and physiology of flower and fruit development and plant growth regulators.

CO2 Describe the basic aspects of the anatomy of plant tissue systems and the structure of plant organs.

CO3 Interpret the ultra-structure of cell and its division, the function of cells at the molecular level and molecular basis of plant life.

CO4 Apply Mendel's principles and acquire knowledge on cytoplasmic inheritance, regulation of gene expression

in eukaryotic organisms.

CO5 Compare the vegetation types and justify different plant formations on the planet and their adaptation.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	3	1							
CO2	3	3	1						1	
CO3	3	3	2						1	
CO4	3	3	2				2			
CO5	3	3	2				2			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Cobley, L. S., & Steele, W. M. (1976). *An introduction to the botany of tropical crops* (2nd ed.). The English Language Book Society and Longman.
2. Dutta, A. C. (2000). *Botany for degree students* (6th ed.). Oxford University Press.
3. Dutta, A. C. (1972). *A class book of botany* (14th ed.). Oxford University Press.
4. Hill, A. F. (1979). *Economic botany*. Tata McGraw-Hill.
5. Huq, A. M. (1986). *Plant names of Bangladesh*. Bangladesh National Herbarium.
6. Sharma, O. P. (1980). *A manual of practical botany* (Vol. 1). PragatiPrakashani.
7. Beck, C. B. (2010). *An introduction to plant structure and development: Plant anatomy for the twenty-first century* (2nd ed.). Cambridge University Press.
8. Evert, R. F., &Eichhorn, S. E. (2013). *Raven biology of plants* (8th ed.). W. H. Freeman and Company.
9. Mauseth, J. D. (2008). *Botany: An introduction to plant biology* (4th ed.). Jones and Bartlett Publishers.

Course No: AMS 05111102	**Credit: 1.0**	**Year: First**	**Semester: First**
Botany Laboratory			**Course Status: Practical**

Course Rationale:

This course focuses on practical applications of Botany in field-level management, aiming to enhance plant productivity to meet growing demands while promoting environmental stability.

Course Objectives:

The objectives of the course are-

- To acquaint students with the basic tools of different Botanical specimens.
- To develop skills to solve the relevant field-level management issues to meet the needs of the country.

Course Content:

1. **Plant Morphology and Anatomy:** Study of special plant organs; collection and identification of fruits and seeds of forest trees; anatomical study of dicots and monocots stem, root and leaf.
2. **Cell Division:** Demonstration of mitosis and meiosis.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Analyze and classify the internal structure of monocot and dicot (root, stem and leaf), secondary thickening (Dicot and Monocot).
CO2 Identify and illustrate specialized plant organs such as modified roots, stems and leaves.
CO3 Compare mitosis and meiosis cell division through the cytological preparation from root tips.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	1					1		1	
CO2	3	1					1		1	
CO3	3	1					1		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Cobley, L. S., & Steele, W. M. (1976). *An introduction to the botany of tropical crops* (2nd ed.). The English Language Book Society and Longman.
2. Dutta, A. C. (1972). *A class book of botany* (14th ed.). Oxford University Press.
3. Huq, A. M. (1986). *Plant names of Bangladesh*. Bangladesh National Herbarium.
4. Pandey, B. P. (1999). *Modern practical botany*. S. Chand & Company.
5. Sharma, O. P. (1980). *A manual of practical botany* (Vol. 1). PragatiPrakashani.

Course No: FES 08211122	**Credit: 1.0**	**Year: First**	**Semester: First**
Fieldwork in Forests (Swamp Forest Ecosystems)			**Course Status: Practical**

Course Rationale:
This course will allow a student to gain hands-on experience of the real forest and forestry sector situation in Bangladesh.

Course Objectives:
The objectives of this course are to-

- Help students understand forest diversity, its trends, applications, role in environmental stability.
- Enable students to apply the acquired knowledge and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Diversity of forest in Bangladesh and its components;
2. The overall situation of different forest, forestry and environmental sector related areas of Bangladesh

3. Introduction to nursery management (theory); Spot demonstration of the nursery practices; Nursery site organization
4. Practical application of all core courses of the current semester will be demonstrated.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Classify the forest diversity, its trends, applications, role in environmental stability.
CO2 Outline the procedure of roles management in forest and forestry sectors.
CO3 Synthesize relevant field level management issues to meet the needs of the country.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3								3	
CO2	3								3	
CO3	3						3		3	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL06	CA03, CA04, CA05
CO2	TL04, TL06	CA03, CA04, CA05
CO3	TL04, TL06	CA03, CA04, CA05

Recommended References:
1 Any documents supplied by the related visited areas' officials.
2 Any documents supplied by the related course teacher(s).

Course No: CHE 05311101	**Credit: 3.0**	**Year: First**	**Semester: First**
General Chemistry			**Course Status: Theory**

Course Rationale:
Students will learn basic concept of atomic structure, will learn to analyze periodic table, will learn how to read and write chemical formulae. This course will teach them concept of acid-base and reaction kinetics.

Course Objectives:
The objectives of this course are to-

- Familiarize the students with the basic concept of electronic structure
- Acquire the knowledge about the properties of elements on the periodic table
- Understand the concept of chemical formula and equation
- Acquire the basics of acid-base concepts and apply them to identify different acids and bases
- Understand gaseous state of matter and their properties
- Introduce preliminary ideas of chemical equilibrium and kinetics
- Acquaint students with the fundamentals of organic chemistry

Course Content:

1. **Introduction to Chemistry in Forestry:** Role of chemistry in soil–plant–environment systems. (CO1)
2. **The periodic Table:** Development of the periodic table, Electron arrangements and the periodic table, Summarized chemical properties of s-block, p-block, d-block and f-block elements; Essential plant nutrients: N, P, K, Ca, Mg, Fe, Zn, Cu. (CO1)
3. **Chemical formulas and equations:** Concept of mole, Solution: Different concentration units, Percent composition from formula, Writing and balancing chemical equations, Mass relations in reactions, Limited reactant and theoretical yield. (CO2)
4. **Organic Chemistry:** Introduction**,** Classification, Nomenclatures, and Properties (Physical & Chemical) of (i) Aliphatic and aromatic hydrocarbons, (ii) Carbonyl compounds, (iii) Carboxylic acids and (iv) Carbohydrates: cellulose and hemicellulose, Lignin Chemistry, Natural Products: Chemistry of resins, gums, tannins, and essential oils. (CO5)
5. **Acids, Bases:** Theories and Modern definition of acids and bases, Dissociation constant, strength, pH, Buffer solution, Mechanism of buffer action. (CO3)
6. **Environmental and Analytical Chemistry:** Acid rain, Water quality parameters, Pulp and paper chemistry (lignin removal processes), Wood preservation chemicals (fungicides, insecticides), Biofuels: chemical conversion of biomass (ethanol, charcoal), Basic analytical techniques: Titration, Spectrophotometry, Chromatography, Coordination chemistry in forestry: Chlorophyll (Mg-complex), Metalloenzymes in plant systems. (CO4)
7. **Chemical Kinetics:** Rate laws, rate constant, equilibrium constant, order of reaction, molecularity, first and second order reactions and their rate expressions. (CO4)

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand electronic structure of atoms, and explain periodic trends in physical and chemical properties of elements, and their roles in soil–plant–environment systems.
CO2 Predict the percent composition of a compound from formula, balance chemical equations, and measure mass relations in reactions including limiting reactant and theoretical yield.
CO3 Define and apply the modern concepts of acids and bases, and measure their strengths including pH and buffer systems.
CO4 Explain chemical kinetics and equilibrium, and interpret environmental chemical processes and basic analytical techniques.
CO5 Describe the structure and properties of aliphatic and aromatic hydrocarbons, carbonyl compounds, carboxylic acids, carbohydrates, and natural products relevant to forestry.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3		3							
CO3	3									
CO4	3		3						1	
CO5	3								1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL03, TL05	CA01, SA01, SA02
CO2	TL01, TL03, TL05	CA01, SA01, SA02
CO3	TL01, TL03, TL05	CA01, CA02, SA01, SA02
CO4	TL01, TL03, TL05	CA02, CA04, SA01, SA02
CO5	TL01, TL02, TL03, TL05	CA02, CA04, SA01, SA02

Recommended References:

1. Haider, S. Z. (n.d.). *Introduction to modern inorganic chemistry*.
2. Haque, M., &Mollah, M. (n.d.). *Physical chemistry*.
3. Morrison, R. T., & Boyd, R. N. (1992). *Organic chemistry* (6th ed.). Prentice Hall.
4. Chang, R. (n.d.). *General chemistry*. McGraw-Hill.

Course No: CHE 05311102	**Credit: 2.0**	**Year: First**	**Semester: First**
Chemistry Laboratory			**Course Status: Practical**

Course Rationale:

This course is going to introduce the students with different analytical techniques. They will learn to identify anions and cations in unknown molecules. Students will also learn to estimate different concentrations of acids and bases.

Course Objectives:

The objectives of this course are to-

- Familiarize students with the qualitative analysis of inorganic salts
- Acquaint students with the qualitative analysis of organic functional groups
- Accumulate practical skill on titration

Course Content:

1. Qualitative analysis of inorganic salts:

- Separation and identification of group I cations
- Separation and identification of group II cations
- Separation and identification of group IIIA and IIIB cations
- Separation and identification of group IV cations
- Separation and identification of group V cations
- Identification of anions

1. Qualitative analysis of organic salts:

- Analysis of the functional groups of organic compounds

2. Quantitative analysis:

- Standardization of NaOH solution with standard oxalic acid solution

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Identifying and determining analytical group cations in the solution of inorganic salt.
CO2 Use an analytical scheme to separate and identify the known ions from a mixture of inorganic salts.
CO3 Apply designed analytical scheme to separate and identify the unknown ions from a mixture of inorganic salts.
CO4 Identify the functional group(s) presence in an organic compound and then perform a confirmatory test.
CO5 Determine the concentration of an unknown solution using a standard solution of known

concentration.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		2						2	
CO2	3		2						2	
CO3	3		2			1			2	
CO4	3		3			2			1	
CO5	3		3						1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL03, TL06	CA05, SA01, SA02
CO2	TL01, TL03, TL06	CA05, SA01, SA02
CO3	TL01, TL03, TL05, TL06	CA05, SA01, SA02
CO4	TL01, TL03, TL06	CA05, SA01, SA02
CO5	TL01, TL03, TL06	CA05, SA01, SA02

Recommended References:

1. Vogel, *Qualitative Inorganic Analysis*
2. A.I. Vogel, *A Text Book of Practical Organic Chemistry*
3. A.I. Vogel, *Elementary Practical Organic Chemistry* (Part 1)
4. Vogel, *Text book of Quantitative Analysis.*

Course No: MAT 05411101	**Credit: 2.0**	**Year: First**	**Semester: First**
Calculus and Differential Equations		**Course Status: Theory**	

Course Rationale:

This course is designed for the students of the department of Forestry and Environmental Science to introduce the concept of calculus and differential equations and its real-world implementations. The fundamental concept of different topics of differential and integral calculus is given along with relevant examples. The formulation of Differential Equations from physical problems is introduced and the solution is limited to first order only. Several methods are discussed to find appropriate solutions of the first order Differential Equations.

Course Objectives:

The objectives of this course are to-

- Introduce students to the fundamental concepts of differential and integral calculus.
- Acquire knowledge of different topics of differential calculus and solve relevant physical problems.
- Teach indefinite and definite integrals through different methods of integrations and apply them to find area and volume of a given function.
- Acquire the concept of ordinary differential equations and how to formulate them from forestry and environmental related problems.
- Teach techniques and methods to solve first order differential equations and its applications.

Course Content

1. **Differential Calculus:** Limit, continuity and differentiation of functions; partial differentiation; Leibnitz's theorem and its applications; maxima and minima of a function of one variable.
2. **Integral Calculus:** Methods of integration; integration by parts; definite integral; areas and volumes.

3. **Differential Equations:** Definition and solution of ordinary differential equations; first order ordinary differential equations.

Course Learning Outcomes (COs)
After the successful completion of the course, students will be able to

CO1 Find the domain range of a function and sketch curve of a piece-wise function.
CO2 Compute limit, continuity and differentiability of a function.
CO3 Determine derivatives of different types of function through several techniques, compute partial derivatives of multivariable functions and solve first order differential equations by using different methods.
CO4 Solve indefinite and definite integrals through different methods, find area and volume of a curve by using integration, apply Leibnitz theorem for successive differentiation and find nth order derivative.
CO5 Determine Increasing functions, decreasing functions, and concavity, relative and absolute extrema of a function.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1			3			1	2			
CO2			3			1	2			
CO3			3			1	2			
CO4			3			1	2			
CO5			3			1	2			

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL02, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL02, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL02, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL02, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL02, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Thomas and Finney: Calculus and Analytic Geometry
2. Stewart, J.: Calculus
3. Das and Mukherjee: Differential Calculus
4. Das and Mukherjee: Integral Calculus
5. Ross, S. L.: Differential Equations
6. Ayres, F.: Differential equation

Course No: ENG 02311101	Credit: 2.0	Year: First	Semester: First
English Language		Course Status: Theory	

Course Rationale:
This course is expected to develop two basic skills i.e. reading and writing. A variety of reading strategies and texts will be used to effectively develop first year students' academic reading skills thereby facilitating their future study. Also, the course focuses on developing the writing skills of students by familiarizing them with grammar rules, providing them with practice and enabling them to demonstrate the accurate use of grammar in their writing.

Course Objectives:
The objectives of the course are-

- To enable students to write with accuracy;
- To facilitate effective and comprehensible writing;
- To raise awareness of common errors that occur in writing;
- To develop students' ability to understand write-ups on issues of general concern;
- To improve the vocabulary of learners for effective communication.

Course Content:
Reading

- Different Reading Strategies
- Guessing Meaning from the Context
- Critical Reading (Analyze)
- Critical Reading (Synthesize)
- Critical Reading (Evaluate)
- Annotation
- Summary Writing

Writing

- Forms and functions of different word categories (noun, verb, adjective, etc.)
- Aspects and uses of tense
- Subject-verb agreement
- Use of infinitive, gerund, present participle, past participle, modals, causatives, conditionals, subjunctives, modals.
- Use of sentence connectors/ cohesion markers/ punctuation
- Effective combination of sentences (simple, complex, compound)
- Developing a paragraph.

Course Learning Outcomes (COs):
After the successful completion of the course, students will be able to

CO1 Apply grammar rules
CO2 Produce grammatically correct meaningful sentences
CO3 Express oneself correctly by using appropriate words, phrases, sentences or ideas
CO4 Critically reflect on a text (grasp abstract ideas and interpret them effectively, arrive at well-reasoned conclusions and solutions)
CO5 Extract information from passages accurately

Mapping of CLOs with PLOs

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1O
CO1	2	3								
CO2						2		2		

CO3						2			3	
CO4				2		2	2			
CO5						2				

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Payle, M. (2019). *Cliff's TOEFL preparation guide* (12th ed.). Cliffs Notes Inc.
2. Soars, L., & Soars, J. (2014). *New Headway upper intermediate student's book*. Oxford University Press.
3. Tibbits, E. E. (Ed.). (2013). *Exercises in reading comprehension*. Longman.
4. Otherresources recommended by course instructors
5. A selection of 08-10 editorials and reports from newspapers/magazines/journals, etc.
6. Reading texts in New Headway Upper Intermediate Student's Book (Current edition).
7. Selected passages from recommended books.
8. A selection of other materials may be supplied as handouts by the instructor as necessary.

Course No: AMS 05221201	**Credit: 2.0**	**Year: First**	**Semester: Second**
Wildlife Biology			**Course Status: Theory**

Rationale of the Course

Wildlife Biology provides the biological foundation for understanding wild animals in forest ecosystems. It introduces students to wildlife diversity, structure, function, behaviour, adaptation, and population characteristics that support conservation and scientific forest management.

Course Objectives

This course aims to:

1. Introduce the basic principles and scope of wildlife biology.
2. Develop understanding of wildlife classification, morphology, physiology, behaviour, and adaptation.
3. Explain wildlife population characteristics, reproduction, and survival strategies.
4. Build the foundation for advanced study in wildlife ecology, conservation, and management.
5. Improve students' ability to interpret wildlife-related biological information scientifically.

Course Content

1. **Introduction to wildlife biology**: scope, importance, and relation to forest ecosystem.
2. **Wildlife classification and diversity**: identification, classification, basic features of major vertebrate groups- fish, amphibians, reptiles, birds, and mammals.
3. **Wildlife morphology and anatomy**: morphology and body structure of wildlife, wildlife forensics.
4. **Wildlife physiology and adaptation:** basic physiological processes and environmental adaptation of major wildlife.

5. **Evolution and co-evolution:** definition, theories of evolution, co-evolution, Lamarckism, Darwinism, speciation, evolution of elephant, tiger, monkey and deer, highlighting their importance.
6. **Reproduction and life history**: breeding systems, growth, development, and survival strategies of major wildlife.
7. **Animal behaviour:** definition, types, learning, instinct; food and feeding habits of deer, elephant, tiger, monkey, wild-boar, porcupine, rat and birds; communication, migration, territoriality, and social behaviour.
8. **Population characteristics**: census, population size, density, natality, mortality, and age structure.
9. **Economic aspects of wildlife**: economically important fauna found in terrestrial and aquatic ecosystems in Bangladesh.
10. **Threats and conservation:** human–wildlife conflict and mitigation, major threats to wildlife and biological basis of conservation.

Course Learning Outcomes (COs)

CO1. Explain the basic concepts, scope, and importance of wildlife biology.

CO2. Describe the classification, diversity, anatomy, physiology, behaviour, and adaptation of major wildlife groups.

CO3. Interpret wildlife population, reproduction, and life-history characteristics in ecological contexts.

CO4. Relate biological knowledge of wildlife to conservation and forest management needs.

CO5. Communicate economic importance and wildlife biology concepts clearly with scientific and ethical awareness.

CO-PO Mapping Matrix

Legend: 3 = Strong, 2 = Moderate, 1 = Low

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CLO1	3	1				1	1			
CLO2	3	1			1	1	1			
CLO3	2	1			1	2	2			
CLO4	2	1			1	2	2			
CLO5	1	3			1	1	1	3	2	

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL02, TL05	CA01, CA03, CA04, CA05, SA01, SA02
CO2	TL01, TL02, TL05	CA02, CA03, CA04, CA05, SA01, SA02
CO3	TL01, TL02, TL05	CA03, CA04,CA05, SA01, SA02
CO4	TL01, TL02, TL05	CA03, CA04,CA05, SA01, SA02
CO5	TL01, TL02, TL05	CA03, CA04,CA05, SA01, SA02

Suggested References

1. Mills, L. S. (2023). Conservation of Wildlife Populations: Demography, Genetics, and Management (3rd ed.). Wiley.
2. Krausman, P. R., & Cain, J. W. (2024). Introduction to Wildlife Management: The History, Ecology, and Practice of Wildlife Management (2nd ed.). Johns Hopkins University Press.
3. Schmidt-Nielsen, K. (2024 reprint). Animal Physiology: Adaptation and Environment. Cambridge University Press.

4. Johnston, A., Matechou, E., & Dennis, E. B. (2023). Outstanding challenges and future directions for biodiversity monitoring using citizen science data. Methods in Ecology and Evolution, 14(1), 103-116.
5. Beaulieu, M., & Masilkova, M. (2024). Plugging biologging into animal welfare: An opportunity for advancing wild animal welfare science. Methods in Ecology and Evolution, 15(12), 2172-2188.
6. Mirante, D., Santini, L., Costantini, D., & Benítez-López, A. (2025). Glucocorticoid responses of wildlife to anthropogenic stressors are influenced by disturbance type and species traits. Functional Ecology, 39(3), 681-697.

Course No: AMS 05221202	Credit: 1.0	Year: First	Semester: Second
Wildlife Biology Laboratory			Course Status: Practical

Rationale of the Course

This laboratory course complements Wildlife Biology by providing hands-on exposure to wildlife identification, biological observation, specimen-based learning, and basic field and laboratory techniques. It helps students connect theory with practical evidence.

Course Objectives

This course aims to:

- Provide hands-on training in identifying major wildlife groups and biological features.
- Develop skills in observing morphology, signs, behaviour, and basic measurements.
- Introduce simple laboratory and field techniques used in wildlife studies.
- Improve recording, reporting, and teamwork skills in practical work.
- Promote safe and ethical handling of wildlife-related materials and data.

Course Content

1. **Identification of wildlife groups:** practical identification of major vertebrate groups.
2. **Study of external morphology:** observation of body form, feathers, fur, scales, beaks, and horns.
3. **Wildlife signs and evidence:** tracks, feces, nests, burrows, and feeding marks.
4. **Study of biological materials:** skulls, feathers, horns, shells, and other diagnostic materials.
5. **Measurement and documentation and reporting:** basic measurements, field notes, specimen collection, diagrams, and short reports..

Course Learning Outcomes (CLOs)

CLO1. Identify major wildlife groups and common wildlife signs using practical characters.

CLO2. Apply basic laboratory and field techniques for wildlife observation, measurement, and recording.

CLO3. Use practical materials to interpret basic biological features of wildlife.

CLO4. Organize and present simple wildlife practical data in tables, notes, and short reports.

CLO5. Perform practical work responsibly with safety, ethics, and teamwork.

CLO-PO Mapping Matrix

Legend: 3 = Strong, 2 = Moderate, 1 = Low

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CLO1	3	1	1			1	1			
CLO2	3	1	2			2	2		2	
CLO3	2	1	1			2	2			
CLO4	1	3	2			1	1			
CLO5	1	1	1			1	1	3	3	

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO2	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO3	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO4	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO5	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02

Suggested References

1. Sutherland, W. J. (2023). Ecological Census Techniques: A Handbook (3rd ed.). Cambridge University Press.
2. Goodenough, A. E., & Perks, S. J. (2025). Using non-invasive and minimally-invasive methods to inform conservation of terrestrial mammals: A review of novel techniques and new applications. Journal for Nature Conservation.
3. Silva-Rodríguez, E. A., Cortés, E. I., Vásquez-Ibarra, V., Gálvez, N., Cusack, J., Moreira-Arce, D., et al. (2025). A protocol for error prevention and quality control in camera trap datasets. Journal of Applied Ecology, 62(4), 773-782.
4. Barroso, P., & Palencia, P. (2025). Pixelated pathologies: Camera trapping as a tool for monitoring wildlife health. Journal of Applied Ecology, 62(11), 2894-2913.
5. Roemer, C., Haquart, A., López-Baucells, A., Besnard, A., et al. (2025). Current frontiers in the passive acoustic monitoring of bats. Methods in Ecology and Evolution, 16(11), 2534-2544.
6. Beaulieu, M., & Masilkova, M. (2024). Plugging biologging into animal welfare: An opportunity for advancing wild animal welfare science. Methods in Ecology and Evolution, 15(12), 2172-2188.

Course No: FES 08211221	**Credit: 3.0**	**Year: First**	**Semester: Second**
Dendrology and Species Silviculture			**Course Status: Theory**

Course Rationale:

A strong foundation in dendrology and species silviculture is vital for sustainable forest management and conservation. This course equips students with essential skills in tree identification, taxonomy, and ecological distribution, focusing on Bangladesh's native species. By examining phenology and silvicultural needs, students gain insights for adaptive management under climate change. Additionally, the course emphasizes the role of herbaria, botanical gardens, and arboreta in biodiversity conservation, supporting ecological literacy and sustainable forestry practices.

Course Objectives:

The objectives of the course are-

- To provide comprehensive knowledge of key forest tree species, including their scientific and common names, taxonomic families, geographic distribution, ecological roles, & economic uses.
- To develop proficiency in tree species identification and classification using taxonomic hierarchy and diagnostic morphological features.
- To examine the ecological and silvicultural requirements of major tree species from Bangladesh's hill forests, Sal forests, mangroves, and social forestry plantations.
- To apply phenological knowledge in silvicultural practices.
- To understand the role of conservation infrastructures (herbaria, botanical gardens, arboreta) in biodiversity preservation, taxonomic research, and forestry education.

Course Content:

Dendrology

1. **Introduction to Dendrology**: Scope and significance of dendrology in forestry; Concepts of Phenology, phenophases.
2. **Plant Taxonomy & Classification**: Definition, goals and fundamentals of taxonomy; taxonomy and systematics, phases of taxonomy, taxonomic hierarchy; classification, short history of plant classification, units of classification, types of classification system; overview of major classification system: Linnaeus system, Bentham and Hooker's system, Engler and Prantl's system, Hatchinson's system, Cronquist's system; nomenclature, binomial system of nomenclature, International Code of Nomenclature for algae, fungi, and plants; tree species identification.
3. **Identifying characters, economic importance and examples of forest tree species of the following families with emphasis on Bangladesh**: Gymnosperm: Cycadales, Gnetales, and Coniferales, Monocots: Graminae (Bamboosoidae) and Palmae, Dicots: Verbenaceae, Dipterocarpaceae, Fabaceae (Leguminosae), Meliaceae, Sterculiaceae, Anacardiaceae, Rhizophoraceae, Moraceae, Rubiaceae, Myrtaceae, Lythraceae (Sonneratiaceae), Loranthaceae, Acanthaceae (Avicenniaceae).
4. **Conservation and Documentation Systems**: Role and functions of Herbarium, Botanical Garden and Arboretum in forestry education, research, and conservation.

Species Silviculture

5. **Fundamentals silviculture:** Basic concepts: stages of tree growth, different plant parts, e.g., bark, bole and branches, arrangement and types of leaves, flowers, and fruits; Importance of phenology in silvicultural applications and assessment of climate change impacts.
6. **Silviculture of important species:** Teak, Sal, Garjan, Gamar, Mahogani, Dhakijam, Chapalish, Telsur, Akashmoni, Neem, Lohakat, Sissoo, Babla, Rain Tree, Eucalyptus, Mangium, Arjun, Jarul, Koroi, Bot, Champa, Amloki, Kadam, Simul, Bamboo, Cane, Patipata, Sundri, Keora, Gewa, Goran, Baen, Kankra, Rhizophora spp, Passur, Dhundal, Golpata.

Course Learning Outcomes (COs):

After the successful completion of the course, students will be able to

CO1 Describe & explain the fundamental concepts of dendrology, species silviculture and phenology.
CO2 Demonstrate the implication of taxonomy, classification, nomenclature in natural science, with emphasis on their relevance to forestry.
CO3 Identify major forest tree species of Bangladesh through key diagnostic features of selected plant families.
CO4 Analyze and evaluate the geographical distribution, habitat requirements, regeneration techniques, economic importance of key species from different forest types of Bangladesh.
CO5 Evaluate the role of different conservation systems in biodiversity preservation, taxonomic research, and sustainable forest management.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					1				
CO2	3					1				
CO3	3					1				
CO4	3					1				
CO5	3					1		3		

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Andrés Bravo-Oviedo, A., Pretzsch, H., del Río, M., & Löf, M. (2018). *Dynamics, silviculture and management of mixed forests* (Vol. 31). Springer.
2. Benson, L. (1957). *Plant classification*. Oxford & IBH Publishing Co.
3. Drury, C. H. (1985). *The useful plants of India*. International Book Distributors.
4. Godfray, H. C. J., Knapp, S., O'Neill, K., Hauser, C., & Watson, M. (2007). The web and the structure of taxonomy. *Systematic Biology, 56*(6), 943–955.
5. Hossain, M. K. (2015). *Silviculture of plantation trees of Bangladesh*. Arannayk Foundation.
6. Hossain, M. (2015). *Handbook of selected plant species of the Sundarbans and the embankment ecosystem*. Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH.
7. Jensen, M. (1995). *Trees commonly cultivated in Southeast Asia* (RAP Publication 1995/38). FAO.
8. Raven, P. H., Berlin, B., & Breedlove, D. E. (1971). The origins of taxonomy. *Science, 174*(4015), 1210–1213.
9. Sakai, S. (2001). Phenological diversity in tropical forest. *Population Ecology, 43*, 77–86.
10. Uddin, S. N., &Hasan, M. A. (2018). *Vascular flora of Chittagong and the Chittagong Hill Tracts* (Vols. 2 & 3). Bangladesh National Herbarium.
11. Zabala, N. Q. (1990). *Silviculture of species* (Field Document No. 14). UNDP/FAO/BGD/85/011, Institute of Forestry, Chittagong University & FAO.

Course No: FES 08211222	**Credit: 1.0**	**Year: First**	**Semester: Second**
Dendrology and Species Silviculture Laboratory			**Course Status: Practical**

Course Rationale:

The Dendrology and Species Silviculture Lab course is designed as a companion to cover the field aspect of theoretical knowledge on Dendrology and Species Silviculture (FES 08211221). This laboratory course focuses on the taxonomic identification of species on the basis of different morphological and phenological knowledge in the forest of Bangladesh. Attention is also given to specific families, traits, habits, and the uses of species.

Course Objectives:

The objectives of the course are-

- To help the students to develop ability in different species identifying keys in relation to taxonomy.
- To facilitate necessary knowledge about morphological, ecological, and phenological traits for identification of woody plants.
- To accumulate basic ideas about the distribution, habitat, ecology, and economic importance of species.
- To enhance the skill on the important method of collection & preservation of species as herbarium.

Course Content:

1. Terminology of taxonomic work.
2. Procedure of using identification key in taxonomic works.
3. Describing a plant following the procedure of identification.
4. Collection, preparation and preservation of herbarium specimens.
5. Recording of phenological characteristics of important tree species.

Course Learning Outcomes (COs):

After the successful completion of the course, students will be able to

CO1 Explain different terms of taxonomy.
CO2 Classify species according to taxonomy.
CO3 Describe plants with respect to different identification keys.
CO4 Prepare & maintain herbarium of species.
CO5 Compare different ecological & phenological characteristics of species

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3									
CO3	3					2				
CO4	3					2				
CO5	3					2				

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO5	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Benson, L. (1957). *Plant classification*. Oxford & IBH Publishing Co.
2. Drury, C. H. (1985). *The useful plants of India*. International Book Distributors.
3. Godfray, H. C. J., Knapp, S., Watson, M., Scoble, M., & House, T. (2007). The web and the structure of taxonomy. *Systematic Biology, 56*(6), 943–955. https://doi.org/10.1080/10635150701777521
4. Jones, B. S., & Luchsinger, A. E. (1979). *Plant systematics*. McGraw-Hill Book Company.
5. Raven, P. H., Berlin, B., & Breedlove, D. E. (1971). The origins of taxonomy. *Science, 174*(4015), 1210–1213. https://doi.org/10.1126/science.174.4015.1210
6. Victor, J. E., Koekemoer, M., Fish, L., Smithies, S. J., & Mössmer, M. (2004). *Herbarium essentials: The southern African herbarium user manual* (SABONET Report No. 25). Southern African Botanical Diversity Network.
7. Wondafrash, M. (2008). *A preliminary guide to plant collection, identification and herbarium techniques*. The National Herbarium, Addis Ababa University.

Course No: FES 08211223	**Credit: 3.0**	**Year: First**	**Semester: Second**
Forest Tree Physiology			**Course Status: Theory**

Course Rationale:

This course is aimed to provide an introduction to basic principles of plant function, primarily covering physical processes in plants, metabolism, secondary products, cell physiology, and introducing principles of growth and development. The emphasis of this course is plant function at the level of the organism.

Course Objectives:

The objectives of the course are to-

- Help them conceptualize basic theories of plants application in forest and homestead, their diverse physiological functions.
- Foster the analytical and critical process to pursue advanced courses in forestry, and to conduct their own plant physiology research.

Course Content:

1. **Concepts:** Definition and scope of plant physiology, physiological processes, Plant Cell Structure and Function, The Cell Wall (Structure, Biogenesis, and Function), Energy and enzymes.

2. **Transport and Translocation of water and solutes:**
    - Water and plant cells: Water transport processes e.g. diffusion, pressure driven bulk flow, osmosis, water potential gradient
    - Water balance of plants: water in the soil, water absorption by roots, water transport through the xylem,
    - Water movement from the leaf to the atmosphere, transpiration, guard cells and stomatal opening and factors affecting transpirations.
    - Plant mineral nutrition,
    - Uptake and transport of solute,
    - Phloem transport
3. **Biochemistry and Metabolism:** Photosynthesis: Pigments and Light, The Light Dependent Reactions, Carbon Metabolism and Regulation of the Calvin Cycle, Photorespiration and CO_2 Concentrating Mechanisms, C3, C4 and CAM plant, respiration and its kinds, mechanism, importance with affecting factors.
4. **Growth and Development:** Embryogenesis, meristems in plant development, leaf development, root development, cell differentiation, senescence and cell death, hormones: auxin, gibberellins, cytokinins, abscisic acid, ethylene.
5. **Stress physiology:** Response of plants to drought, heat and other stresses.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand key concepts in plant physiology.
CO2 Explain the physiological mechanisms involved in the uptake and transport of water and solutes, and the translocation of food by plants.
CO3 Become familiar with biochemical processes in plant metabolism and their interaction with physiological and ecological considerations.
CO4 Investigate plant growth and development, and the roles of plant hormones.
CO5 Analyse the responses of plants to environmental stressors and understanding the physiological mechanisms that allow plants to cope with adverse conditions.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					3	3		1	
CO2	3	1	1			3	3		1	1
CO3	3	1	2			3	3		2	1
CO4	3	1	1			3	3		2	1
CO5	3	1	2			3	3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Hopkins, W. G. (1995). *Introduction to plant physiology* (1st ed.). John Wiley & Sons, Inc.
2. Taiz, L., &Zeiger, E. (2010). *Plant physiology* (5th ed.). Sinauer Associates, Inc.
3. Pandey, S. N., & Sinha, B. K. (2009). *Plant physiology*. Vikas Publishing House.
4. Devlin, R. M., & Witham, F. H. (1983). *Plant physiology 4th edition*.PWS Publishers, Boston

Course No: FES 08211224	Credit: 1.0	Year: First	Semester: Second
Forest Tree Physiology Laboratory			Course Status: Practical

Course Rationale:
There is a huge variety and complexity of the functional characteristics of plants that needs to be demonstrated. Thus, this course is aimed at students of 1st year 2nd semester to help them better understand the structure and function of plants by conducting experiments and making observations designed to illustrate certain biological phenomena and principles. By providing them with an opportunity to work with their hands it is hoped that students will gain some insight into the principles and methodology of experimental biology.

Course Objectives:
The objectives of the course are to-

- Help them conceptualize basic scientific tools in investigating plant phenomenon
- Make the students understand the various water relations with plants.
- Provide the knowledge of physiological stress.
- Understand basic on how plants manufacture biological molecules

Course Content:

1. Analysis of chlorophyll pigments in tree leaves; measuring stomatal conductance, photosynthesis, water use efficiency, sap flow.
2. Measurement of the solute potential of plant cells, osmolytes, antioxidants, Amino acid and phenolic compounds.
3. Measurements of respiration rate of various plant parts.
4. Electrolytes leakage from leaf
5. Mineral nutrition and nutrition deficiency symptoms
6. Nodulation Assessments.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand how carbon assimilation and photosynthetic pigments of plants differ with respect biotic and abiotic stresses.
CO2 Explain the water status of plants and Mineral nutrition
CO3 Evaluate the responses of plants with respect to biotic and abiotic factors.
CO4 Examine the mechanism of carbohydrate production and storage with response to biotic and abiotic factors.
CO5 Analyze the seed germination and viability test for reproduction of plants.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		1			3	3		2	
CO2	3		1			3	3		2	
CO3	2		1			3	3		1	
CO4	2		1			3	3		1	
CO5	3		3			3	3		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO5	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Bajracharya, D. (1999). *Experiments in plant physiology: A laboratory manual*. Narosa Publishing House.
2. Hopkins, W. G., &Huner, N. P. A. (2004). *Introduction to plant physiology* (3rd ed.). Wiley.
3. Law, D., Malek, L., & Henderson, J. (2006). *Plant physiology and biotechnology laboratory manual*.

Course No: FES 08211225	**Credit: 2.0**	**Year: First**	**Semester: Second**
Non-Wood Forest Products			**Course Status: Theory**

Course Rationale:

In order to conserve and expand forest resources, forestry graduates need to have sound knowledge on the crucial role of NWFPs for forest conservation. This course will provide adequate knowledge of the basic concepts of NWFPs, their importance, problems and prospects; special reference to Bangladesh. Moreover, students will have a sufficient understanding regarding the contribution of NWFPs in the improvement of rural livelihoods and socioeconomic development as well as the basis for managing forests in a more sustainable way. Thus, study of NWFPs is fundamental to get complete knowledge of forestry.

Course Objectives:

The objectives of the course are to -

- Help the students conceptualize the types, concepts, propagation, management and utilization of the commercially important NWFPs of Bangladesh.
- Understand the role of NWFPs in reducing the depletion of our major forest resources.
- Provide knowledge on the crucial role of NWFPs in sustainable forest management.
- Facilitate necessary knowledge on important NWFPs collections and marketing methods.

Course Content:

1. **Introduction:** Definition and major groups of non–wood products, importance of non-wood forest products of the world with special reference to Bangladesh**.**
2. **Fibers and flosses**: Fibers from stems and leaves of forest species trees and shrubs producing silky flosses from fruits, grasses; status and uses; Wild fruits and vegetables; Forest species for paper making, fodder, mating, ropes, thatching, status and uses; Oils: Grass oils, wood oils, leaf oils from different parts of forest species through distillation and extraction, status and their uses.
3. **Bamboos, canes, patipata, golpatta, water hyacinth, sea grass and kans grass**: Distribution, identification, propagation and cultivation; seasoning, drying and preservation; status and uses in Bangladesh and abroad.
4. **Khair**: Distribution of khair yielding species, country method of manufacturing khair, status and its uses; Tans and dyes: Common Tans and dyes yielding forest species and their parts used; distribution of important species; status and uses; Gum, resins and oleoresin: Definition; gum and resin yielding forest species and their parts used; distribution of important species, status and uses.
5. **Agar**: Distribution of agar yielding species; propagation, cultivation and management of agar yielding species, extraction and processing**.**
6. **Medicinal plants**: Identification, distribution, status, their cultivation, and extraction from natural forests, useful portions of the plants and diseases in which they are used.
7. **Charcoal**: Utility of making charcoal, methods, properties and uses of charcoal, advantages and disadvantages of charcoal over fire wood.
8. **Honey:** Definition, status, method of collection from natural forests; artificial cultivation and uses; Lac: Definition, history and status of lac cultivation in Bangladesh, host plant and their grading, lac crop

cultivation; collection and storage of lac; manufacturing process of shellac and their uses; Silk: Definition, Important silkworms and their artificial culture, propagation, cultivation and management of mulberry and their common diseases and pests; silk production process.

9. **Markets and marketing of non-wood products**: Marketing (supply & value chains) of non-wood forest products, problems and prospects of non-wood products' marketing in Bangladesh;
10. **Management of NWFP**: Importance of NWFPs in sustainable forest management; Sustainable management strategy for non-wood products in Bangladesh.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Identify the major NWFPs as well as their distribution, habitat, propagation and collection methods.
CO2 Discuss the problems and prospects of NWFPs in sustainable forest management.
CO3 Compare the utilization procedure of different NWFPs found in Bangladesh.
CO4 Criticize the major problems in managing NWFPs and discuss the possible solutions.
CO5 Evaluate the important NWFPs of Bangladesh and their contribution in the rural economy.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2								1
CO2	3	1					2			
CO3	2	2								
CO4	1	3					3			
CO5	1	2	1							

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Belcher, B. M. (2003). What isn't an NTFP? *International Forestry Review*, *5*(2), 161–168.
2. Cocksedge, W. (2006). *Incorporating non-timber forest products into sustainable forest management: An overview for forest managers*. British Columbia Ministry of Forests and Range, Forest Practices Branch.
3. Khan, N. A., Alam, M. K., Khisa, S. K., &Millat-e-Mustafa, M. (2002). *Farming practices and sustainable development in the Chittagong Hill Tracts*. CHTDB and VFFP-IC.
4. Latif, M. A. (2008). *Bamboo species, their distribution, habitat, rehabilitation and uses in Bangladesh*. Bangladesh Forest Research Institute.
5. Rashid, M. H. (1984). *Medicinal plants of Bangladesh*. Forest Research Institute.
6. Roshetko, J. M., & Bose, S. K. (2001). *Propagation and cultivation of rattan and patipata in Bangladesh*. Training courses at BFRI, Chittagong.
7. Van Rijsoort, J. (2000). *Non-timber forest products (NTFPs), their role in sustainable forest management in the tropics*. Forests, Forestry and Biological Diversity Support Group, National Reference Centre for Nature Management (EC-LNV), International Agricultural Centre (IAC).
8. Manzoor Rashid, A. Z. M., Khan, N. A., & Mahmood, H. (Eds.). 2022. *Non-wood forest products of Asia: Knowledge, conservation and livelihood*. Springer.

Course No: AMS 07311201	Credit: 2	Year: First	Semester: Second
Survey and Settlement			Course Status: Theory

Course Rationale:
This course will provide adequate knowledge of forest surveying and map-making, both of which are essential for forest management, conservation, monitoring, and inventory. A well-prepared map can offer a comprehensive overview of forest types, features, boundaries, and vacant areas within a forested landscape. Maps are frequently used to support future planning for plantations, the establishment of structures, the demarcation of felling zones, and the identification of suitable sites for specific purposes. To produce an accurate and effective map, land surveying is a critical prerequisite. By engaging in this course on forest land measurement and mapping, young graduates will gain valuable skills in land assessment and map preparation.

Course Objectives:
The objectives of the course are

- To develop foundational skills in forest land surveying techniques essential for accurate data collection, boundary demarcation, and spatial planning in forest management.
- To equip students with practical knowledge and technical expertise in map-making for effective visualization and analysis of forest features, land use, and conservation planning.

Course Content:

1. **Objectives and scope**: Introduction, scope, and importance of surveying in forestry; Digital Surveying; Land history, laws and rules.
2. **Scales:** Definitions, RF; Plain, diagonal, and vernier scales.
3. **Maps and map reading:** Classification of maps; Orientation, location of point(s); Enlargement, reduction, and copying of plans and maps.
4. **Chain Survey:** Instruments and procedures, Errors and corrections, Recording and plotting.
5. **Traverse Survey:** Angular instruments, types, bearings. Field procedures, plotting, Closing errors, and adjustments. Theodolites, measurement of horizontal and vertical angles, measurement of bearings, Azimuths.
6. **Plane Table Survey:** General instrumentation and principle; Methods and procedure - setting up the table and orientation, radiation, intersection, traversing, and resection.
7. **Levels and Leveling:** Instrumentation and principles of leveling; Classification of leveling-direct and indirect leveling, profile leveling; common errors in leveling.
8. **Topographical Surveying:** Contours-contouring, characteristics of contours, methods of contouring, drawing of topographical maps; GPS.
9. **Area computation:** Computation of area by planimeter, and square methods.
10. **Land settlement procedures in Bangladesh**: Land administration and Management; Steps in land survey and records. Costing of forest survey.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, the student will able to

CO1 Describe the scope and importance of surveying in forestry, interpret use of map and scale in surveying.
CO2 Recognize different instruments, understand recording procedures, Learn to perform different types of surveying, understand field procedures, errors and correction.
CO3 Analyze different methods of leveling and contouring.
CO4 Perform different methods of computing areas such as grid method, planimeter, GPS.
CO5 Evaluate the land administration and land settlement procedure in Bangladesh.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		2			1	1			
CO2	3		1							
CO3	3		1							

CO4	3		2							
CO5	3	1				1	1		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Bannister, A., & Raymond, S. (1977). *Surveying*. Pitman Publishing Limited.
2. Brinker, R. C., & Wolf, P. R. (1984). *Elementary surveying* (7th ed.). Harper & Row Publishers, Inc.
3. Coover, L. S. (1966). *Drawing and blueprint reading* (3rd ed.). McGraw-Hill Book Company.
4. Kjellström, B. (1976). *Be expert with map and compass: The orienting handbook* (4th ed.). Charles Scribner's Sons.
5. Moffitt, F. H., & Bouchard, H. (1982). *Surveying* (7th ed.). Harper & Row Publishers, Inc.
6. Punmia, B. C., Jain, A. K., & Jain, A. K. (2016). *Surveying* (Vol. 1). Laxmi Publications.
7. Roy, S. K. (2011). *Fundamentals of surveying*. Prentice-Hall of India Pvt. Ltd.
8. Shahjahan, M., & Aziz, M. A. (2016). *A textbook of surveying* (4th ed.). University Campus Publishers.

Course No: AMS 07311202	**Credit: 1.0**	**Year: First**	**Semester: Second**
Survey and Settlement Laboratory			**Course Status: Practical**

Course Rationale:
This lab course provides practical knowledge on conducting surveying, drawing, and map-making operations in the field. It demonstrates various methods and technical procedures for recording the positions of objects on the Earth's surface and producing the associated drawings. The course enables students to carry out both plane and elevation surveys, as well as to correct errors that may arise from physical or environmental factors.

Course Objectives:
The objectives of the course are-

- To facilitate essential knowledge and hands-on experience in land measurement procedures, instrumentation, map-making, and map interpretation at the field level.
- To foster analytical thinking and problem-solving skills related to field-level challenges, using various surveying methods and techniques.

Course Content:

1. Demonstration of drawing equipment.
2. Lettering practice.
3. Construction of different types of scale (Ex. Plain scale, Diagonal scale, Vernier scale)
4. Chain survey: Demonstration of measuring instruments, reconnaissance, field procedure, recording and plotting of details.
5. Traverse Surveying: Demonstration of measuring instruments, reconnaissance, field procedure, recording and plotting of details.
6. Plane Table Surveying: Demonstration of measuring instruments, reconnaissance, field procedure, recording and plotting of details.
7. Levelling and Contouring procedure, preparing contour map.
8. Area calculation, demonstration of different ways of area calculation.
9. Map reading, copying, enlargement and reduction of maps.

10. Digital survey.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, the student will able to

CO1 Demonstrate equipment, lettering and different types of scales used in surveying
CO2 Perform plane table survey, chain survey, and traverse survey.
CO3 Apply different methods of leveling and contouring operations in undulating lands.
CO4 Generate maps; calculate area on-field and maps to perform necessary enlargement.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3						2	
CO2	3		2						2	
CO3	3		3			1			2	
CO4	3		3			2			1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Bannister, A., & Raymond, S. (1977). *Surveying*. Pitman Publishing Limited.
2. Brinker, R. C., & Wolf, P. R. (1984). *Elementary surveying* (7th ed.). Harper & Row Publishers, Inc.
3. Coover, L. S. (1966). *Drawing and blueprint reading* (3rd ed.). McGraw-Hill Book Company.
4. Kjellström, B. (1976). *Be expert with map and compass: The orienting handbook* (4th ed.). Charles Scribner's Sons.
5. Moffitt, F. H., & Bouchard, H. (1982). *Surveying* (7th ed.). Harper & Row Publishers, Inc.
6. Parkash, R. (1983). *Forest surveying*. International Book Distributors.
7. Shahjahan, M., & Aziz, M. A. (2016). *A textbook of surveying* (4th ed.). University Campus Publishers.

Course No: FES 08211226	**Credit: 1.0**	**Year: First**	**Semester: Second**
Fieldwork in Forests (Forest-Protected Areas)			**Course Status: Practical**

Course Rationale:
This course will allow a student to gain hands on experience of the forest-protected areas and forestry sectors in Bangladesh. Additionally, this course will cover basic knowledge related to nursery and plantation management.

Course Objectives:
The objectives of the course are-

- Make the students understand the diversity of forest-protected areas, its trends, applications, role in environmental stability.
- Helping the students to develop ability in identifying plant species and soil properties;
- Getting idea about the professional roles in forest and forestry sectors;
- Apply the acquired knowledge to develop forest nursery and to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Plant diversity of forest-protected areas in Bangladesh and its composition.
2. Identification of plant species and forest soils.
3. Properties of different types of forest soils.
4. The overall situation of different forest-protected areas, cultural values of forest ecosystem, forestry and environmental sector related areas of Bangladesh.
5. Nursery and plantation management (Seed collection, seed treatment, germination test, Tree planting, nursing, reporting etc.).
6. Practical application of all core courses of the current semester will be demonstrated.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Describe the diversity of forest-protected areas, different types of forest soil, its trends, applications, role in environmental stability.
CO2 Identify cultural values of wetland ecosystem and relate the professional roles in forest management.
CO3 Categorize the relevant field level management problems to meet the needs of the country.
CO4 Develop a nursery and support plantation management (Tree Planting, Nursing, Reportingetc.).

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3						3		3	
CO3	3						3		3	
CO4	3									

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL06	CA03, CA04, CA05
CO2	TL04, TL06	CA03, CA04, CA05
CO3	TL04, TL06	CA03, CA04, CA05
CO4	TL04, TL06	CA03, CA04, CA05

Recommended References:

1. Any documents supplied by the related visited areas' officials.
2. Any documents supplied by the related course teacher (s).

Course No: SOC 03141201	**Credit: 3.0**	**Year: First**	**Semester: Second**
Sociology for Forestry		**Course Status: Theory**	

Course Rationale:

This course is organized in such a way that even students without prior knowledge in sociology can acquire an interest in the subject and follow it. It is an introduction to the foundations of the scientific study of human social life, to theories and methods of Sociology, and to such basic concepts as culture, society, social organization, social stratification, and social change. Finally, it also helps students develop the skills to analyze environmental issues in light of sociological perspectives and theories.

Course Objectives:
The objectives of the course are-

- To provide knowledge on the origins of sociology as a social science, and to become familiar with the major theoretical approaches and methods in sociology.
- To define and explore the basic sociological concepts, including culture, society, community, association, institution, socialization, and social structure.
- To provide knowledge on sociological concepts of stratification and social mobility, as well as inequality based on class, status, power, race, gender and sexuality.
- To analyze the cultural aspect of the environment.
- To provide knowledge on ethnic communities living in the forest areas of Bangladesh.
- To focus on global environmental issues and population growth.
- To understand change as a characteristic aspect of human societies.

Course Content:

1. **Nature and Scope of Sociology:** Origin and development of sociology as a separate discipline, relationship between sociology and other social and natural sciences.
2. **Sociological Perspectives**: Functionalist, conflict and interactionist perspectives in sociology.
3. **Basic Concepts:** Society, Community, Association, Institution, and Group.
4. **Culture**: Definition of culture. Components of culture: norms, values, folkways, and mores. Cultural unity and diversity, cultural lag, cultural shock, and cultural relativism.
5. **Ethnicity and Ethnic Groups in Bangladesh:** Definition and theoretical debates on ethnicity. Major ethnic communities in Bangladesh – Khasi, Chakma, Marma, Garo etc. Socioeconomic and cultural backgrounds of these ethnic communities.
6. **Culture, Environment, and Environmental Policy:** language politics, discourse, knowledge, environmental policies, cultural understanding and reflection.
7. **Globalization, Modernization and Environment:** impact of globalization, influence of capital economy, marginalization.
8. **Types of Society**: From early hunting and gathering to industrial development and globalization.
9. **Socialization:** Socialization, agents of socialization, and theories of socialization,
10. **Social Stratification and Class Structure:** Definition of social stratification, systems of social stratification, and functionalist and conflict perspectives of social stratification and social mobility,
11. **Population and Environment:** Population growth, ecological balance, ecosystem, and threats to the global environment.
12. **Interplay between Social and Environmental Change:** Social change and its factors, theories of social change. Relationship between environmental and social change.
13. **Sustainable Development:** Perspectives, issues on sustainability, and environmentalist and humanitarian perspectives.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Generalize the key sociological concepts and theories to reality, social institutions and other social issues.
CO2 Define the core concepts of sociology and the environment.
CO3 Apply sociological perspectives or theories in explaining the environment and environmental issues
CO4 Analyze the importance of studying global environmental change and population issues.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3			3	3					1
CO2				3	3	1				
CO3	3			3	3					2
CO4	2		3	3	2	2				2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01

Recommended References:

1. R.T. Schaefer and R.P. Lamm, *Introducing Sociology*
2. E.C. Cuff, W. Sharrock and D.W. Francis, *Perspectives in Sociology*
3. Micheal P. Robbins, *Organizational Behavior*
4. T.B. Bottomore, *Sociology: A guide to problems and literature*
5. Metta Spencer, *Foundation of Modern Sociology*
6. G. Lenski, J. Lenski and P. Nolan, *Human Societies*
7. F.R. Scarpitti and M.L. Andersen, *Social Problems*
8. G. T. Miller, *Living in the Environment*

Course No: MAT 05411201	**Credit: 2.0**	**Year: First**	**Semester: Second**
Matrix and Vector Analysis		**Course Status: Theory**	

Course Rationale

This course is designed for the students of the department of Forestry. It is the standard complete introduction to the concepts and methods of matrix and vector analysis. The emphasis is given to teach the understanding of using matrix algebra and real valued vector functions. The applications of matrices and vector functions for the related discipline will be discussed.

Course Objectives

The objectives of this course are to-

- Introduce students to the fundamental concepts of matrices and vectors.
- Teach techniques and methods to solve a system of linear equations using matrix properties and its applications.
- Acquire knowledge of behavior of vector functions by studying its derivatives and integrations.
- Expose students to mathematical applications of matrices and vector functions to handle diverse problems which occur in forest life situations.

Course Content:

1. **Matrix:** Definition; types of matrices; operations on matrices; inverse of a matrix; elementary properties and solution of system of linear equations with the help of matrices.
2. **Vector Analysis:** Scalars and vectors; algebraic operations on vectors; scalar and vector product of two vectors; derivative of vectors with respect to scalars; vector operators' gradient, divergence and curl; outlines of line, volume and surface integration.

Course Learning Outcomes

After the successful completion of the course, students will be able to

CO1 Compute matrix algebra.
CO2 Find the solution set of a system of equations and compute the area and volume formed by the position vectors.
CO3 Discuss the nature of vector functions using derivatives.
CO4 Find the length of a curve line, surface area and volume of some models.
CO5 Apply the mathematical knowledge of matrix and vectors to their related topics.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1			3			1	2			
CO2			3			1	2			
CO3			3			1	2			
CO4			3			1	2			
CO5			3			1	2			

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL02, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL02, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL02, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL02, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL02, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Howard Anton and Chris Rorres: *Elementary linear algebra with applications*, 9th edition
2. Speigel, M R.: *Vector analysis*

Course No: FES 05322121	**Credit: 3.0**	**Year: Second**	**Semester: First**
Geology and Soil Science			**Course Status: Theory**

Course Rationale:

Geology and forest soils have great importance in understanding the Earth processes including Earth's surface environments, different types of rocks and soil, and soil formation processes. This course is designed to provide basic knowledge on geology and forest soils that helps students to select suitable tree species based on soil characteristics. It also provides information on different soils zones in a country and site index.

Course Objectives:

The objectives of the course are to-

- Get idea about the earth structure & history of the commencement of life form on earth according to geological time scale
- Accumulate basic ideas about common structures & features of Igneous, Metamorphic & Sedimentary rock
- Provide the knowledge of identifying types of fossils commonly found in rocks

- Enhancing the skill on identifying the factors of soil formation & describe how soil forming factors may interact to produce soil
- Apply the knowledge of soil properties & their significance in forest management

Course Content:

Geology

1. **Introduction:** Scope of geology and soil science, origin and interior of the earth.
2. **Mineralogy:** Definition and physical characteristics of minerals, study of essential and accessory minerals.
3. **Petrology:** Classification of rocks into types and their formation, igneous, metamorphic and sedimentary rocks structures and textures, description of common rocks and soils developed from them.
4. **Structural Geology:** Strike, dip, folding, anticlines, synclines, faults, stratification and lamination.
5. **Forms and uses of fossils, geological time scale.**
6. **Geology of Bangladesh**.

Soil Science

1. **Basic concept of Soil :** Definition and concepts of soil, forest soils, forest soil vs agricultural soil.
2. **Weathering and Soil Formation:** Factors and process of soil and forest soil formation; physical, chemical, and biological agents and their actions and role in weathering of rocks and minerals.
3. **Soil Properties:** Physical, chemical and biological properties of forest soils and their significance in silvicultural practices and tree growth.
4. **Soil Profile:** Definition, master horizon designation, their characteristics and subdivisions, soil profile development: calcification, laterization, podzolization, humification.
5. **Soil Organic Matter:** Organic matter and humus and their influence on physical, chemical and biological properties of soils.
6. **Soil Fertility:** Macro and micronutrients in the soil and their function in tree growth.
7. **Bangladesh Soil Zones:** Location and characteristics, classification of soils of Bangladesh and their bearing on forests, forest soils of Bangladesh, problem soils in Bangladesh.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Understand the fundamental concepts of geology and forest soils, including their scope and significance.

CO2 Identify and describe minerals, rocks, fossils, and geological structures.

CO3 Explain the processes of soil formation and development, including the influence of physical, chemical, and biological weathering in forest settings.

CO4 Analyze the properties, profiles, and fertility of forest soils, and assess their impact on silviculture and tree growth.

CO5 Evaluate the geological and soil characteristics of Bangladesh, including regional soil zones and their implications for forest management and sustainability.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2								
CO2	3	2								
CO3	3	2	1							
CO4	2	2	1							
CO5	3	3								

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01

CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Armson, K. M. (1979). *Forest soils: Properties and processes*. University of Toronto Press.
2. Bangar, M. M. (1981). *A textbook of geology*. Standard Publishers Distributors.
3. Binkley, D., & Fisher, R. F. (2019). *Ecology and management of forest soils* (5th ed.). Wiley.
4. Brady, N. C. (1996). *The nature and properties of soil*. Prentice Hall of India.
5. Brammer, H. (1971). *Soil survey project Bangladesh: Soil resources*. AGL; SF/PAK.
6. Fitzpatrick, E. A. (1974). *An introduction to soil science*. Oliver & Boyd.
7. Osman, K. T. (2013). Forest disturbances and soil degradation. In *Forest soils: Properties and management* (pp. 157–171). Springer.
8. Sarkar, R. S., &Chandul, P. S. (1988). *Plant ecology and soil science*. S. Chand & Co. (Pvt.) Ltd.

Course No: FES 05322122	**Credit: 1.0**	**Year: Second**	**Semester: First**
Geology and Soil Science Laboratory			**Course Status: Practical**

Course Rationale:

Geology and Forest Soils Laboratory is designed as a companion to cover the field aspect of theoretical knowledge on Geology & Forest Soils.This laboratory course focuses on the study of physical properties of rocks & minerals, which are the basic identifying keys of parent materials. Moreover, field study of soil profile with associated soil horizons gives a clear depiction of an area for plantation.

Course Objectives:

The objectives of the course are to-

- Acquaint students with the basic tools of identifying common rocks & minerals in hand samples & field exposure, using visual clues & sample test
- Facilitate necessary knowledge about visual identification of different soil horizons & understand the basic differences
- Enhancing the skill on the procedure of soil sampling method for further investigating
- Apply the knowledge of field demonstration of soil organic matter & soil texture and structure
- Foster the analytical and critical knowledge on different approaches of soil analysis

Course Content:

1. Study of physical properties of rock and mineral specimens in the laboratory.
2. Identification of texture.
3. Identification of litter and humus (students will observe and identify physically).
4. Collection of soil from different forest zones and their identification (with reference to H. Brammer).
5. Method of soil sampling.
6. Study of soil profile in the field, identifications of soil horizons and textural classification.
7. Visits to different soil resources research centers to demonstrate mechanical analysis of soil by hydrometer method; determination of soil organic matter by wet oxidation method; estimation of total nitrogen; determination of water holding capacity; determination of soil P^H and different physical and chemical properties of soil.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Identify rocks & minerals in the field.
CO2 Evaluate practical knowledge about physical and chemical properties of soil.
CO3 Outline top soil in field visit.

CO4 Accumulate knowledge on different forest soils of Bangladesh.
CO5 Apply knowledge of soils on soil examination.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2	1			
CO2	3	2	1			2				
CO3	3									
CO4	3	3				2				
CO5	3					3	1			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO5	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Armson, K. M. (1979). *Forest soils: Properties and processes*. University of Toronto Press.
2. Bangar, M. M. (1981). *A textbook of geology*. Standard Publishers Distributors.
3. Brady, N. C. (1996). *The nature and properties of soil*. Prentice Hall of India.
4. Brammer, H. (1971). *Soil survey project Bangladesh: Soil resources*. AGL; SF/PAK.
5. Fitzpatrick, E. A. (1974). *An introduction to soil science*. Oliver & Boyd.
6. Sarkar, R. S., &Chandul, P. S. (1988). *Plant ecology and soil science*. S. Chand & Co. (Pvt.) Ltd.

Course No: FES 05212121	**Credit: 3.0**	**Year: Second**	**Semester: First**
Forest Ecology			**Course Status: Theory**

Course Rationale:
As global climate change increasingly impacts natural systems, the conservation and sustainable management of forest resources have become more critical than ever. To address these challenges, forestry graduates must possess a solid understanding of ecological processes and their interrelationships with the natural environment. This course is designed to equip students with essential ecological knowledge, enabling them to comprehend the dynamics of nature and appreciate the significance of ecosystem interactions in forest management and conservation.

Course objectives:
This course aims to:

- Provide foundational knowledge of forest ecology and its relevance to forest management.
- Enable students to conceptualize core ecological theories and understand the complexity of ecosystems.
- Encourage the practical application of ecological principles for the conservation and sustainable use of natural environments.
- Develop analytical and critical thinking skills regarding forest structure, composition, and community dynamics.

Course Content:

1. **Introduction:** Fundamental concepts of forest ecology; the role of forest ecology in silviculture and forest management.

2. **Ecosystems:** Structure and function; ecological interactions; mineral nutrition.
3. **Site Factors:** Influence of light, water, temperature, soil, wind, and nutrients on forest ecosystems.
4. **Population Ecology:** Population characteristics; distribution and abundance; population dynamics and growth patterns.
5. **Community Ecology (Synecology):** Qualitative and quantitative attributes of communities; global biomes; community study methods; community dynamics.
6. **Forest Succession:** Concepts, causes, and types of succession; applications and retrogression.
7. **Forest Nutrient Dynamics:** Macro- and micronutrients; nutrient cycles; roles of soil flora and fauna in nutrient cycling.
8. **Ecological Modeling:** Principles of ecological modeling; model types, applications, and limitations.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Interpret the basic concepts of the ecosystem.

CO2 Understand the relationship between species distribution and different site factors.

CO3 Synthesize how the biogeochemistry (nutrient cycling) interacts in different ecosystems.

CO4 Assess the underlying mechanisms of species diversity in an ecosystem.

CO5 Describe the procedure of evolving different species in an ecosystem.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	3	1	2	2	3	3	3	2	1
CO2	3	3	1		2	3	2	3	1	1
CO3	3		2			3	3			
CO4	3		3			3	3		2	
CO5	3		3			3	2			2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Chapman, J. L., & Reiss, M. J. (1992). *Ecology: Principles and application*. Cambridge University Press.
2. Frelich, L. E. (2016). Forest dynamics. *F1000Research, 5*, 183.
3. Grant, W. E., &Swannack, T. M. (2008). *Ecological modeling*. Blackwell Publishing.
4. Medina, E., & Cuevas, E. (1989). IV. Mineral nutrition: Humid tropical forest. *Progress in Botany, 55*, 115–121.
5. Molles, M. C. Jr. (2008). *Ecology: Concepts and applications* (4th ed.). McGraw Hill.
6. Morin, P. J. (2011). *Community ecology*. Wiley-Blackwell.
7. NCERT. (n.d.). *Biology (Ecosystem, Chapter 4)*. National Council of Educational Research and Training.
8. Odum, E. P. (2004). *Fundamentals of ecology* (4th ed.). Brooks Cole.
9. Shukla, R. S., &Chandel, P. S. (1996). *Plant ecology* (8th ed.). S. Chand and Company Ltd.
10. Zabala, N. Q. (1990). *Forest ecology*. Institute of Forestry, University of Chittagong & FAO.

Course No: FES 05212122	**Credit: 1.0**	**Year: Second**	**Semester: First**
Forest Ecology Laboratory			**Course Status: Practical**

Course Rationale:

The sustainable management and conservation of forest ecosystems require forestry graduates to possess strong practical skills in ecological fieldwork. A clear understanding of natural interactions and the ability to conduct field-based research are essential for evaluating forest structure and dynamics. This laboratory course is designed to build foundational skills in vegetation analysis, preparing students to observe, measure, and interpret ecological patterns in natural forest environments.

Course Objectives:

This laboratory course aims to:

- Develop practical knowledge of ecological interactions within natural systems.
- Enable students to design and implement vegetation sampling plots for field investigations.
- Strengthen students' skills in analyzing the quantitative structure and composition of forest communities.

Course Content:

1. **Vegetation Survey:**
   Introduction to quantitative sampling techniques; selection of appropriate methods based on research objectives and site conditions.
2. **Quantitative Analysis of Vegetative Communities:**
   Calculation and interpretation of key structural parameters, including:
   - Frequency
   - Density
   - Dominance
   - Average plant size
   - Relative frequency
   - Relative density
   - Relative dominance
   - Relative importance
   - Importance Value Index (IVI)
   - Tree density
   - Basal area

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to:

CO1 Interpret the basic concepts of the ecosystem.

CO2 Outline the different methods for species and vegetation survey.

CO3 Analyze the basic structure and function of any particular species in an ecosystem.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3		2	3	3		2	
CO2	3		3			3	3			
CO3	3		3		2	3	3		2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Magurran, A. E. (1988). *Ecological diversity and its measurement*. Princeton University Press.
2. McCune, B., & Grace, J. B. (2002). *Analysis of ecological communities*. MjM Software Design.
3. Paradise, C. J. (2003). Patterns and processes, density and dispersion. In C. J. Paradise (Ed.), *Biology 321: Ecology laboratory and field manual* (pp. 6–12). Davidson College.
4. Phillips, E. A. (1959). *Methods of vegetation study*. Henry Holt & Co.
5. Shukla, R. S., &Chandel, P. S. (1996). *Plant ecology* (8th ed.). S. Chand and Company Ltd.

Course Code: FES 05112121	**Credit: 2**	**Year: Second**	**Semester: First**
Forest Entomology		**Course Status: Theory**	

Course Rationale:

This course will allow students to acquire basic knowledge on various insect species that play a vital role by regulating pollination, affecting plants and wild lives, and maintaining the food chain. Besides, insects act as disintegrators of biotic components. Management of beneficial and harmful insects to an optimum level can ensure healthy ecosystems, increase forest productivity and conserve the natural balance of biodiversity.

Course Objectives:

The objectives of the course are-

- To provide knowledge on diversity, classification and population dynamics of insects.
- To facilitate an insight on how forest ecosystems are interlinked with insects and vice-versa.
- To acquaint students with the tools of pest control/ management to ensure healthy forest/produce.
- Diagnose common insect damage and other common damaging agents in the field

- To apply skills and knowledge in identifying the beneficial and harmful insects.
- To foster appreciation for the significance of entomological research in the context of forest management.

Course Content:

1. **Introduction:** Scope of forest entomology; importance of insects in forestry; their roles in forest health,
2. **Insect Morphology and Physiology**: Insect structure and functions, reproduction and metamorphosis. Different types of insect larvae and pupae.
3. **Insect classification:** Taxonomy and classification of forest-related insects up to order, beneficial and harmful insects of forestry.
4. **Insect ecology:** Concept of insect ecology and population dynamics. Ecological factors influencing population fluctuations and outbreaks of pests. Relative susceptibilities of different forest types to insect damage.
5. **Methods of pest management**: Mechanical, Chemical, Biological, Integrated Pest Management (IPM),
6. **Common Forest Pests:** Biology, nature, the extent of damage and control of the major pests of forests and forest products. Nursery pest- Cutworms, Cockchafers, Termites and Ants, Plantation pests - Teak defoliator and skeletonizer, Sal heartwood borer, Mahogany shoot borer, Gamar defoliator and Keora bee hole borer, Wood and timber pests - Ghoon beetle, Ambrosia beetle and Termites, Pests of forest tree seeds
7. **Commercial Entomology**: Biology and management of honey bee, silkworm, lac insects to increase forest productivity.

Course Learning Outcomes (COs) (3-5)

After the successful completion of the course, students will be able to

CO1 Recognize the importance of insects to maintain forest health.
CO2 Describe and classify insects based on their morphological, physiological and developmental characteristics.
CO3 Understand the basic information about how insects interact with their ecosystem.
CO4 Identify major beneficial and harmful insects of forest, their life cycle and management.
CO5 Apply the knowledge to develop new sources for economic benefits.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3		1			2	1			
CO 2	3		1			2				
CO 3	3		1			3				
CO 4	3		1			3	3			
CO5	3		1			3				

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Baksha, M. W. (2008). *Insect pests of forests of Bangladesh* (Forest Entomology Series, Bulletin 8). Bangladesh Forest Research Institute.
2. Chapman, R. F., Simpson, S. J., & Douglas, A. E. (2013). *The insects: Structure and function*. Cambridge University Press.
3. Fenemore, P. G., & Prakash, A. (2004). *Applied entomology*. New Age International (P) Limited.
4. Gullan, P. J., & Cranston, P. S. (2010). *The insects: An outline of entomology*. Wiley.

5. Nair, K. S. S. (2007). *Tropical forest insect pests: Ecology, impact, and management*. Cambridge University Press.
6. Radcliffe, E. B., Hutchison, W. D., &Cancelado, R. E. (Eds.). (2009). *Integrated pest management: Concepts, tactics, strategies and case studies*. Cambridge University Press.
7. Shields, V. D. C. (Ed.). (2017). *Insect physiology and ecology*. IntechOpen.
8. William, C. (2011). *Forest entomology: A global perspective* (1st ed.). Wiley-Blackwell.

Course No: **FES 05112122**	**Credit: 1**	**Year: Second**	**Semester: First**
Forest Entomology Laboratory		**Course Status: Practical**	

Course Rationale:
This course is the practical engagement of the students to various forest insects, their identification and management measures. This course will enable students to have an in-hand study of pest control methods that are very crucial for the foresters in managing and increasing forest productivity.

Course Objectives:
The objectives of the course are

- To explain and familiarize the students with various insect identification tools and methods.
- To allow the students to identify some common beneficial and harmful insects.
- To facilitate them with insect collection and preservation methods.
- To provide students with an insight on how pests can be controlled in nurseries and plantations.

Course Content:

1. **Examine the Insect Parts:** Orthoptera, Hemiptera, Coleoptera Lepidoptera, Diptera, Hymenoptera, Odonata, Isoptera, Dermaptera
2. **Diagnosis of symptoms**: Diagnosis of the insect attack by symptoms
3. **Identification of insect pests**: Identifications of the common forest insect pests, teak defoliator &skeletonizer, sal heartwood borer, teak canker grab mahogany (*Swietenia*spp.) shoot borer, gamar (*Gmelinaarborea*) defoliator, keora (*Sonneratiaapetala*) defoliator, keora bee hole borer, sissoo (*Dalbergiasissoo*) defoliator and cane (rattans) top shoot borer.
4. **Collection of insects**: Methods of insect collection, preservation and dispatch to Entomologist for advice, insecticide formulations
5. **Application of pesticides**: Pesticide application equipment and safety precautions.

Course Learning Outcomes (COs) (3-5)
After the successful completion of the course, students will be able to

CO1 Identify important insect pests of forest nurseries and plantations
CO2 Use methods to collect and process insect samples for future study
CO3 Design management system to control pests in the forest nurseries and plantations

Mapping Course Learning Outcomes (COs) with the POs

	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
CO vs. POs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3		1			2	1			
CO 2	3		1			2	1			
CO 3	3					2	3			

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Berryman, A. A. (2012). *Forest insects: Principles and practice of population management*. Springer Science & Business Media.
2. Chapman, R. F. (2013). *The insects: Structure and function* (5th ed.). Cambridge University Press.
3. Mathews, G. A. (2000). *Pesticide application methods* (3rd ed.). Blackwell Science.
4. Nair, K. S. S. (2007). *Tropical forest insect pests: Ecology, impact, and management*. Cambridge University Press.
5. Singh, S., Singh, K. P., & Forest Research Institute (FRI), Dehradun. (2012). Forest entomology and National Forest Insect Collection (NFIC). In *Forestry in service of nation: ICFRE technologies* (ICFRE Publication No. 137, pp. 368–398). Indian Council of Forestry Research and Education (ICFRE).

Course No: FES 08212122	**Credit: 1.0**	**Year: Second**	**Semester: First**
Fieldwork in Forests (Wetland Ecosystem)			**Course Status: Practical**

Course Rationale:
This course will allow a student to gain hands-on experience of the wetland ecosystem and forestry sectors in Bangladesh. Additionally, this course will cover basic knowledge related to nursery and plantation management.

Course Objectives:
The objectives of the course are to-

- Make the students understand the diversity of the wetland ecosystem, its trends, applications, and role in environmental stability.
- Help the students to develop ability in identifying insect species and characterize them
- Get idea on the professional role on forest and non-wood forest product sectors
- Apply the acquired knowledge to develop forest nursery and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Diversity of the wetland ecosystem in Bangladesh and its components.
2. Identification of insect species and their grouping in wetland.
3. The overall situation and cultural values of different wetland ecosystem and environmental sector related areas of Bangladesh.
4. Nursery and Plantation management (Nursery bed preparation; soil preparation; poly bag filling of different sizes; seed sowing, tree planting, nursing, reporting etc.)
5. Practical application of all core courses of the current semester will be demonstrated.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Describe the diversity of the wetland ecosystem, different types of non-wood products, insect species and their grouping.
CO2 Identify cultural values of the wetland ecosystem and relate the professional roles in wetland management.
CO3 Recognize the relevant field level forest management issues to address the challenges.
CO4 Design a nursery and manage plantations (Nursery bed preparation, tree planting, nursing, reporting etc).

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3						3		3	

CO3	3						3		3	
CO4	3						3		3	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL06	CA03, CA04, CA05
CO2	TL04, TL06	CA03, CA04, CA05
CO3	TL04, TL06	CA03, CA04, CA05
CO4	TL04, TL06	CA03, CA04, CA05

Recommended References:

1 Documents provided by the related visited areas' official/s.
2 Documents provided by the related course teacher/s.

Course No: PHY 05332101	**Credit: 3.0**	**Year: Second**	**Semester: First**
General Physics			**Course Status: Theory**

Course Rationale:

This course will help the students to understand basic physics and their application in normal life. They will be introduced with the different properties of matters and how they are related to life. It will also increase the curiosity of students to know more about the advanced physics.

Course Objectives

The objectives of this course are:

- To acquire the knowledge of physics in different areas.
- To apply the physics knowledge in their related field.
- To facilitate basic ideas about modern physics.

Course Content

1. **Mechanics:** motion in two dimensions, projectile and circular motion, work and work-energy theorem, conservation of energy, centre of mass, conservation of linear momentum, angular momentum and its conservation, simple harmonic motion, Kepler's laws
2. **Thermodynamics:** heat and temperature, zeroth law of thermodynamics, principles of thermometry, first and second law of thermodynamics, Carnot's cycle, Carnot's heat engine, absolute scale of temperature, entropy, black-body radiation.
3. **Optics:** nature of light, electromagnetic theory and spectrum, speed of light and refractive index, laws of reflection, refraction and absorption, total internal reflection, scattering of light.
4. **Properties of Matter:** (i) Elasticity: elastic modulii and their inter-relations, cantilever. (ii) Surface Tension: molecular forces, surface energy, capillary rise. (iii) Viscosity: laminar flow, critical velocity, Poiseuille's equation.
5. **Modern Physics:** radioactivity and radioactive decay, half and mean life, radioactive dating, uncertainty principle, two slit experiment.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Apply Newton's laws, contrast between force and acceleration, interpret dynamics of angular momentum, describe central force and planetary motion and gain basics on applying Newtonian physics to the general problem of central force motion.

CO2 Compare between heat and temperature, apply laws of thermodynamics, explain black-body radiation and

understand wave nature of light, describe nature of light, relevant experiments to illustrate different light characteristics

CO3 Explain elastic properties of matter and select suitable material when required, apply equation related to viscosity and surface tension.

CO4 Identify different atomic models and the basic concept of radioactivity for understanding related physics.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3			1	1			
CO2	3		2				1			
CO3	3		2							
CO4	3		3			2	2		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA 04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Halliday, D. and Resnick, R.: *Physics* (Vol. I and Vol II)
2. Puri, S. P.: *Fundamentals of Vibrations and Waves*
3. Jenkins, F.A. and White, H.A.: *Fundamentals of Optics*
4. Beiser, A.: *Perspective of Modern Physics*

Course No: ECO 03112101	**Credit: 3.0**	**Year: Second**	**Semester: First**
Principles of Economics			**Course Status: Theory**

Course Rationale:

This course provides an introduction to the main ideas and concepts involved in modern economics and attempts to provide students with an understanding of how the economy works, what type of problems economists attempt to solve, and how they set about trying to solve them. The course is primarily concerned with the analysis of individual decision-making agents, the behaviour of firms and industries in the economy (microeconomics), on the economy as a whole (macroeconomics) and the inherent problems facing underdeveloped and developing countries (economic development).

Course Objectives:

The objectives of this course are:

- To make student able to understand economic theories.
- To analysis in the field of development economics.
- To deal with a selection of issues and problems facing the developing economies.

Course Content:

1. **Introduction to Microeconomics:** Definition and scope; basic concepts and tools—PPF and circular flow model; fundamental economic problems and solution systems; Concepts of demand, supply and equilibrium; Concepts of elasticity, different types of elasticities, their applications; Concepts of total and marginal utility; Concepts of production, cost and profit, characteristics of different types of markets.
2. **Introduction to Macroeconomics**: Key macroeconomic indicators and their performance measurement - GNP, GDP, inflation, unemployment; money, functions of money, function of commercial and central bank, monetary policy; fiscal policy and structure of govt. budget.
3. **Development and related issues:** Growth and development; concept of poverty and poverty measures; HDI; key human-socio-economic development indicators of Bangladesh, Sustainable Development Goals (SDG).

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Analyze the individual decision-making agents, the behaviour of firms and industries in the economy

CO2 Generalize the concept of elasticity, markets and macroeconomics, and interpret the behavior of key macroeconomic variables

CO3 Evaluate monetary and fiscal policy and Government budget

CO4 Synthesize main issues confronting underdeveloped and developing countries

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	2			3	3					
CO2	3			3	3					1
CO3	2			3	3					
CO4	2			3	3					

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Arnold, R. A. (2014). *Economics* (11th ed.). South Western Publishing Company.
2. Mankiw, N. G. (2012). *Principles of economics* (6th ed.). Thomson South Western Publishing.
3. Samuelson, P. A., &Nordhaus, W. D. (2009). *Economics* (19th ed.). McGraw-Hill.
4. Todaro, M. P., & Smith, S. C. (2012). *Economics of development in the Third World* (11th ed.). Longman.

Course No: STA 05422101	**Credit: 2.0**	**Year: Second**	**Semester: First**
Statistics			**Course Status: Theory**

Course Rationale:

The primary goal of this course is to acquire basic concepts of different statistical techniques to explore data**.**

Course Objectives:

The objectives of this course are:

- Provide the knowledge on fundamental concepts of statistical methods,
- Acquaint students with the basic tools of exploratory data analysis,
- Facilitate necessary knowledge about bivariate data analysis.

Course Content:

1. Statistics: its nature and scope. Nature of statistical data. Attributes and variables, population and sample, collection and condensation of data. Frequency distribution. Graphical representation of data. Probability: concept of probability and its law, a study of χ^2,test for r x c contingency tables, statistical student's t and F distributions, and drawing of different kinds of samples, total and proportion.

2. Measures of central tendency and location: arithmetic mean, median, mode, geometric mean, harmonic mean, quadratic mean, quartiles, deciles and percentiles.

3. Measures of dispersion: range, mean deviation, standard deviation, variance, quartile deviation, coefficient of variation, moments and cumulants of a distribution, skewness and kurtosis.

4. Probability and Probability Distributions: Basic concepts of probability and its laws; Probability Distributions: basic study of the Binomial, Poisson, and Normal distributions. Application should be focused to forestry data.

5. Regression and correlation: Bivariate data. Relationship between the variables. Methods of least squares, regression line. Correlation and regression coefficients. Rank correlation.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Explain basic concepts of statistics, interpret statistical parameters, describe various statistical tools and illustrate data graphically

CO2 Understand and performmeasures of central tendency of data

CO3 Develop and interpret different measures of dispersion, location, data distribution and shape characteristics

CO4 Analyze bivariate data, analyze relationship between data variables and interpret the results

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3							
CO2	3		3				1			
CO3	3		3			2	1			
CO4	3		3			3	3		2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Gupta, S. C., & Kapoor, V. K. (2000). *Fundamentals of mathematical statistics* (10th ed.). Sultan Chand and Sons.
2. Hoel, P. G. (n.d.). *Introduction to mathematical statistics* (5th ed.). John Wiley.
3. Islam, M. N. (2006). *Introduction to statistics and probability* (3rd ed.). Books World.
4. Mostafa, M. G. (1989). *Methods of statistics*. Karim Press and Publication.
5. Shill, R. N., &Debnath, S. C. (2001). *An introduction to the theory of statistics*. Dhaka.
6. Wonnacott, T. H., &Wonnacott, R. J. (1990). *Introductory statistics* (5th ed.). Wiley.

Course No: FES 07222221	Credit: 3.0	Year: Second	Semester: Second
Wood Structure and Properties			Course Status: Theory

Course Rationale

Understanding wood structure and properties has great importance on tree growth and functions in prevailing environmental conditions, and to determine quality of wood for versatile uses including wood identification. This course is designed to provide basic knowledge on wood structure (anatomical) and properties (physical, mechanical and chemical) of wood for identification, understanding tree growth and functions, and their efficient utilization.

Course Objectives:

The objectives of the course are to:

- Provide knowledge on anatomical properties of wood.
- Help students conceptualize basic theories of wood formation.
- Apply the knowledge on natural defects of wood and their effects on wood utilization.
- Make the students understand the wood-water relationships, wood density and different mechanical properties of wood.
- Acquaint students with the ideas on chemical properties of wood.
- Foster the analytical and critical understanding of the effects of environment and silvicultural practices on wood structure and properties.

Course Content:

1. **Introduction:** Definition and characteristics of wood; importance of studying wood structure and properties; chrematistics of woody and non-woody plants; softwood and hardwood; wood formation process.
2. **Wood structure:** Macroscopic structure (hardwood and softwood): bark, phloem, cambium, xylem, pith, sapwood and heartwood, growth ring, growth ring anomalies, earlywood and latewood;
   Microscopic structure (hardwood and softwood): vessels and tracheid, arrangement of vessels and tracheid, fibre, parenchyma, ray, cell wall, epithelial cell, resin canals, tyloses, and pits.
3. **Natural defects**: Juvenile vs. mature wood; reaction wood; compression and tension wood; knots: characteristics, classification and causes of formation.
4. **Physical properties:** Colour, lusture, odour, taste, grain, texture, figure; wood-water relationship: water in wood, types of water in wood, moisture content, equilibrium moisture content (EMC), fiber saturation point (FSP), Shrinkage and Swelling; Wood density and specific gravity; Thermal, electrical and acoustic property of wood.
5. **Mechanical Properties:** Behavior of wood in relation to stresses and strains; Hook's law, definitions and fundamental aspects of mechanical properties: stiffness and elasticity, tensile strength, compressive or crushing strength, shearing strength, transverse or bending strength, toughness, hardness, cleavability; MOE and MOR; factors affecting the mechanical properties of wood.
6. **Chemical properties:** Chemical constituents of wood; cellulose, hemicelluloses, lignin and extractives and their influences on wood properties; cell walls and their chemical composition; microfibrils; stable isotopes in wood.
7. **Environmental and silvicultural aspects of wood properties:** Effects of silviculture and environment on wood properties; application of growth-ring concept in forestry and environmental science.

Course Learning Outcomes (COs) (3-5):

After successful completion of the course, students will be able to-

CO1 Basic concept on woody, non-woody plants, hardwood and softwood
CO2 Analyze the basic concept of wood and its formation process
CO3 Identify different macro- and microscopic structure of wood
CO4 Compare the properties of reaction wood and normal wood, and relate its effect on wood quality
CO5 Recognize the physical, chemical and mechanical properties of wood and the factors affecting those properties.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO 1	3					2				
CO2	3									
CO3	2					2				
CO4	2									
CO5	2					1				

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01,CA02,SA01
CO2	TL01, TL05	CA01,CA02,SA01
CO3	TL01, TL05	CA01,CA02,SA01
CO4	TL 01, TL05	CA01,CA02,SA01
CO5	TL01, TL05	CA01,CA02,SA01

Recommended References:

1. Forest Products Laboratory. (2021). *Wood handbook: Wood as engineering material* (General Technical Report FPL-GTR-282). USDA Forest Service, Forest Products Laboratory.
2. Shmulsky, R., & Jones, P. D. (2019). *Forest products and wood science: An introduction* (7th ed.). Wiley-Blackwell.
3. Gardiner, B. (2014). *The biology of reaction wood*. Springer-Verlag.
4. Sjöström, E. (2013). *Wood chemistry: Fundamentals and applications* (2nd ed.). Elsevier.
5. Schweingruber, F. H. (2010). *Wood structure and environment*. Springer-Verlag.
6. Desch, H. E., &Dinwoodie, J. M. (1996). *Timber: Structure, properties, conversion and uses* (7th ed.). MacMillan Press Ltd.
7. Panshin, A. L., & de Zeeuw, C. (1982). *Textbook of wood technology* (Vol. 1). McGraw-Hill Book Company.
8. Kollmann, F. F. P., & Cote, W. A., Jr. (1968). *Principles of wood science and technology: Solid wood* (Vol. 1). Springer-Verlag.
9. Siegwolf, R. T. W., Brooks, J. R., Roden, J., &Saurer, M. (Eds.). (2021). *Stable isotopes in tree rings: Inferring physiological, climatic and environmental responses*. Springer Nature.

Course No: FES 07222222	**Credit: 1.0**	**Year: Second**	**Semester: Second**
Wood Structure and Properties Laboratory			**Course Status: Practical**

Course Rationale:
This course is aimed to provide practical knowledge on wood structure and properties to increase basic understanding of wood and wood products.

Course Objectives:
The objectives of this course are to-

- Enhance the skills for identification of different macroscopic and microscopic features of wood.
- Develop skills for wood identification of different species using wood identification key.
- Apply the knowledge for analyzing physical and mechanical properties of wood.

Course Content:

1. **Macroscopic features**: Identification of different macroscopic structure of wood.
2. **Slide preparation**: Sectioning, staining and mounting of woody tissues for microscopic study.
3. **Minute features**: Microscopic identification of different cell under microscope

4. **Species identification**: Identification of wood of different tree species growing in Bangladesh using Wood Identification Key.
5. **Physical property**: Inspection of wood physical properties on wood samples.
6. **Mechanical Property:** Determination of compressive and bending strength of wood.

Course Learning Outcomes (COs) (3-5):
After successful completion of the course, students will be able to-

CO1 Identify different surfaces of wood, different gross and microscopic features of wood
CO2 Prepare microscopic slides for anatomical study
CO3 Recognize wood of different species using wood identification key
CO4 Compute wood density, moisture content, shrinkage and swelling, strength properties of wood

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2				
CO2	3					2				
CO3	1					1				
CO4	3					2				

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA01, CA03, CA04, CA05, SA01

Recommended References:

1. IAWA Committee. (2007). *IAWA list of microscopic features for hardwood identification* (4th print ed.). Leiden.
2. IAWA Committee. (2016). *IAWA list of microscopic bark features*. Leiden.
3. Schweingruber, F. H. (2010). *Wood structure and environment*. Springer-Verlag, Berlin and Heidelberg.
4. Forest Products Laboratory. (2013). *Wood handbook: Wood as engineering material* (Centennial ed.). USDA Forest Service.

Course No: FES 05112221	**Credit: 3**	**Year: Second**	**Semester: Second**
Mycology, Forest Pathology and Forest Protection		**Course Status: Theory**	

Course Rationale: This course will provide students with the knowledge on major biotic agents (fungi, bacteria, viruses and nematodes) that are responsible for the diseases of trees. Three factors such as the pathogen, the host and the environment are necessary for the diseases to occur. The knowledge on these factors will assist students to understand causes and processes of diseases in forest trees and explore methods of disease diagnosis, prevention, and control in forest ecosystems.

Course Objectives:
The objectives of the course are to-
● Facilitate necessary knowledge on fungi, their reproduction, mode of nutrition, taxonomy, beneficial and harmful role of fungi in forestry

● Help students conceptualize basic knowledge in different diseases in forest nurseries and plantations as well as their management
● Apply the knowledge in managing different biotic and abiotic agents injurious to forest ecosystems.

Course Content:

1. **Mycology:** Definition and scope of mycology, importance of fungi in Forestry, characteristics of fungi and their growth condition, classification of fungi up to classes and description of major classes of fungi. Vegetative structure of fungi, ultra-structure of fungi, reproduction of fungi. Mode of nutrition and survival of fungi and their dispersal. Major fungal and fungal-like diseases of plants. Life history of selected fungi representing major groups and associated with tree diseases: *Phytophthora/Phthium spp., Nectriahaematococca, Rhizophusartocarpi, Ganodermalucidum, Fusariumsolani, Phomopsishevae*.
2. **Bacteria, viruses and nematodes:** General account of plant bacteria, viruses and nematodes.
3. **Forest Pathology:** Definition and scope of forest pathology, historical perspectives of forest pathology, causes of forest diseases and biotic and abiotic agents of tree diseases, plant quarantine. Pathogenicity and stages of disease development in plants, classification of plant diseases. Disease diagnosis and symptoms identification: leaf spot, leaf blight, canker, gall, rot etc.
4. **Principles of forest disease management and control:** Ecology of disease. The disease cycle, disease triangle and control triangle. Natural resistance against diseases.
5. **Diseases in the forest nurseries:** Symptoms, causes, management and/or control.
6. **Study of the important forest tree diseases:** Detailed study of major diseases in forests and plantations - top dying of Sundri, root rot of Pynkado, heart rots in trees, bamboo blight in village groves, mistletoe infestation of Gamar, dieback of Keora, massive mortality of Keora, sap and/or blue stain in timber.
7. **Forest protection:** Classification of agencies injurious to forests. Nature and extent of damage due to forest fire, wildlife, grazing and human interference and measures to prevent these damages.

Course Learning Outcomes (COs) (3-5)

CO1 Identify different species of fungi and their reproduction, structure and nutrition, distribution and dissemination of fungi
CO2 Classify fungi based on their morphology, physiology and molecular traits
CO3 Determine beneficial and harmful fungi, symptoms caused by them and other biotic agents like viruses and bacteria on trees
CO4 Develop methods to manage diseases caused by fungi in nurseries, forests and plantations
CO5 Describe agencies injurious to forests and measures to mitigate harmful effects of the injurious agents

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3		1			2				
CO 2	3		2			1				
CO 3	3		1			2				
CO 4	3		1			1	3	1	1	
CO 5	3		1			1	3	1		

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Agrios, G. N. (2005). *Plant pathology* (5th ed.).
2. Manion, P. D., &Lachance, D. (1992). *Forest pathology*.
3. Tainter, F. H., & Baker, F. A. (1996). *Principles of forest pathology*.

4. Alexopoulos, C. J., Mims, C. W., & Blackwell, M. M. (1996). *Introductory mycology* (4th ed.). John Wiley & Sons, New York, USA.
5. Bakshi, B. F. (1976). *Forest pathology: Principles and practice in forestry*. Government of India.
6. Erwin, D. C., &Ribeiro, O. K. (1996). *Phytophthora disease worldwide*. APS Press, St. Paul, Minnesota, USA.
7. Manion, P. D. (1981). *Tree disease concepts*. Prentice Hall Inc., New Jersey.
8. Rahman, M. A. (1990). *Major diseases in the nurseries and forests of Bangladesh: Lecture guide on forest pathology*. Institute of Forestry, University of Chittagong, Chittagong.
9. Sinclair, W. A., & Lyon, H. H. (2005). *Diseases of trees and shrubs* (2nd ed.). Comstock Publishing Associates.

Course No: **FES 05112222**	**Credit: 1**	**Year: Second**	**Semester: Second**
Mycology, Forest Pathology and Forest Protection Laboratory		**Course Status: Practical**	

Course Rationale: Practical knowledge is essential for students to get an idea on how things work. To conserve and expand forest resources in the face of global climate change, forestry graduates need to have the practical knowledge on various pathogens and injurious agents responsible for the damages of forests as well as their protection.

Course Objectives:
The objectives of the course are to-

- Acquaint students with the practical knowledge on the isolation and identification of fungi as well as their collection and preservation
- Apply the knowledge in identification of disease symptoms in forest nurseries and plantations as well as their management

Course Content:

1. **Collection and identification of disease samples:** Identification of major disease symptoms associated with trees and field visit to forest sites with disease occurrences
2. **Collection of fungi:** Methods of collection and preservation of fungi
3. **Isolation and identification of fungi:** Techniques of isolation and identification of fungi.

Course Learning Outcomes (COs) (3-5):

CO1 Identify fungi from different plant parts such as leaves, stems and roots
CO2 Use methods to collect and preserve fungi and diseased samples
CO3 Design control techniques to manage fungal diseases in the forest nurseries and plantations

Mapping Course Learning Outcomes (COs) with the POs

	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
Course Learning Outcomes (CO)	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3		1			2				
CO 2	3		1			2				
CO 3	3		1			3	3			

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy

CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Ashrafuzzaman, M. H. (1976). *Laboratory manual of plant pathology*. Chapagar, Mymensingh.
2. Bakshi, B. F. (1976). *Forest pathology: Principles and practice in forestry*. Government of India.
3. Blanchard, R. O., &Tattar, T. A. (1981). *Field and laboratory guide to tree pathology*. Academic Press.

Course No: FES 08212221	**Credit: 3.0**	**Year: Second**	**Semester: Second**
Principle and Practice of Silviculture			**Course Status: Theory**

Course Rationale:

This course is designed to provide students with foundational knowledge in silviculture, including the morphological adaptations of trees, forest regeneration processes, and the principles and practices involved in manipulating forest vegetation to enhance forest stand productivity.

Course Objectives:

The objectives of the course are to-

- Introduce fundamental terminology related to silviculture and tree morphological adaptations.
- Help students understand the key factors influencing regeneration across different forest ecosystems.
- Provide a theoretical foundation for conducting various forest tending operations.
- Deliver knowledge on different silvicultural systems and their effects on forest ecosystem functioning.

Course Content:

Principles of Silviculture

1. Introduction: Introduction to Silviculture; Historical Development of Silviculture; Objectives of Silviculture; Importance of Silviculture in Forestry Practices; Role of Silviculture in Forest Resource Management

2. Morphology and Adaptation: Stages of tree growth; Adaptation of trees in different forests. Classification of root, stem and crown; Application of growth processes in Silviculture

3. Classification of forests and community structure: Ecological, structural and functional bases; Stand Structure; Stand dynamics/stand development. Classification of tropical forest species in relation to light environment.

4. Natural regeneration: Natural regeneration by seed, coppice and root sucker; Natural regeneration in Mangrove forests, Hill forests, Sal forests and Bamboo; Factor affecting natural regeneration; Gap phase dynamics; Concept of secondary forest; Competition of tree seedling/sapling for different site factors, shade-light responses of forest tree seedlings, recruitment processes and assessment of growth of recruits; Assisted Natural Regeneration (ANR).

Practice of Silviculture:

5. Tending Operations: Weeding, cleaning, climber cutting, girdling, pollarding, release cutting, salvage cutting. Thinning - objectives of thinning , methods of thinning; factors influencing thinning and, thinning carryout in various forests of Bangladesh.

6. Silvicultural systems: Low forest methods, High forest methods; Application of different silvicultural systems, impact of silvicultural practices on forest environment.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Define and apply fundamental silvicultural terms and concepts in forest management contexts.
CO2 Analyze morphological adaptations of tree species across various site conditions and forest ecosystems.

CO3 Evaluate ecological processes influencing natural regeneration and propose strategies to enhance regeneration in different forest types.
CO4 Demonstrate competence in performing tending operations to improve forest stand health and productivity.
CO5 Design and assess silvicultural systems tailored to specific forest ecosystems, considering both productivity and environmental impact.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3									
CO3	3									
CO4	3						2			
CO5	3					2	1			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Ashton, M. S., & Kelty, M. J. (2018). *The practice of silviculture: applied forest ecology*. John Wiley & Sons.
2. Duan, J., & Abduwali, D. (2021). Basic Theory and Methods of. *Silviculture*, 77.
3. Shepherd, K. R. (2012). *Plantation silviculture*. Springer Science & Business Media.
4. Nyland, R. D. (2016). *Silviculture: concepts and applications*. Waveland Press.
5. Zabala, N. Q. (1990). Principles and practice of silviculture (Field Document No. 25). UNDP/FAO/BGD/85/011. Institute of Forestry, Chittagong University, Chittagong & FAO.
6. Zabala, N. Q. (1991). Silvicultural systems (Field Document No. 27). UNDP/FAO/BGD/85/011. Institute of Forestry, Chittagong University, Chittagong & FAO.

Course No: FES 08212222	**Credit: 1.0**	**Year: Second**	**Semester: Second**
Fieldwork in Forests (Tropical Hill Forest)			**Course Status: Practical**

Course Rationale:
This course will allow students to gain hands-on experience on the tropical hill forest ecosystem, forest diseases and nurseries in Bangladesh.

Course Objectives:
The objectives of the course are

- To make the students understand the diversity of tropical hill forest ecosystem, its trends, applications, role in environmental stability.
- To develop skills for practicing silvicultural management and species silviculture.
- Helping the students to develop ability in identifying the symptoms and pathogens of different diseases and their characteristics.
- To get idea about the professional role in forest and forestry sectors.
- To apply the acquired knowledge to develop forest nursery and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Diversity of tropical hill forest in Bangladesh and its components.
2. Silvicultural practices and species silviculture.
3. Identification of diseases.
4. The overall situation and cultural values of different hill forest, forestry and environmental sector related areas of Bangladesh.
5. Nursery and Plantation management (Preparation of compost, taking care of the previous semester seedlings, Planting seedlings, nursing, reporting etc.).
6. Practical application of all core courses of the current semester will be demonstrated.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Interpret the diversity of hill forest, its trends, applications, role in environmental stability.
CO2 Apply knowledge of silvicultural systems and practices.
CO3 Distinguish different types of fungi and relevant diseases.
CO4 Evaluate the cultural values of hill forest and relevant field level management problems and challenges
CO5 Develop plans for nursery and Plantation Management (Tree Planting, Nursing, Reporting etc.).

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3						3		3	
CO3	3						3		3	
CO4	3						3		3	
CO5	3						3		3	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL06	CA03, CA04, CA05
CO2	TL04, TL06	CA03, CA04, CA05
CO3	TL04, TL06	CA03, CA04, CA05
CO4	TL04, TL06	CA03, CA04, CA05
CO5	TL04, TL06	CA03, CA04, CA05

Recommended References:

1. Documents provided by the related visited areas' official/s.
2. Documents provided by the related course teacher/s.

Course No: BUS 04112201	**Credit: 2.0**	**Year: Second**	**Semester: Second**
Cost and Management Accounting			**Course Status: Theory**

Course Rationale:

This course will help the students to understand the different cost concept and techniques and develop their skill to apply the knowledge for better decision making. In professional sector they will apply their managerial ability to motivate others for achieving goals.

Course Objectives:

The objectives of this course are:

- To facilitate necessary knowledge about the cost concepts, cost behavior, and cost accounting techniques that are applied to manufacturing and service businesses.
- To be capable of interpreting cost accounting statements
- To provide the students with the capability to apply theoretical knowledge in decision making.
- To be able to analyze and evaluate information for cost ascertainment, planning, control of business operations.
- To apply the various techniques available to measure managerial performance and to motivate employees toward organizational goals.
- To identify and analyze both qualitative and quantitative standards to formulate best control methods

Course Content

1. Introduction to Cost Accounting: Definition of Cost Accounting, Comparison of Cost Accounting and Financial Accounting; The role of Cost Accounting; Methods and Techniques of Cost Accounting; Characteristics of an Ideal Cost Accounting System

2. Cost Concepts, Classifications and Statements: Cost Object; Expenditures, Cost, Expense and Loss; Cost Classifications; Cost Data and Uses; the Chart of Accounts; Statement of Cost of Goods Manufactured and Sold; Cost Statement or Cost Sheet

3. Costing and Control of Materials: Classification of Materials; Accounting for Materials; Store ledger(FIFO & WAM) method; Inventory Planning; Ordering Cost, Holding Cost and EOQ; Effect of Quantity Discounts on EOQ; Safety Stock and Reorder Point; Material Control Methods; Materials Requirement Planning System. Practical problem.

4. Costing and Control of Labour: Productivity and Labour Costs; Costs included in Labour; Accounting for Labour; Time Keeping, Computation of total payroll and Allocation of Payroll Costs; Different incentive plan; Labour Cost Control, Labour Turnover and Control of Labour Turnover; Learning Curve Theory. Practical problem & solution

5. Costing and Control of Manufacturing Overhead: Manufacturing Overhead Costs; Actual Vs. Normal Costing of Manufacturing Overhead; Production Capacity, Predetermined Overhead Rates; Departmental vs. Plant-wise Overhead Rates; Separating Mixed Costs. Scatter-graph; High-low Method and Regression Analysis; Accounting for Manufacturing Overhead; Analysis and Disposition of Under-applied-and Over-applied Overhead

6. Contract Costing: Determination of profit of completed and incomplete contracts.

7. Introduction of Management Accounting: Definition-process of Management Accounting, characteristics of Management Accounting, scope of Management Accounting, purpose and objectives of Management Accounting, Comparison of Management Accounting and Financial Accounting

8. Cost Terms, Concepts and Classifications: Cost Behaviour (Analysis and Use): General Cost classifications- product Costs versus period Costs- Cost classifications on Financial Statements. Types of Cost behaviour patterns- the Analysis of Mixed Costs, High-low method

9. Cost-Volume-Profit Relationships: The basics of CVP analysis- Break -even analysis- Break-even chart- Sales Mix. Business application and mathematical problem of CVP analysis

10. Budget: Define Budget, Types of Budget, Cash budget, purchase budget, sales budget, flexible budget and related problems

11. Standard Costing: Meaning and Objectives- Types of ratios. Standard Costing and its uses for making business decision. Variance calculation, Decision making process from these calculation.

Course Learning Outcomes (COs):

CO1 Generalize the basic concepts of costs, uses of cost accounting knowledge in decision making, performance evaluation and modern economic environment.

CO2 Evaluate how materials, labor and overhead costs are added to a product at each stage of the production cycle.

CO3 Synthesize the skills necessary for carrying out effective management decision-making and strategic management planning and selection of the costs according to their impact on business and society

CO4 Interpret the impact of the selected costs method.

CO5 Design management control process in different business areas.

3: Strong 2: Moderate 1: Weak

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10

CO 1	3	-	-	3	3	-	-	-	-	-
CO 2	3	-	-	3	3	2	2	-	-	-
CO 3	3	-	-	3	3	2	2	-	-	-
CO 4	3	-	-	3	3	2	2	-	-	-
CO5	3			3	3	2	2			

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Basu, A., & Das, P. *Cost accounting*: Volume 1.
2. Garrison, R. H., & Noreen, E. W. *Managerial accounting*.
3. Mutz, U., Uzry, K., & et al. *Cost accounting*.

Course No: GEE 05322201	**Credits:2.0**	**Year: Second**	**Semester: Second**
Physical Geography			**Course Status: Theory**

Rationale of the Course: The primary goal of this course is to study the unique characteristics of earth and atmosphere and controlling factors, along with the origin and causes and processes which influence them. In addition, it will introduce different spheres like hydrosphere, biosphere, lithosphere, and atmosphere. It will also provide a deeper insight into climate change and its impact both globally and regionally.

Course Objectives:

The objectives of this course are –

1. To make students able to understand the origin and evolution of earth.
2. To elaborate the solar system and internal structure of the earth.
3. To compare different spheres (lithosphere, atmosphere, biosphere, and hydrosphere).
4. To acquire knowledge on microclimate.
5. To give learners a proper concept on climate induced hazards which they can apply to discuss the hazards of Bangladesh.

Course Contents:

1. **The Universe and Earth in Space:** The earth in space its origin and evolution. Heavenly Bodies-Stars, Constellations, Galaxies. Heliocentric and Geocentric Theories.
2. **Solar System and Earth:** Definition, Revolution and Rotation. Perihelion and Aphelion. Earth's Orbit, Solstice and Equinox, Shape and Size.
3. **Lithosphere:** Composition of Earth Crust: Rocks and Minerals. Endogenetic Processes-Diastrophism and Volcanism. Exogenetic Processes-Weathering, Mass-wasting, Erosion and Deposition.
4. **Hydrosphere:** Ocean and their Location. Composition of the Sea Water and Their Circulation. Ocean Currents and Their Causes. Man and Oceans.
5. **Biosphere:** The Definition of Biosphere, Ecosystems, Cycling of Metter, and Flow of Energy.
6. **Atmosphere:** Composition and Structure of the Atmosphere. Weather and Climate: Factors and Elements, Insolation, Temperature and Heat Budgets. Atmospheric Pressure, Planetary Wind System. Humidity. Clouds, Precipitation, Air mass, Classification of Climates (Elementary).
7. **Microclimate:** Urban and Forest Climate, Changes in different climatic parameters (Humidity, Temperature, Rainfall etc.) and it impute.

8. **Climate induced hazards:** Cyclones and Anticyclones. Storms and Thunderstorms. Atmospheric hazards. Application of Climatological data. Climate of Bangladesh, Climate Change.

Course Learning Outcomes (COs):
After the successful completion of the course, students will be able to –

CO 1. Identify the physical principles and processes governing the circulation and characteristics of the atmosphere and climates on Earth.
CO 2. Compare and contrast ecology and ecosystems.
CO 3. Interpreting and analyzing different endogenetic and exogenetic processes.
CO 4. Illustrate the interactions between the atmosphere, ocean and land in the Earth's climate system and apply the knowledge of climate induced hazards to understand the different hazards of Bangladesh.

Mapping Course Learning Outcomes (COs) with the POs:

3: Strong 2: Moderate 1: Weak

CO / PO	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	2								
CO 2						3	3		2	
CO 3		3	1			3				
CO 4		2			3	2		2		

Mapping Course Learning Outcomes (COs) with the Teaching Learning & Assessment Strategy:

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Strahler, A. (2017). *Introducing physical geography* (6th ed.). John Wiley & Sons.
2. Reynolds, S. J., &Rohli, R. V. (2015). *Exploring physical geography* (1st ed.). McGraw-Hill.
3. Holden, J. (2017). *An introduction to physical geography and environment* (4th ed.). Pearson Education Limited.
4. Barry, R. G., & Chorley, R. J. (2003). *Atmosphere, weather and climate* (8th ed.). Routledge.
5. Brammer, H. (1996). *The geography of the soils of Bangladesh* (1st ed.). University Press.

Course No: CSE 06122216	**Credit: 3.0**	**Year: Second**	**Semester: Second**
Database Management and Programming Laboratory			**Course Status: Practical**

Rationale:
Database System LAB course will concentrate on the design and implementation of a database system and applying SQL query.

Objectives:
The objectives of this course are to-

- Introduce the basic concepts of database.

- Developing a real world database application.
- Learn the design of a database starting from the conceptual design
- Implementation of database schemas and user interfaces to a database.

Course Contents:

1. **Introduction:** MySQL, Oracle, SQL, Data types, SQL / PLSQL, Oracle Software Installation, User Type, Creating User, Granting.
2. **Parts of Speech in SQL:** Creating Newspaper Table, Select Command (Where, order by), Creating View, Getting Text Information & Changing it, Concatenation, Cut & paste string (RPAD, LPAD, TRIM, LTRIM, RTRIM, LOWER, UPPER, INIT, LENGTH, SUBSTR, INSTR, SOUNDEX).
3. **Playing The Numbers:** Addition, Subtraction, Multiplication, Division, NVL, ABS, Floor, MOD, Power, SQRT, EXR, LN, LOG, ROUND, AVG, MAX, MIN, COUNT, SUM, Distinct, SUBQUERY FOR MAX, MIN.
4. **Grouping things together:** Group By, Having, Order By, Views Renaming Columns with Aliases.
5. **Query dependence:** Union, Intersect, Minus, Not in, Not Exists.
6. **Changing Data:** INSERT, UPDATE, MERGE, DELETE, ROLLBACK, AUTOCOMMIT, COMMIT, SAVEPOINTS, MULTI TABLE INSERT, DELETE, UPDATE, MERGE.
7. **Creating and Altering tables & views:** Altering table, Dropping table, Creating view, Creating a table from a table.
8. **Authority management:** Creating User, Granting User, Password Management.
9. **Introduction to PL/SQL**: Implement a few problems using PL/SQL (e.g. Prime Number, Factorial, Calculating Area of Circle, etc.).
10. **Introduction to Trigger and Procedure:** Implement few problems using Trigger and Procedures.
11. **Introduction to Indexing:** Implement indexing using a large database and observe the difference of Indexed and Non-Indexed databases.

Course Learning Outcomes (COs):

After the successful completion of the course, the student will be able to-

CO1 Demonstrate the knowledge in projects with a commercial relational database system (Oracle) and design a team-based project.

CO2 Utilize the database design principles, SQL and PL/SQL.

CO3 Demonstrate the relational database theory and be able to write relational algebra expressions for queries.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1			3				3			
CO2	3					3				
CO3			3				3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching Learning & Assessment Strategy:

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Silberschatz, A., Korth, H. F., &Sudarshan, S. (2002). *Database system concepts* (4th ed.). McGraw-Hill.
2. Smith, P. D., & Barnes, G. M. (1987). *Files and databases: An introduction*. Addison-Wesley.
3. Ramakrishnan, R., &Gehrke, J. (2003). *Database management systems* (3rd ed.). McGraw-Hill.

Course No: AMS 05222201	Credit: 2.0	Year: Second	Semester: Second
Wildlife Ecology and Management			Course Status: Theory

Rationale of the Course

Wildlife Ecology and Management helps students understand how wildlife populations interact with habitats, resources, disturbances, and human activities. The course links ecological principles with practical approaches needed for conservation, protected area planning, habitat improvement, and conflict mitigation.

Course Objectives

This course aims to:

- Introduce the ecological principles governing wildlife populations and habitats.
- Explain habitat relationships, population dynamics, community interactions, and carrying capacity.
- Develop understanding of wildlife management approaches, conservation planning, and human-wildlife interactions.
- Enable students to analyze wildlife management problems using ecological evidence.
- Build the basis for field-based wildlife assessment and management planning.

Course Content

1. **Introduction to wildlife ecology and management:** concepts, scope, and importance of wildlife management, wildlife resources of Bangladesh and their diversity.
2. **Ecological requirements of wildlife:** food, water, cover, space, habitat, and ecological niches.
3. **Landscape ecology**: landscape components affecting wildlife, edge effects.
4. **Population ecology:** census/counting, growth, distribution, regulation, and carrying capacity.
5. **Species interactions and communities:** predation, competition, mutualism, and community relationships.
6. **Wildlife monitoring:** conventional and advanced survey methods; digital monitoring tools; camera trap- principles and applications of camera trapping, acoustic monitoring- principles of passive acoustic monitoring, acoustic indices, advantages and limitations of acoustic monitoring; citizen science- public participation in wildlife monitoring.
7. **Human-Wildlife conflict management:** causes, impacts, and common mitigation approaches, volunteer organizations and their roles; ERT - Elephant Response Team, TRT - Tiger Response Team.
8. **Habitat fragmentation and management:** habitat fragmentation, characteristics of species vulnerable to fragmentation, habitat quality evaluation, management of fragmented habitat.
9. **Wildlife conservation and management:** red list index of IUCN, smart patrolling, wildlife territory, corridors, site specific plan (SSP), challenges in conservation planning, wildlife management system in Bangladesh, international organizations (WWF, IUCN, CITES and WCPA) and their role in wildlife conservation and management.
10. **Laws, acts, policy, and ethics**: wildlife laws, acts, policy issues, ethics in protection and conservation.

Course Learning Outcomes (CLOs)

CLO1. Explain major ecological principles relevant to wildlife populations, habitats, and communities.

CLO2. Analyze wildlife-habitat relationships and common management issues in forest ecosystems.

CLO3. Design basic wildlife management strategies using ecological and conservation principles.

CLO4. Evaluate wildlife conservation problems in relation to policy, ethics, and human dimensions.

CLO5. Apply wildlife ecology and management ideas clearly in professional contexts.

CLO-PO Mapping Matrix

Legend: 3 = Strong, 2 = Moderate, 1 = Low

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CLO1	3	1			1	1	1			
CLO2	2	1	2		1	2	3		1	
CLO3	2	1			1	3	3		2	
CLO4	2	1	1		2	2	3	3	1	
CLO5	1	3			1	1	2	1	3	

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL02, TL05	CA01, CA03, CA04, CA05, SA01, SA02
CO2	TL01, TL02, TL05	CA02, CA03, CA04, CA05, SA01, SA02
CO3	TL01, TL02, TL05	CA03, CA04,CA05, SA01, SA02, SA02
CO4	TL01, TL02, TL05	CA03, CA04,CA05, SA01, SA02
CO5	TL01, TL02, TL05	CA03, CA04,CA05, SA01, SA02

Suggested References

- Liu, J., et al. (2024). Challenges and opportunities for sustainable wildlife population recoveries. BioScience, 74(12), 874-889.
- Rattner, B. A., et al. (2024). Wildlife ecological risk assessment in the 21st century. Integrated Environmental Assessment and Management, 20(3), 725-744.
- Lemus, K., Barrasa, S., & McCall, M. K. (2024). Wildlife and cultural landscape management from a local knowledge approach: A review. Human Dimensions of Wildlife, 30(1), 1-13.
- Johnston, A., Matechou, E., & Dennis, E. B. (2023). Outstanding challenges and future directions for biodiversity monitoring using citizen science data. Methods in Ecology and Evolution, 14(1), 103-116.
- Ruck, A., van der Wal, R., Hood, A. S. C., Mauchline, A. L., Potts, S. G., WallisDeVries, M. F., & Öckinger, E. (2024). Farmland biodiversity monitoring through citizen science: A review of existing approaches and future opportunities. Ambio.
- Barnes, A. E., et al. (2025). Not just species recording: The potential of citizen science for habitat monitoring. Landscape Ecology.

Course No: AMS 05222202	Credit: 1.0	Year: Second	Semester: Second
Wildlife Ecology and Management Laboratory			Course Status: Practical

Rationale of the Course

This laboratory course provides practical training in wildlife ecology and management methods. It enables students to apply ecological concepts through habitat assessment, wildlife survey techniques, data analysis, and interpretation for management decisions.

Course Objectives

This course aims to:

- Train students in practical methods of wildlife and habitat assessment.
- Develop skills in collecting, organizing, and interpreting ecological data.
- Introduce simple analytical tools used in wildlife ecology and management.
- Strengthen practical decision-making, reporting, and presentation skills.

- Promote teamwork, ethics, and responsible field practice.

Course Content

1. **Conventional wildlife survey methods**- direct and indirect observation, transects, point counts, and quadrats, sign surveys and habitat assessment.
2. **Field and digital monitoring tools**- mobile-based application and data collection.
3. **Camera trap:** camera trap installation, Species detection.
4. **Acoustic monitoring:** applications in bird, amphibian, insect, and mammal monitoring, acoustic recorders and sound data collection.
5. **Citizen science:** species monitoring, community-based data collection and ecological observation.
6. **Ecological data analysis and reporting:** data entry and management, species richness, abundance, and diversity indices, basic analysis, visualization, habitat mapping, and report writing.

Course Learning Outcomes (CLOs)

CLO1. Apply basic wildlife survey and habitat assessment techniques in field and laboratory settings.

CLO2. Use simple quantitative and computer-based tools to summarize and interpret wildlife ecological data.

CLO3. Interpret wildlife signs, camera trap outputs, and acoustic records for ecological understanding.

CLO4. Prepare basic management-oriented conclusions from field observations and practical datasets.

CLO5. Perform field and laboratory tasks with teamwork, ethics, and professional responsibility.

CLO-PO Mapping Matrix

Legend: 3 = Strong, 2 = Moderate, 1 = Low

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CLO1	3	1	2			3	2		2	
CLO2	2	1	3			2	2			
CLO3	2	1	2		1	2	2			
CLO4	2	2	2			2	3		2	
CLO5	1	1					1	3	3	

Mapping Course Learning Outcomes (COs) with the Teaching-Learning& Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO2	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO3	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO4	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02
CO5	TL02, TL03, TL06	CA03, CA04,CA05, SA01, SA02

Suggested References

1. Barroso, P., & Palencia, P. (2025). Pixelated pathologies: Camera trapping as a tool for monitoring wildlife health. Journal of Applied Ecology, 62(11), 2894-2913. DOI: 10.1111/1365-2664.70171.
2. Bruce, T., Amir, Z., Allen, B. L., Alting, B. F., et al. (2025). Large-scale and long-term wildlife research and monitoring using camera traps: A continental synthesis. Biological Reviews, 100(2), 530-555.
3. Silva-Rodríguez, E. A., Cortés, E. I., Vásquez-Ibarra, V., Gálvez, N., Cusack, J., Moreira-Arce, D., et al. (2025). A protocol for error prevention and quality control in camera trap datasets. Journal of Applied Ecology, 62(4), 773-782. DOI: 10.1111/1365-2664.70010.

4. Roemer, C., Haquart, A., Lopez-Baucells, A., Besnard, A., et al. (2025). Current frontiers in the passive acoustic monitoring of bats. Methods in Ecology and Evolution, 16(11), 2534-2544. DOI: 10.1111/2041-210X.70157.
5. Martins, F. C., et al. (2025). Acoustic detection and occupancy models: A systematic review with insights for future monitoring programs. Ecological Indicators.
6. Hoefer, S., Allen-Ankins, S., McKnight, D. T., Nordberg, E. J., & Schwarzkopf, L. (2025). Sensors versus surveyors: Comparing passive acoustic monitoring, camera trapping and observer-based monitoring for terrestrial mammals. Methods in Ecology and Evolution, 16(11), 2603-2621.

Course No: FES 08213121	Credit: 3.0	Year: Third	Semester: First
Forest Mensuration and Inventory			Course Status: Theory

Course Rationale:
This course offers a comprehensive understanding of the fundamental principles of Forest Mensuration and Inventory, emphasizing their practical applications in field-level management. Learners will gain insight into modern forest measurement techniques, inventory processes, and related environmental challenges.

Course Objectives:
The objectives of the course are to-

- Facilitate necessary knowledge about the principles and governing rules of measurements of individual trees and stands, sampling, aerial photography and remote sensing.
- Develop skills for the field level assessment of forest resources through modern instruments and techniques to meet the needs of sustainable forest management.
- Acquaint students with the basic tools for the detailed procedure and methodology of forest inventory.

Course Content:
Mensuration

1. **Introduction:** Definition, objectives, scope and importance of mensuration, definition of different terms related to forest mensuration and inventory.
2. **Measurement of single trees:** Age, Diameter, Basal Area, Height, Bark Thickness, Tree Volume & Biomass, Tree Crown and Foliage, Leaf surface area, leaf weight and sapwood area.
3. **Volume, Log Classes and Weight of Felled Trees:** Volume, Log Rules, Grades and Classes, and Weight, Volume Table.
4. **Measurement of Forest Stands:** Definition of Forest Stand, Age, Diameter distribution, mean, and other parameters of the diameter distribution, Height distribution, relationship between diameter and height, mean and top heights, Stand density, Diversity and spatial structure, Volume and biomass, Site index, site class, or yield class, Present and future growth, Stand quality and vitality, Yield, and Damages.
5. **Trees stem form**: Stem form and shapes of trees, Taper functions and equations, Form factor, Form height, Form quotient, Methods of measuring form.
6. **Tree and Stand Biomass Estimation**: Biomass Components, Tree Level Regression Models, Additivity of Biomass Components, Dummy-Variables for Tree Species, Ratio Estimators and Cluster Sampling.
7. **Growth and Yield:** Definitions. Growth of Single Trees (Growth parameters, Stem analysis, Single-tree models); Site Class and Site Index (Definition, Site index curves, Site index equations); Growth of Stand (Estimating stand growth based on actual measurements, Stand table projection, Recent developments).

Inventory

1. **Introduction:** Definition, Objectives and types of forest inventory.
2. **Sampling for Forest Inventories:** random, systematic, cluster, two-phase, 3-P, multistage and stratified sampling designs, statistical concepts and their application in inventory, relative efficiency of

sampling designs, subjective and objective sampling designs, principles and application of point sampling.

3. Choice of plot size and shape, permanent sample plots and recurrent forest inventory.
4. **Carbon Inventory:** Sampling, above and below ground carbon assessment, REDD+
5. **Inventory planning and reporting:** Forest inventory planning and procedure; application of remote sensing and GIS techniques in forest inventory planning and design; collection of data from the field, different types of errors in inventory; processing and report writing.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Describe the scope and importance of mensuration, and be familiar with different terms related to forest mensuration and inventory.

CO2 Apply the principles and methods for measuring diameter, height of trees, shapes of trees, taper functions, form factor, tree volume, biomass, and tree growth.

CO3 Operate the principles and uses of different instruments and the selection of appropriate sampling techniques for field-level assessment.

CO4 Interpret the growth curve and yield table, and understand the use of GIS in forest inventory.

CO5 Use knowledge of mensuration in forest inventory, field data collection and report writing.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2	2		1	
CO2	3		3			1				
CO3	3					2				
CO4	3								2	
CO5	2			1		3			2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Kershaw, J. A., Jr., Ducey, M. J., Beers, T. W., &Husch, B. (2017). Forest mensuration (5th ed.). John Wiley & Sons.
2. Van Laar, A., &Akça, A. (2007). Forest mensuration. In K. van Gadow, T. Pukkala, & M. Tomé (Eds.), Managing forest ecosystems (Vol. 13). Springer.
3. Chaturvedi, A. N., & Khanna, L. S. (1982). Forest mensuration. International Book Distributors.
4. Temu, A. R. (1990). Forest mensuration (Field Document No. 3). UNDP/FAO/BGD/85/011, Institute of Forestry, Chittagong University.
5. Mahmood, H., Siddique, M. R. H., Abdullah, S. M. R., Akhter, M., & Islam, S. M. Z. (2016). Field measurement protocol on tree allometric equations for estimating above-ground biomass and volume in Bangladesh. Food and Agriculture Organization of the United Nations & Forest Department, Government of the People's Republic of Bangladesh.
6. Kangas, A., &Maltamo, M. (2007). Forest inventory: Methodology and applications. In K. van Gadow, T. Pukkala, & M. Tomé (Eds.), Managing forest ecosystems (Vol. 13). Springer.
7. Matthews, R. W., & Mackie, E. D. (2006). Forest mensuration: A handbook for practitioners. Forestry Commission.

Course No: FES 08213122	Credit: 1.0	Year: Third	Semester: First
Forest Mensuration and Inventory Laboratory			Course Status: Practical

Course Rationale:
This course will provide practical knowledge on forest mensuration and inventory in the context of forest resource assessment and management.

Course Objectives:
The objectives of this course are to-

- Help the students to develop the ability to operate instruments commonly used for forest mensuration and inventory.
- Provide knowledge on data collection from forest plots.
- Foster the analytical and critical knowledge to prepare an inventory plan.

Course Content:

1. Demonstration of instruments commonly used in forest mensuration and inventory.
2. Use of different instruments to measure diameter and height, slope correction.
3. Field exercises in tree and stand measurements.
4. Construction of growth curve, form factor, biomass equation and volume table.
5. Demonstration and practice of different sampling techniques followed in forest inventory including- point sampling, cluster sampling, transects,
6. Preparation and execution of an inventory plan.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Apply instruments commonly used in forest mensuration and inventory.
CO2 Operate different instruments in tree and stand measurements.
CO3 Construct growth curve, form factor, biomass equation and volume table.
CO4 Prepare an inventory plan.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3						2			
CO2	3		1				2			
CO3			3			2				
CO4				2		3			1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Kershaw, J. A., Jr., Ducey, M. J., Beers, T. W., &Husch, B. (2017). Forest mensuration (5th ed.). John Wiley & Sons.
2. Van Laar, A., &Akça, A. (2007). Forest mensuration. In K. van Gadow, T. Pukkala, & M. Tomé (Eds.), Managing forest ecosystems (Vol. 13). Springer.
3. Kangas, A., &Maltamo, M. (2007). Forest inventory: Methodology and applications. In K. van Gadow, T. Pukkala, & M. Tomé (Eds.), Managing forest ecosystems (Vol. 13). Springer.
4. Mahmood, H., Siddique, M. R. H., Abdullah, S. M. R., Akhter, M., & Islam, S. M. Z. (2016). Field measurement protocol on tree allometric equations for estimating above-ground biomass and volume in Bangladesh. Food and Agriculture Organization of the United Nations & Forest Department, Government of the People's Republic of Bangladesh.
5. Matthews, R. W., & Mackie, E. D. (2006). Forest mensuration: A handbook for practitioners. Forestry Commission.

Course No: FES 08213123	Credit: 2.0	Year: Third	Semester: First
Plantation Silviculture			Course Status: Theory

Course Rationale:
Plantation activities are essential for maintaining the balance of forest ecosystems. Effective and sustainable forest plantations require sound scientific understanding, practical experience, and familiarity with global silvicultural systems. This course introduces students to the science of silviculture specific to plantation forestry, using both theoretical and practical approaches.

Course Objectives:
The objectives of the course are to-

- Help the students to know the basic of the silvicultural system and practices; need and types of plantation
- Facilitate necessary knowledge about tree seeds, seed production area, seeds preservation, processing for plantation.
- Enhance the skill of choosing species and nursery sites for plantation, establishment of nursery site and other cultural practices prior to any plantation activity.
- Apply the knowledge of silviculture for better forest management.

Course Content:

1. **Introduction:** Silviculture, silvicultural system and silvicultural practices, need for forestation in Bangladesh, scope of plantation silviculture in Bangladesh.
2. **Types of Plantation:** Mono and mixed plantation, different forms of mixture in a mixed plantation.
3. **Forest Tree Seeds:** Sources of seeds, seed collection time and procedure, extraction, drying, cleaning, storing procedure of forest tree seeds, pre-treatments of seeds, seed testing, seed importing, seed quarantine and seed certification, use of seed from seed orchards.
4. **Seed Production Area:** Requirements for seed production area, criteria for selection of trees to be retained, thinning and management of seed production area.
5. **Choice of Species:** Criteria for selection of species, species-site matching.
6. **Nursery and Cultural Practices:** Importance and types of forest nursery, site selection, nursery establishment, sowing and potting, care and maintenance - seedling counting in the nursery, watering, shading, mulching, weeding, root pruning, and hardening off, pest control, morphological grading, transporting seedlings, seedling raised in nursery bed, polybag and/or polybag substitute and containers/root trainers, role of women in the nursery management and estimation.
7. **Plantation Establishment, Estimation and Management:** Planting survey, site preparation, out planting procedures, seedling transportation– labour required, guard required (man days/wheat required), cultural practices - mulching, growing of cover crops and nurse crops, watering, use of fertilisers, method and season of application and environmental implications, under planting, enrichment plantation, assisted natural regeneration (ANR), pure and mixed plantation development and management, protection and harvesting methods.

8. **Coastal afforestation:** History, objectives, and impacts of coastal afforestation in Bangladesh; nursery and plantation techniques of major species.
9. **Sustainable plantation forestry:** Concepts and practices.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Explain fundamental silvicultural practices and their application in plantation forestry.
CO2 Compare and evaluate different types of plantations, including mono and mixed systems.
CO3 Illustrate the procedures for forest tree seed collection, processing, storage, and certification.
CO4 Develop a comprehensive plan for establishing and managing a forest nursery.
CO5 Formulate strategies for plantation establishment, management, and sustainable forestry practices.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3						1			
CO2	2						2			
CO3	2					2	2			
CO4	3			2		2	3		2	1
CO5	3			1			3		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Chowdhury, M. R. (1995). *The forest nursery and plantation manual*. Forest Department, Government of the People's Republic of Bangladesh.
2. Davidson, J. (1985). *Species and sites: What to plant and where to plant* (Field Document No. 5). UNDP/FAO Project BGD/79/017.
3. Evans, S. (1982). *Plantation forestry in the tropics*. Clarendon Press.
4. Zabala, N. Q. (1990). *Plantation silviculture* (Field Document No. 19). UNDP/FAO/BGD/85/011, Institute of Forestry, University of Chittagong, & FAO.
5. Zabala, N. Q. (1990). *Silviculture of species* (Field Document No. 14). UNDP/FAO/BGD/85/011, Institute of Forestry, University of Chittagong, & FAO.
6. Evans, J., & Turnbull, J. W. (2004). *Plantation forestry in the tropics: The role, silviculture and use of planted forests for industrial, social, environmental and agroforestry purposes* (3rd ed.). Oxford University Press.
7. Shepherd, K. R. (2014). *Plantation silviculture*. Springer.

Course No: FES 08213124	**Credit: 1.0**	**Year: Third**	**Semester: First**
Plantation Silviculture Laboratory			**Course Status: Practical**

Course Rationale:
Plantation activities play crucial roles in maintaining and establishing forest ecosystems. Such plantation activities require proper scientific knowledge, experience and practice. This course aims to provide practical

knowledge of silvicultural practices of raising a plantation. This course also provides options for practical experience on seed treatment and seedling rising of different species.

Course Objectives:
The objectives of the course are to-

- Help students to develop skill in collection and treatment of different seeds
- Apply the knowledge to prepare nursery bed and raise seedlings
- Foster the analytical and critical analysis to prepare a plantation plan for a specific area

Course Content:
1. Seed collection & storage
2. Seed testing and their uses
3. Treatments and germination of major tree species according to forest types
4. Preparation of a seed bed, sowing and nursing, care and maintenance of seedlings in the nursery
5. Compost preparation
6. Participation in plantation activities and maintenance of plantation

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Describe and explain the collection and storage procedure for seeds of different species
CO2 Perform different seed testing procedures and explain their uses
CO3 Prepare seed bed and raise seedlings; Maintain and care for the seedlings
CO4 Design different plantation activities

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2	2		3	
CO2	3					2	2		3	
CO3	3					2	2		3	
CO4	3					2	2			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1 Chowdhury, M. R. (1995). *The forest nursery and plantation manual*. Forest Department, Government of the People's Republic of Bangladesh.
2 Davidson, J. (1985). *Species and sites: What to plant and where to plant* (Field Document No. 5). UNDP/FAO Project BGD/79/017, Assistance to Forestry Sector.
3 Evans, J. (1982). *Plantation forestry in the tropics*. Clarendon Press.
4 Zabala, N. Q. (1990). *Plantation silviculture* (Field Document No. 19). UNDP/FAO/BGD/85/011, Institute of Forestry, University of Chittagong & FAO.
5 Zabala, N. Q. (1990). *Silviculture of species* (Field Document No. 14). UNDP/FAO/BGD/85/011, Institute of Forestry, University of Chittagong & FAO.

Course No: FES 08113121	Credit: 2.0	Year: Third	Semester: First
Production and Management of Tea, Coffee and Rubber			Course Status: Theory

Course Rationale:
Tea, coffee, and rubber are vital to Bangladesh's economy, contributing to GDP and rural employment. This course teaches scientific cultivation, processing, and management of these crops, from nursery practices to pest control and product grading. It emphasizes sustainable agro-techniques, climate-resilient farming, and eco-friendly processing for global standards. Students will analyze their socio-economic impact and learn modern methods for enhancing yield and quality. The course prepares graduates to lead in agribusiness, policy, or research, boosting Bangladesh's plantation sector.

Course objectives:
The objectives of the course are

- To facilitate the students to acquire necessary knowledge about the management, plantation techniques and different agro-types of tea, coffee and rubber.
- To provide the knowledge of the current situation of management of these crops and analyze the problems in management programs.
- To enhance students' skill on improving different varieties of these crops.
- To help the students apply the knowledge for different nursery techniques for improving crop quality.

Course Content:

1. **Tea Estate and Plantation Overview:** Tea estate in Bangladesh: structure, growth, national economic importance. History of tea cultivation. Tea ecosystem: climatic and edaphic factors.
2. **Tea Plant Biology and Agro-types:** Botanical description of tea plant. Tea agro-types in Bangladesh. Important VP clones of BTRI
3. **Agro-techniques of Tea Culture:** Propagation by seeds and vegetative parts. Selection of mother bush from seed barie and nucleus clone plot. Nursery bed preparation for seeds and VP nursery. Planting techniques and maintenance of nursery plants.
4. **Tea Crop Management and Protection:** Land preparation. Planting techniques for tea, shade trees, and green manuring crops. Bringing up young tea: pruning, plucking, manuring and fertilization. Shade trees and ancillary crops management. Pest, disease, and weed control.
5. **Tea Manufacture:** Plucking techniques and leaf handling. Withering, rolling, fermentation (oxidation), drying, sorting, grading. Manufacturing of green tea and black tea. Tea grades and packaging.
6. **Coffee Cultivation and Processing:** Importance and future of coffee in Bangladesh. Botanical description of coffee plant. Climatic and edaphic requirements. Coffee processing and product grades.
7. **Rubber Plant Biology and Ecosystem:** Origin and botanical description of Hevea brasiliensis. History and development of rubber plantations in Bangladesh. Climate and soil requirements for rubber cultivation.
8. **Rubber Cultivation and Nursery Management:** Site selection and land preparation. Seed collection; germination bed, nursery bed, bud bank development; bud grafting; transfer of ramet to poly bags and maintenance. Important clones
9. **Latex Harvesting and Rubber Processing:** Techniques and rules of tapping. Latex collection and handling. Processing of latex leading to crude rubber production. Grades of final rubber products.

Course Learning Outcomes (COs) :
After the successful completion of the course, students will be able to
CO1: Explain the economic, ecological, and social importance of tea, coffee, and rubber to Bangladesh and the global plantation industry.
CO2: Apply management and plantation techniques for tea, coffee, and rubber cultivation, including nursery establishment, planting, pruning, and pest control
CO3: Compare agro-types of tea, coffee, and rubber, and select appropriate nursery techniques (seed barie, V.P. nursery, bud grafting) to improve crop quality.
CO4: Evaluate the current management systems of tea, coffee, and rubber plantations in Bangladesh and propose improvements to address identified problems in cultivation, pest control, and yield.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2		3						
CO2	2					2	3			
CO3	3		1			2	2			
CO4	2	3		1		3	3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Eden, T. (1965). *Tea* (2nd ed.). Longman's Green.
2. Harler, C. R. (1967). *The culture and marketing of tea* (3rd ed.). Oxford University Press.
3. Ho, C. T., Lin, J. K., &Shahidi, F. (2008). *Tea and tea products: Chemistry and health-promoting properties*. CRC Press.
4. Sharma, V. S., & Gunasekare, K. (2018). *Global tea science: Current status and future needs*. Burleigh Dodds Science Publishing.
5. Wintgens, J. N. (Ed.) (2009). Coffee: Growing, Processing, Sustainable Production (2nd ed.). Wiley-VCH.
6. Priyadarshan, P. M. (2017). Biology of Hevea rubber (2nd ed.). Springer.
7. DaMatta, F. M., & Ramalho, J. D. C. (2006). Impacts of drought and temperature stress on coffee physiology and production: a review. Brazilian Journal of Plant Physiology, 18(1), 55–81.
8. Sethuraj, M. R., & Mathew, N. M. (Eds.) (1992). Natural Rubber: Biology, Cultivation and Technology (Developments in Crop Science, Vol. 23). Elsevier Science Publishers, Amsterdam.

Course No: FES 08113122	**Credit: 1.0**	**Year: Third**	**Semester: First**
Production and Management of Tea, Coffee and Rubber Laboratory			**Course Status: Practical**

Course Rationale:
This course will provide practical knowledge on important aspects of plantation and management of tea, coffee and rubber. Direct visit and engagement in preparatory seedling and plantation raising activities are essential for the better understanding of tea, coffee and rubber gardening.

Course Objectives:
The objectives of the course are to-

- Acquaint students with the basic tools of management and plantation techniques of tea, coffee and rubber.
- Help them conceptualize basic theories in different agro-types, cones and seedlings of tea, coffee and rubber.
- Provide knowledge in improving different varieties of these crops.
- Help the students accumulate basic ideas about different nursery techniques for improving crop quality.

Course Content:

1. Various methods of propagation of tea, coffee and rubber.
2. Establishment of nursery of rubber, tea and coffee, techniques of tapping rubber.
3. Participation in plantation activities and maintenance of plantation of tea, coffee and rubber.
4. Visiting a T eamanufacturing factory.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1	Experiment methods of propagation of tea, coffee and rubber.
CO2	Prepare propagation materials and plan the plantation techniques and maintenance of tea, coffee and rubber.
CO3	Visualize the outline, design and formation of the nursery .
CO4	Design the tapping technique of rubber.
CO5	Manage cultural operations of tea, coffee and rubber plantation.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3									
CO3	3									
CO4	3									
CO5	2						1			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO5	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Grammer, P. J. S. (1957).*Review of literature of coffee research in Indonesia.* Inter-American Institute of Agricultural Research, Miscellaneous Bulletin No. 15. Turrialba, Costa Rica.
2. Verheye, W. (2010). *Growth and production of rubber*. In W. Verheye (Ed.), Land use, land cover and soil sciences (Vol. 1). UNESCO-EOLSS

3. Eden, T. (1965). *Tea* (2nd ed.). Longman's Green.
4. Priyadarshan, P. M. (2017). *Biology of Hevea rubber* (2nd ed.). Springer.
5. Zhongmao, C. (1996). *Evolution and development of China as a key tea-producing and consuming country*. Asia International Conference Proceedings, Singapore, 253–264.
6. Harler, C. R. (1967). *The culture and marketing of tea* (3rd ed.). Oxford University Press.
7. Harper, A. E. (1962). *Modern coffee production* (2nd ed.). Leonard Hill.

Course No: FES 08213126	**Credit: 1.0**	**Year: Third**	**Semester: First**
Fieldwork in Forests (Deciduous Sal Forest)			**Course Status: Practical**

Course Rationale:
This course will allow a student to gain hands-on experience of the deciduous Sal forest, forestry sector situation and nursery management in Bangladesh.

Course Objectives:
The objectives of the course are to-

- Make the students understand the diversity of the deciduous Sal forest ecosystem, its trends, applications, role in environmental stability.
- Develop skills for practicing silviculture, GIS, Agroforestry, Forest Industries, Community development.
- Get ideas about the professional roles in forest and forestry sectors.
- Apply the acquired knowledge to develop forest nursery and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Diversity of deciduous Sal forest ecosystem in Bangladesh and its components.
2. Plantation silviculture, GIS, Agroforestry, Forest industries and Community development.
3. The overall situation and cultural values of different deciduous Sal forest, forestry and environmental sector related areas of Bangladesh.
4. Nursery and Plantation Management (Nursery bed preparation and seedling raising; stem cutting; grafting; nursery disease management; irrigation management; plantation disease management; Tree Planting; Nursing; Reporting etc.)
5. Practical application of all core courses of the current semester will be demonstrated.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Discuss diversity of deciduous Sal forest ecosystem, its trends, applications, role in environmental stability.
CO2 Generalize knowledge about silviculture, GIS, Agroforestry, Forest Industries, Community development, and nursery management.
CO3 Evaluate the cultural values of the deciduous Sal forest ecosystem.
CO4 Criticize the relevant field level management problems and challenges to meet the needs of the country.
CO5 Develop plans for nursery and Plantation Management (Tree Planting, Nursing, Reporting etc.).

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	3								

CO2	3	3								
CO3	3	3					3		3	
CO4	3	3					3		3	
CO5	3	3					3		3	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL06	CA03, CA04, CA05
CO2	TL04, TL06	CA03, CA04, CA05
CO3	TL04, TL06	CA03, CA04, CA05
CO4	TL04, TL06	CA03, CA04, CA05
CO5	TL04, TL06	CA03, CA04, CA05

Recommended References:

1. Documents provided by the related visited areas' official/s.
2. Documents provided by the related course teacher/s.

Note: In case of unavoidable situations , this course will be exchanged with course no**. FES 08213222**

Course No: FES 08113123	**Credit: 3.0**	**Year: Third**	**Semester: First**
Agroforestry and Upland Farming System			**Course Status: Theory**

Course Rationale:

This course focuses on importance, technical aspects and feasibility of agroforestry to the plain land and upstream and qualifies students with the sense of species composition, budget, design and management of ecologically and economically viable agroforestry projects. As agroforestry is a highly technical practice that combines agriculture and forestry, all forestry students should have a sound knowledge of this specialized production system to ensure both the food crop and forest products to meet the challenges of limited land and growing demand for food.

Course Objectives:

The objectives of this course are to-

- Enhance the knowledge of agroforestry practices in Bangladesh and abroad.
- Increase the skills of planning and management of various agroforestry projects.
- Develop skills on the financial and economic aspects of agroforestry projects.
- Provide the knowledge of different agroforestry research tools and designs.
- Apply the knowledge of upland farming activities, problems, prospects and consequences.

Course Content:

1. **Introduction:** Concepts, components, beneficiaries' roles, the potential role of agroforestry; Agroforestry for soil conservation.
2. Importance of people's awareness for agroforestry.
3. **Basic Principles in Agricultural Crop Production:** Classification of agronomic crop based on use; Classification of horticultural crops and their cultural management; Pest and diseases control, harvesting and post production technology.
4. **Tree-Crop Interactions:** Competitive and complementary relationships in agroforestry systems; Effects on growth, yield, and sustainability.
5. **Agroforestry Planning:**Biophysical and socio-economic survey by diagnosis and design method; Evaluation of species adaptability for agroforestry system.

6. **Financial and Economic Analysis of Agroforestry Project:**Aspects of project preparation and analysis; Identifying project Costs and benefits; Indicators of profitability and examples of financial analysis of agroforestry; Demonstration to agroforestry farms.
7. **Agroforestry Practices in Bangladesh**: Agroforestry in homesteads, croplands and forest lands of Bangladesh; Cropping intensity and cropping pattern; Different models of agroforestry.
8. **Agroforestry in Other Countries:** Agroforestry practices in Ghana, Kenya, Nigeria, Trinidad, Malaysia, Indonesia, Thailand, India, Sri Lanka and Philippines.
9. **Upland farming system:** Concepts, characteristics and components.; Interaction of enterprise, on-farm and off-farm activities; Physical, biological and economical determinants of upland farming systems, traditional and modern upland farming systems-Tanguaya, shifting cultivation, SALT, contour.
10. **Modern Technologies:** Contour farming, Sloping Agricultural Land Technology (SALT), terracing, and other soil and water conservation techniques.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand the basic concepts of agroforestry practices.
CO2 Evaluate agroforestry projects.
CO3 Create financial and economic aspects of agroforestry projects.
CO4 Design an upland farming systems.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2		2	2	2				
CO2	3	2	3	3	3	3	3	3	3	3
CO3	3	2	3	3	3	3	3	3	3	3
CO4	3	2	3	3	3	3	3	3	3	3

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Huxley, P. A. (1983). *Plant research and agroforestry*. International Centre for Research in Agroforestry (ICRAF).
2. Nair, P. K. R., Scholz, F. G., & Duryea, M. L. (1990). *Agroforestry systems* (Vol. 12). Kluwer Academic Publishers.
3. Raintree, J. B. (1987). *D & D user's manual: An introduction to agroforestry diagnosis and design*. ICRAF.
4. Zabala, N. Q. (1990). Agroforestry (Field Document No. 1). UNDP/FAO/BGD/85/011, Institute of Forestry, Chittagong University & FAO.
5. International Centre for Research in Agroforestry. (Various years). *Agroforestry today*. ICRAF.
6. Nair, P. K. R. (1993). *An introduction to agroforestry*. Kluwer Academic Publishers in cooperation with ICRAF.
7. Nair, P. K. R., Kumar, B. M., & Nair, V. D. (2009). *Agroforestry as a strategy for carbon sequestration*. Journal of Plant Nutrition and Soil Science, 172(1), 10–23.
8. Raj, A. J., &Lal, S. B. (2014). *Agroforestry theory and practices*. Scientific Publishers.

Course Code: FES 08113125	Credit: 3	Year: Third	Semester: First
Forest and Environmental Economics		**Course Status: Theory**	

Rationale of the Course: This course will provide an adequate knowledge of the basic concepts of microeconomics, techniques of microeconomic analysis and application in forestry and environmental decision-making problems. The main emphases of this course are to increase land productivity to meet the increasing demand and restore the environmental stability.

Course Objectives: The objectives of this course are

- To facilitate necessary knowledge about microeconomics.
- To develop skills on economics of wood and non-wood forest products.
- To apply the knowledge for valuation of land, wildlife, forest recreation, other forest products and ecosystem services.
- To foster the analytical and critical knowledge of economics for pollution control.

Course Content:

1. **Introduction**: Concepts of Economics, Forest Resource Economics, Environmental Economics, Opportunity costs, Production Possibility Curve, Value Added; and Common Property Resource; Forest Economics aid to capital budgeting and resource allocation, scope and nature of Forest and Environmental Economics.
2. **Economics of wood production:** Types of decisions taken by a forest manager; Time as an important factor in timber production; Long term and short term investment; Faustmann's soil expectation value formula, Forest Rent. Application of economic tools in silvicultural decisions- planting, species selection, site improvement, pruning, thinning and felling decisions.
3. **Multiple use**: Global environmental crisis; the challenge to forestry; multiple use in forestry; an economic model of multiple use and joint production; application problems.
4. **Public production and welfare**: Concepts of welfare economics; objectives of public budget policy; public production; production approach; the usual benefit/cost approach; major application problems of benefit/cost approach.
5. **Water and watershed management**: Economic analysis and watershed management; water benefits; water costs; measuring benefits.
6. **Forest recreation**: Economic characteristics of forest recreation; factors affecting demand; factor affecting supply; a demand model for forest campground; travel cost method; modified travel cost method.
7. **Wildlife**: General aspects of wildlife economics; wildlife as a common property resource; the input approach; the capital growth approach.
8. **Forest sector to GDP:** Contribution of forestry sector to GDP; computing methods; valuation of forest goods and services.
9. **Forestry for employment promotion:** Employment opportunities in forestry; creating forestry jobs and its challenges.
10. **Economics of pollution control**: A pollutant taxonomy; efficient allocation of pollution; market allocation of pollution; efficient policy responses.

Course Learning Outcomes (COs)

CO1 Generalize the basic concepts and interpret the importance of microeconomics in forest and environmental decision making problems

CO2 Evaluate the economics of wood and non-wood forest products

CO3 Compute the values of land, water, wildlife, forest recreation and other forest products and ecosystem services

CO4 Synthesize the knowledge of economics of pollution control

Mapping Course Learning Outcomes (COs) with the POs

3: Strong 2: Moderate 1: Weak

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10

CO 1	3	-	-	3	3	-	-	-	-	-
CO 2	3	-	-	3	3	-	-	-	-	-
CO 3	3	-	2	3	3	-	2	-	-	-
CO 4	3	-	-	3	3	1	-	-	-	-

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Ahmed, M. H. (2004). *Principles of environmental economics*. Routledge.
2. Gregory, G. R. (1972). *Forest resource economics*. John Wiley & Sons.
3. International Institute for Environment and Development (IIED). (2003). *Valuing forests*. Environmental Economics Programme, IIED.
4. Nair, C. T. S., &Rutt, R. (2009). *Creating forestry jobs to boost the economy and build a green future*. Unasylva, 60(233), 3–10.
5. Pant, M. M. (1986). *Forest economics and valuation*. Medhawi Publishers.
6. Saha, N., &Mukul, S. A. (2022). *Visitor's willingness to pay for cultural ecosystem services in Bangladesh: An assessment for Lawachara National Park, a biodiversity hotspot*. Small-scale Forestry, 21, 185–201.
7. Tietenberg, T., & Lewis, L. (2018). *Environmental and natural resource economics* (11th ed.). Routledge.
8. Williams, M. R. W. (1990). *Decision making in forest management* (2nd ed.). International Book Distributors.

Course No:FES 07313131	**Credit: 3.0**	**Year: Third**	**Semester: First**
Forest Based Industries and Community Development			**Course Status: Theory**

Course Rationale:

The forests of Bangladesh harbor a wide range of timber and non-timber resources. Effective utilization of these resources can help reduce pressure on forest ecosystems. This course is designed to enhance undergraduate students' basic understanding of lumber processing, the manufacturing of various composite materials, and the production of pulp and paper. Additionally, it aims to provide knowledge on small-scale industrial utilization and the marketing of manufactured forest-based products.

Course Objectives:

The objectives of the course are to:

- Provide an overview of different types of forest-based and forest-based rural industries, and their roles in community development.
- Acquaint students with the raw materials used in forest-based and rural industries, including their sources and associated challenges.
- Enhance skills related to sawmilling, saw doctoring, composite material manufacturing, pulping, and papermaking.
- Foster the application of entrepreneurship in forest-based industries.

Course Content:

1. **Introduction:** Major Forest based industries in Bangladesh, their role in the socio-economic development of Bangladesh and issues related to raw material supply.
2. **Sawmilling:** Types of saws and their advantages and disadvantages, saw teeth geometry, sawing methods, saw doctoring, sawmill selection and layout, and timber grading.
3. **Composites:** Plywood, Laminated Wood, Particleboard and fiberboard--Definition, importance, raw materials, manufacturing process, and uses.

4. **Pulp & Paper:** Planning and layout of a paper mill, pulping raw materials, pre-treatment of raw materials, pulping and bleaching processes, paper properties, characteristics of different paper grades.
5. **Overview of forest-based rural industries/enterprise:** Definition and characteristics of small-scale forest based rural enterprises-technology, location, size, entrepreneurship, and labor. The role of these enterprises in community development, employment, income generation, and other socio-economic impacts.
6. **Types of Forest Based Rural Industries:** Wood-based rural enterprises - types/classifications, current availability, processing facilities, markets, social benefits, and their impact on rural economy. Industries based on agar, bamboo, rattan (cane) and patipata,wild fruits and nuts, medicinal plants, mushrooms, tannins, gums, essential oils, resin, and sericulture.
7. **Market development Strategies:** Market identification and development, financing, promotion of technology and technical skills, improvement of management capabilities, institutional support for small-scale enterprises, extension services, and promotion of community-based small-scale forest processing enterprises.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Generalize the basic concept of forest-based industries and forest-based rural industries, and identify the challenges involved in their development.
CO2 Apply the knowledge of sawmilling and saw doctoring processes in lumber processing industries.
CO3 Develop appropriate strategies to maximize the utilization of wood and forest resources in developing forest-based industries.
CO4 Design and manage forest based rural industries and explain their importance in community development.
CO5 Evaluate and apply knowledge and skills in forest-based industries to develop entrepreneurship and maximize resource utilization.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3						2			
CO3	3						2			
CO4	3								2	
CO5	3								2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Hetemäki, L., &Hurmekoski, E. (2016). F*orest products markets under change: Review and research implications.*Current Forestry Reports, 2, 177–188.
2. Ansell, M. P. (2015). *Wood composites*. Woodhead Publishing.
3. Davim, J. P., & Aguilera, A. (2017). *Wood composites: Materials, manufacturing and engineering*. De Gruyter.
4. Anon. (1970). *Indian forest utilization* (Vol. I). Forest Research Institute (FRI).
5. Britt, K. W. (1984). *Handbook of pulp and paper technology* (1st ed.). CBS Publishers & Distributors.
6. FAO. (1987). *Small-scale forest-based processing enterprises* (FAO Forestry Paper No. 79). Food and Agriculture Organization.

7. FAO. (1986). *Appropriate forest industries* (FAO Forestry Paper No. 68). Food and Agriculture Organization.
8. FAO. (1981). *Small and medium sawmills in developing countries* (FAO Forestry Paper No. 28). Food and Agriculture Organization.

Course No: FES 05323221	Credit: 3.0	Year: Third	Semester: Second
Remote Sensing and Geographic Information System			Course Status: Theory

Course Rationale:
Current forest management, conservation monitoring and inventory system is hugely assisted by aerial science. For an applicable expertise on remote sensing, it is necessary to gather the basic academic knowledge of this science. This may help students to learn about the basics and the possible use of the remote sensing and GIS science for the betterment of our forested areas.

Course Objectives:
The objectives of this course are
- Outline an introduction to the science of Remote Sensing and also acquaint the students with Geographic information System.
- Let the students understand how sensor and spatial photograph works, how it can be used to identify different earth features.
- Provide a general demonstration on electromagnetic energy and satellite, allowing the students analyze satellite images and energy interaction.
- Introduce students with some of the active and passive remote sensing systems.

Course Content:
Remote Sensing (RS):
1. Introduction: Fundamental concepts of remote sensing: History and key applications of remote sensing in resource and environmental management.
2. Fundamentals of Electromagnetic Radiation: Nature and propagation of electromagnetic energy: Interaction of electromagnetic radiation (EMR) with the atmosphere including scattering, reflection, absorption, atmospheric windows, irradiance, and exitance; Interaction of EMR with matter: spectral properties and reflectance characteristics of vegetation, soil, and water; Concepts of spectral signature, red edge, and vegetation indices.
3. Remote Sensing Platforms and Sensors: Overview of major space-borne systems: Sentinel, Landsat, IRS, SPOT, ENVISAT, Terra, Aqua, etc.; Sensor types and characteristics: airborne vs. space-borne sensors, platforms, and their applications; Satellite orbits, orbital altitudes, and types of satellites; Basics of satellite communication.
4. Aerial Photography and Photogrammetry: Terminology: drift, tilt, crab, displacement, overlap, photo centers; Types of aerial photography: scale, quality, aerial cameras, films, and filters; Mosaic development, photography planning, ground control, and ground truthing; Concepts of stereoscopy, parallax, and depth perception; Applications of aerial photography in forestry and natural resource assessment.
5. Satellite Imagery and Digital Image Processing: Understanding digital satellite imagery: display, enhancement, and false color composites (FCC); Image preprocessing techniques: Cosmetic operations, radiometric calibration, atmospheric and geometric corrections (geo-referencing); Image resolution: spatial, spectral, temporal, and radiometric; Image classification: supervised, unsupervised and mixed methods. Classification accuracy assessment.
6. Image Interpretation and Surface Property Estimation: Principles and elements of image interpretation: Estimating earth surface properties, especially vegetation parameters, from remote sensing imagery.
7. Microwave Remote Sensing: Principles and applications of microwave remote sensing in earth observation.
8. Global Positioning System (GPS): Principles, components, and applications of GPS in spatial data collection.

Geographic Information System (GIS)
1. Introduction to GIS: Basic concepts, components, functional elements, and applications.
2. Data Structures and Operations in GIS: Spatial and non-spatial data; Raster and vector data models; data layers and attribute information; topological relationship; basic geometric vector operations; buffer zones and

overlay techniques; post-overlay cleanup operations including clip, update, dissolve, merge, and eliminate; interpolation methods; spatial and attribute data queries; spatial analysis and mapping; map projection system; coordinate system and georeferencing; spatial data visualization.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Outline historical evolution of remote sensing and describe possible applications of the science.
CO2 Analyze variation in electromagnetic waves, energy interactions and categorizes spectral properties of earth objects.
CO3 Distinguish different types of airborne system, satellites and their function.
CO4 Define aerial photography and Photogrammetry, apply basic procedures of digital image interpretation and processing for field application.
CO5 Explain maps and geospatial data, apply geometric operations and evaluate the application of GIS.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	2	1				1	1	1	1	
CO2	3		1			1	1		1	
CO3	3		2			1				
CO4	3	1	3			3	2			
CO5	3	2	3			2	3		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Campbell, J. B. (2006). *Introduction to remote sensing* (4th ed.). Taylor and Francis.
2. Chuvieco, E. (2020). *Fundamentals of satellite remote sensing: An environmental approach*. CRC Press.
3. Colwell, R. N., Esters, I. C., & Thorley, G. A. (Eds.). (1983). *Manual of remote sensing, volume 2: Interpretation and application*. American Society of Photogrammetry.
4. Heywood, I., Cornelius, S., & Carver, S. (2006). *An introduction to geographical information systems* (3rd ed.). Pearson Education Limited.
5. Jensen, J. R. (2007). *Remote sensing of the environment: An earth resource perspective* (2nd ed.). Pearson Prentice Hall.
6. Köhl, M., Magnussen, S., &Marchetti, M. (2006). *Sampling methods, remote sensing, and GIS multiresource forest inventory*. Springer.
7. Lillesand, T., Kiefer, R. W., &Chipman, J. W. (2008). *Remote sensing and image interpretation* (6th ed.). John Wiley & Sons.
8. Paine, D. P. (1981). *Aerial photogrammetry and image interpretation for resource management*. John Wiley & Sons.
9. Shellito, B. A. (2012). *Introduction to geospatial technologies*. W.H. Freeman and Company.
10. Wing, M. G., &Bettinger, P. (2008). *Geographic information systems: Applications in natural resource management.* Oxford University Press.
11. Canada Centre for Remote Sensing. (2010). *Fundamentals of remote sensing*. Natural Resources Canada.
12. NASA Earth Observatory. (2013). *How to interpret a satellite image*: Five tips and strategies. NASA.

Course No: FES 05323222	Credit: 1.0	Year: Third	Semester: Second
Remote Sensing and Geographic Information System Laboratory			Course Status: Practical

Course Rationale:
This course will provide practical exposure of remote sensing technology to students as well as provide an adequate knowledge of the basic concepts of Remote Sensing and Geographic Information System in the contest of Forest Land-Use Change and environmental engineering.

Course Objectives:
The objectives of this course are

- Analyze land use change over time and its impact on the surrounding environment.
- Make well informed decisions in solving environmental problems utilizing combined technology of Remote Sensing and Geographic Information System.
- Understand current research, technology and policy developments in the Remote Sensing and Geographic Information System and their potential applications to environmental and sustainability issues.

Course Content:

1. **Introduction to Tools**: Overview of RS and GIS software; raster and vector data structure.
2. **Image Interpretation**: Visual interpretation of remote sensing image and land cover features; use of false color composites.
3. **Image Preprocessing**: Geo-referencing, band combination, and basic enhancement.
4. **Image Classification**: Supervised and unsupervised classification; accuracy assessment.
5. **GPS Mapping**: Waypoint collection and field data handling.
6. **GIS Operations**: Layer creation, digitization, buffering, overlay, and spatial queries.
7. **Map Layout and Output**: Thematic map generation with legends, scale, and title.
8. **Application Exercise**: Small project using RS-GIS-GPS for land use or vegetation mapping.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Interpret local and regional GIS data sets as well as satellite data (Landsat MSS, TM, ETM, SPOT, ASTER, and MODIS).
CO2 Analyze vegetation mapping and change analysis, land cover/land use change, predict flood risk, erosion risk, interpret wetland monitoring, and evaluate coastal vulnerability.
CO3 Acquire skills to use hardware and software for Remote Sensing and Geographic Information System tools, techniques and modeling.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	2		3			3	2			
CO2	3		3			2	2		1	
CO2	3		3			3	3		1	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Liu, J. G., & Mason, P. J. (2016). *Image processing and GIS for remote sensing: Techniques and applications*. Wiley-Blackwell.
2. Richards, J. A. (2013). *Remote sensing digital image analysis: An introduction*. Springer.
3. Chanda, B., &Majumder, D. D. (2011). *Digital image processing and analysis*. Prentice Hall India Learning Private Limited.

Course No: FES 07223221	Credit: 2.0	Year: Third	Semester: Second
Wood Seasoning and Wood Preservation			Course Status: Theory

Course Rationale:
Increasing the durability of wood and wood products is one of the important ways to reduce the demand of timber leading to deforestation. This course is designed for undergraduate level to provide a basic understanding on principles and practices of wood seasoning and wood preservation to increase the durability of wood and wood products in services.

Course objectives:
The objectives of the course are to-

- Help students understand the basic theories of natural durability, bio-deterioration, seasoning and preservation of wood.
- Provide knowledge on the mechanisms of wood drying and wood preservation.
- Enhance student's skills on different methods of wood seasoning and preservation for better utilization of wood in various end uses.
- Facilitate students in understanding the economic analysis of different treatment methods to increase the service life of wood and wood products.

Course content:

1. **Introduction**: Wood anatomical and physical characteristics related to wood seasoning and preservation; natural durability of wood; hygroscopicity and biodegradability; factors affecting biodegradation of wood; service life of wood in a tropical environment.
2. **Wood seasoning**: General principles of wood seasoning; objectives of seasoning; wood drying mechanism: mechanisms of moisture movement in wood, moving forces and drying pathways; factors of wood drying rate: temperature, relative humidity, air circulation, species, initial moisture content, grain direction and thickness of timber; Methods of seasoning: air seasoning, kiln seasoning and solar seasoning; principles of air seasoning; factors affecting air seasoning; necessity of timber stacking; staking methods for planks, poles and railway sleepers; basic principles of kiln seasoning; types of seasoning kiln; Kiln schedule: types and deciding factors; principles and mechanism of solar seasoning; advantages and disadvantages of each seasoning method; seasoning defects: types, causes and prevention.
3. **Wood preservation:** Definition, objectives and principles of wood preservation; general characteristics of wood preservatives; types of preservatives: oil borne, water borne and organic solvent type; water repellents and fire retardants; retention: definition and calculation; factors influencing the effectiveness of wood preservatives; treating wood before preservative treatment, Non-pressure methods of wood preservation; sap displacement methods; factors affecting penetration and absorption; pressure methods: full cell, empty cell, oscillating pressure and vacuum processes; efficiency of wood preservation and their impacts on wood properties (vice-versa); economics of wood preservation; environmental and human health hazards; precaution in application of preservatives; .

Course Learning Outcomes (COs) (3-5):
After successful completion of the course, students will be able to

CO1 Understand the mechanisms of wood bio-deterioration, wood seasoning and wood preservation process.

CO2 Develop and modify different wood seasoning techniques and their applications

CO3 Select various types of wood preservatives and assess their suitability and efficacy
CO4 Determine service life of treated wood and wood-based products along with economic and environmental benefits.
CO5 Design cost-effective and environment friendly wood preservation plants.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3					2	2		1	
CO3	3									
CO4	3		2					2		
CO4	3					2	3			

3: Strong **2: Moderate** **1: Weak**

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Pandey, C. N., & Jain, V. K. (1992). *Wood seasoning technology*. Indian Council of Forestry Research and Education.
2. Milton, F. T. (1995). *Wood preserving resource conservation and recovery act compliance guide: A guide to federal environmental regulation*. U.S. Environmental Protection Agency.
3. Eaton, R. A., & Hale, M. D. C. (1993). *Wood: Decay, pests & protection*. Chapman & Hall.
4. Findlay, G. W. (Ed.). (1986). *Preservation of timber in the tropics*. Kluwer Academic Publishers.
5. Desch, H. E. (1973). *Timber: Its structure and properties* (5th ed.). McMillan Publisher.
6. Brown, H. P. (1980). *Textbook of wood technology* (Vol. 1). McGraw-Hill Book Company.
7. *Indian forest utilization* (Vol. 1). (n.d.). Forest Research Institute.

Course No: FES 07223222	**Credit: 1.0**	**Year: Third**	**Semester: Second**
Wood Seasoning and Wood Preservation Laboratory			**Course Status: Practical**

Course Rationale:
Proper application of the knowledge acquired in the theory course of Wood seasoning and Wood Preservation requires practical training on the relevant methods and techniques. This course is aimed at students of undergraduate level to provide practical knowledge on how to apply various wood seasoning and preservation methods.

Course objectives:

The objectives of the course are

1. Make the students understand the seasoning behavior of wood in the field and laboratory condition
2. Help the students achieve practical knowledge on traditional and widely used seasoning methods
3. Provide knowledge on the performance and suitability of various types of wood preservatives and preservation methods

Course Content:

1. Collection of wood samples of different species and from different sources.
2. Determination of wood density, moisture content, shrinkage and swelling of wood
3. Compare the performance of air seasoning and kiln seasoning methods
4. Identification of different kinds of rots, stains and damage caused by different deteriorating agents.
5. Analysis of decayed wood for physical and chemical properties.
6. Conducting experiments to test how preservation methods, types of preservatives, species and sawing patterns affect the efficacy of wood preservation.
7. Visit to different industries to have practical experiences on seasoning and preservation methods

Course Learning Outcomes (COs) (3-5):

After successful completion of the course, learners will be able to

CO1 Quantify and predict the magnitude of dimensional changes of wood due to seasoning.

CO2 Select the appropriate seasoning method for a timber species in a particular environmental condition.

CO3 Control the drying processes to minimize seasoning defects.

CO4 Examine wood decaying processes and types of wood decay along with their impact on the physical and chemical properties of wood.

CO5 Choose appropriate preservatives and preservation methods for preservation treatment considering both the economic feasibility and service life of wood and wood products.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2				
CO2	2					2				
CO3	3					2				
CO4	1					1				
CO5	2					2				

3: Strong **2: Moderate** **1: Weak**

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO5	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Brown, W. H., & Brown, W. (1989). *The conversion and seasoning of wood: A guide to principles and practice.* Paperback.
2. Hunt, G. M., & Garratt, G. A. (1953). *Wood preservation* (3rd ed.). McGraw-Hill.
3. Thompson, R. (1991). *The chemistry of wood preservation* (1st ed.). Woodhead Publishing.
4. Thomasson, G., Capizzi, J., Dost, F., Morrell, J., & Miller, D. (1989). *Wood preservation and wood products treatment training manual*. Oregon State University Press.

Course No: FES 05223223	Credit: 2.0	Year: Third	Semester: Second
Water Resources, Watershed and Wetland Management			Course Status: Theory

Course Rationale:
This course is designed to present a comprehensive approach to water resources management including watersheds, wetlands, streams, lakes, groundwater, and coastal waters by integrating forestry and environment science (geology, soils, and hydrology) knowledge. In addition, students will gain a core understanding of the practical applications of watershed planning and management as a tool to manage land, water, and ecosystem resources.

Course Objectives:
The objectives of the course are-

- To provide an insight of watershed, wetland, and water resources of Bangladesh.
- To help the students to develop ability in addressing policy implications of watershed planning and management at the international, national, and local levels and what has been most effective in managing water.
- To develop student's skills on raising awareness among the students about the resource and biodiversity of wetlands.
- To enhance the skill on the prospect to work as a part of an interdisciplinary team.

Course Content:
Watershed Management

1. **Introduction:** Definition, concept, objectives of watershed management; importance of watershed management in Bangladesh.
2. **Hydrological Cycle:** Hydrological cycle of forests; precipitation, interception, surface flow, infiltration, percolation, ground water, stream flow.
3. **Erosion:** Geologic and accelerated erosion; agents, types and major causes of erosion; forms of water erosion; basic concepts of estimating soil loss; overview of the Universal Soil Loss Equation (USLE); erosivity and erodibility; classification of gullies; wind erosion and its control.
4. **Soil Conservation:** Aim of soil conservation; mechanical, vegetative and agronomic measures appropriate to watershed areas; land management-based erosion control practices; processes and impacts of desertification**.**
5. **Planning for Watershed Management:** Need for planning and procedure of watershed management; data requirements for integrated planning.
6. **Integrated Watershed Management:** Watershed classification; institutional and organizational coordination in watershed initiatives; background and approach; integrated management strategies.

Water Resources

1. **Basic Concepts:** National Water Policy of Bangladesh (1998); major river systems of Bangladesh; land classification according to water depth; interrelation among water resources, forestry and fisheries.
2. **Criteria of Water and its Supply System:** Types and quality of water, impurities; general overview of water supply systems (sources, collection, estimation, distribution); basic types of water infrastructure such as pipes and pumps; emphasis on sustainable use of surface and groundwater.
3. **Water Management Principles:** IWRM concepts and principles; water management and sustainable development; basin-wide approaches and water-sharing practices; water rights and conflicts; sectoral water demands and allocation; stakeholder participation.

Wetland Management

1. **Introduction:** Definition, importance and scope of wetland management in forestry.
2. **Major Wetlands in Bangladesh:** Sundarbans, HakalukiHaor, ChalanBeel, BeelDakatia.
3. **Biodiversity and Management:** Ecological and biodiversity importance of wetlands; wetland productivity; current challenges in wetland conservation.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Describe watershed, its importance and need for management
CO2 Demonstrate a strong background in the hydrological cycle and processes
CO3 Evaluate the causes and consequences of soil degradation problems along with their conservation
CO4 Analyze the consequence of water and water related issues in a global, economic, environmental, and societal context.
CO5 Justify an individual wetland's contribution to the aquatic system within its watershed and apply these concepts to wetland conservation, and restoration

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3			2						
CO2	3					2				
CO3	3			3		3				
CO4	3			3		3		2		2
CO5	3			2		3	2	2	3	2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Samantaray, S., Sahoo, A., &Ghose, D. K. (2021). *Watershed management and applications of AI*. CRC Press.
2. Chen, S.-C. (Ed.). (2022). *Soil-water conservation, erosion, and landslide*. MDPI.
3. ASEAN-US Watershed Project. (1988). *Abstracts of watershed management research and related studies in the ASEAN region* (Vol. 1). ASEAN-US Watershed Project.
4. Beattie, B. B. (1979). *Watershed conditions and watershed research needs, Chittagong Hill Tracts* (Project Report No. 2). UNDP/FAO/72/005, Forest Research Institute.
5. Hudson, N. (1971). *Soil conservation*. B. T. Batsford.
6. Randhir, T. O. (2006). *Watershed management: Issues and approaches*. IWA Publishing.
7. Heathcote, I. W. (2009). *Integrated watershed management: Principles and practices* (2nd ed.). Wiley.
8. Morgan, R. P. C. (2004). *Soil erosion and conservation* (3rd ed.). Wiley.
9. Lal, R. (2019). *Integrated watershed management in the global ecosystem*. CRC Press.
10. Lenton, R., & Muller, M. (2012). *Integrated water resources management in practice: Better water management for development*. Routledge.
11. Borchardt, D., Bogardi, J. J., &Ibisch, R. B. (Eds.). (2016). *Integrated water resources management: Concept, research and implementation.* Springer.

Course No: FES 08213221	**Credit: 3.0**	**Year: Third**	**Semester: Second**
Forest Harvesting and Transportation Engineering			**Course Status: Theory**

Course Rationale:
Forest harvesting and transportation engineering is a critical component of sustainable forest management, balancing economic viability with environmental conservation. This course provides students with comprehensive knowledge of timber harvesting systems, road and bridge engineering, log extraction methods, and transportation logistics. It emphasizes Reduced Impact Logging (RIL) techniques to minimize ecological damage while maintaining operational efficiency. Students will learn to design harvest plans, assess economic

feasibility, and implement safety and ergonomic best practices, preparing them for real-world challenges in harvesting operations.

Course Objectives:
The objectives of the course are to-

- Understand the principles, components and functions of forest harvesting systems.
- Develop an operational timber harvesting plan considering safety, silvicultural needs, and environmental impacts.
- Provide ideas on designing forest roads, bridges and other constructions with proper alignment, stability, and cost-efficiency.
- Provide knowledge on various techniques of felling, bucking, extraction, transportation and storage of log.
- Enhance the skill of conducting basic economic analyses on timber harvesting operations.
- Develop student's skills to implement Reduced Impact Logging (RIL) strategies to promote sustainable forest management.

Course Content:

1. **Introduction:** Definition of harvesting and logging; Scope of forest harvesting; Efficacy of forest ban on the natural forests of Bangladesh.
2. **Pre-harvesting field preparation:** Tree marking; exclusion areas; location of log landing, demarcation of logging area.
3. **Logging preparation:** Road construction-obligatory points, alignment of plain and hill roads, preliminary survey, location and demarcation, labour supply and preparation of the camp.
4. **Retaining walls and breast walls:** Materials used, forces acting upon them, conditions of stability, rules of thumb for design of walls.
5. **Culverts and Bridges:** Types of forest culverts and bridges, planning and construction; waterways selection for small culverts and bridges, economic spans, thumb rules in designing culverts and bridges.
6. **Felling and conversion operations:** Preparation and methods of felling; equipment required for felling operation, procedure of delimbing, topping and bucking operations and equipment required in the operations.
7. **Log extraction:** Methods of log extraction – skidding and yarding; log landing – its location and characteristics; skid trail patterns; traditional, semi-mechanized and mechanized system of skidding, assortment, tree length and whole tree method of skidding; methods of cable yarding systems and their merits and demerits.
8. **Log transportation:** Types of land, water and aerial transportation of logs; traditional and mechanized system of land and water transportation; types of loading and unloading of logs at landing and mill sites.
9. **Road and equipment maintenance:** General wear of roads, surface maintenance, improving existing roads – renewal of surface, road waterbed and polo drains, road signs; importance of equipment maintenance; Felling, bucking and extraction equipment; maintenance of petrol/diesel transports.
10. **Log storage and depot:** Dry and wet storage of logs with their merits and demerits, types of depot, maintenance of depot records, different timber forms and registers used in Forest Department, stock taking at depot, methods of stacking, measures for protection of timber in depot.
11. **Logging plans and costs:** Preparation of logging plan, procedures of production and cost calculation of logging operation, production planning in logging, production standard.
12. **Safety and ergonomics:** Safety requirements in logging operations, protective dress, ergonomic problems in harvesting activities.
13. **Reduced Impact Logging (RIL):** Definition; context of standard logging practice; objectives of RIL; what RIL includes; pre-requisite for effective RIL planning; challenges and opportunities of RIL; sustainable forest management and RIL.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Explain forest harvesting systems, their components, and environmental considerations.
CO2 Plan timber harvest operations, including log landing, forest roads, bridges, and culverts design and construction using engineering principles and cost-effective methods.
CO3 Apply technical skills in felling, bucking, extraction, transportation and log storage.
CO4 Perform basic economic analysis for evaluating designs for a harvesting operation based on equipment and timber characteristics
CO5 Evaluate the concept of safety, ergonomics and RIL principles in harvesting operations.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3							1		
CO3	2						2			
CO4	2									
CO5	3									1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Brown, N. (1949). *Logging*. John Wiley & Sons.
2. Conway, S. (1982). *Timber cutting practices* (3rd ed.). Miller Freeman Publishers.
3. Heinrich, R. (1982). *Logging in mountain forests* (FAO Forestry Paper 33). Food and Agriculture Organization.
4. Staff, A., & Wiksten, N. A. (2010). *Tree harvesting techniques*. Springer. (Reprint of 1984 ed.)
5. Sarker, S. K., Deb, J. C., & Halim, M. A. (2011). *A diagnosis of existing logging bans in Bangladesh*. International Forestry Review, 13(1), 56–63.
6. Castaneda, F. (n.d.). *Forest harvest and transportation*.
7. Schiess, P., & Krogstad, F. (n.d.). *Forest harvest and transportation*. University of Washington.
8. Barry, P., & Wychreschuk, A. L. (2011). *Forestry transport: Requirements, current practices and opportunities for more sustainable harvesting*. The 58th CASI Aeronautics Conference.
9. Castaneda, F. (1989). *Forest transportation engineering*. UNDP/FAO/BGD/85/011, Institute of Forestry, Chittagong University.
10. Castaneda, F. (1991). *Maintenance of forest engineering equipment*. UNDP/FAO/BGD/85/011, Institute of Forestry, Chittagong University.
11. Food and Agriculture Organization. (1977). *Planning forest roads and harvesting systems* (FAO Forestry Paper No. 2). FAO.
12. Mobley, K., & Higgins, L. (2008). *Maintenance engineering handbook* (7th ed.). McGraw-Hill Professional

Course No: FES 08213221	**Credit: 3.0**	**Year: Third**	**Semester: Second**
Mangrove Forestry			**Course Status: Theory**

Course Rationale:

Mangrove forests dominate the world's tropical and subtropical coasts. Mangroves are important, not only for sustaining biodiversity but also for their ecosystem services. Despite their ecological and economic significance, mangroves are vanishing rapidly. Throughout the tropics, they are threatened directly and indirectly by

commercial development. Increased industrial development as well as destruction of inland tropical forests for agricultural development indirectly impact coastal and island mangrove swamps through water pollution from industrial effluent, oil spills, sewer discharge, and agricultural and sediment runoff. This course will provide an adequate knowledge of the ecology, biodiversity, silviculture and management of mangroves.

Course Objectives:
The objectives of the course are

- To help the student understand the biological complexity of mangrove ecosystems.
- To provide knowledge on the ecological roles, economic importance, and cultural significance of mangrove ecosystems.
- To help the students recognize the structural and functional attributes of the Sundarbans mangrove forest which will help to deliver appropriate management as well as conservation initiatives therein.

Course Content:

1. **Introduction:** Origin of mangroves. Mangrove floristics. Mangrove biodiversity and biogeography. Ecosystem services and global conservation status of mangroves. Mangrove forests' distribution in Bangladesh.
2. **Coastal environment and plantations:** Coastal area of Bangladesh; delta formation, delta morphology, delta and river dynamics. Objectives of Coastal Plantation, Species Composition in the Coastal Afforestation Programs in Bangladesh.
3. **Mangrove Adaptations:** Morphological, Physiological and Anatomical adaptations in mangroves.
4. **Physio-chemical and Climatic Factors and Mangrove Performance:** Nutrition of Mangroves, Impacts of climate change on mangrove ecosystems. Impacts of Sediment Burial on Mangroves. Impacts of environmental and climatic factors on mangrove growth and development.
5. **Biotic Interactions, Human disturbance, and Mangrove Performance:** Plant-plant interactions. Plant-animal interactions. Mangrove-Human interactions.
6. **Mangrove Structure and Classification:** Succession and Retrogression, Classification of mangrove forests. Zonation in mangroves.
7. **The Sundarbans - Mangrove Ecology and Biodiversity:** Mangroves Habitat Preference, Spatio-temporal Changes in Sundarbans' Biodiversity, Factors Regulating Sundarbans' Biodiversity, Spatial Variations in Functional Traits and Biomass Productivity in the Sundarbans, Management History.
8. **The Sundarbans - Mangrove Resource Management and Conservation:** Management History, Forest Fire, Top-dying of Sundri, Raised land management, biological invasion, oil spill control, NTFPs Management, Wildlife Conservation, Aquatic Resource Management.
9. **Mangrove Restoration:** Concepts of restoration, Steps of mangrove restoration, Community-based Mangrove Ecosystem Restoration.
10. **Bangladesh Sundarban delta vision 2050 and Bangladesh Delta Plan (BDP), 2100.**

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Define mangrove flora, mangrove biodiversity and ecosystem services
CO2 Understand the origin of mangroves, global conservation status of mangroves, species composition in natural (The Sundarbans, Bangladesh) and planted mangrove forests, mangrove adaptations under changing environment
CO3 Demonstrate and predict the influences of abiotic, biotic and climatic factors on mangrove structure, biodiversity, functions and productivity
CO4 Analyze succession and retrogression, classify mangrove forests
CO5 Design mangrove restoration plans based on the knowledge gained on mangrove

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10
CO1	3									
CO2	3					3				
CO3	2					2				
CO4	2					2				
CO5	2			2		3		2	2	1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Bangladesh Forest Department. (2010). *Integrated resources management plans for the Sundarbans: 2010 - 2020.* Dhaka.
2. Global Mangrove Alliance. (2021). *The state of the world's mangroves 2021*.
3. Hossain, M., Saha, S., Saha, C., Faisal, A. M., &Hoque, M. M. (2016). *Nursery raising, planting techniques, and plantation management of climate resilient mangrove species of Bangladesh*. In Integrating community-based adaptation into afforestation and reforestation (ICBAAR) programs in Bangladesh. Bangladesh Forest Department Ministry of Environment, Forest and Climate Change.
4. Mahmood, H. (2015). H*andbook of selected plant species of the Sundarbans and the embankment ecosystem.* Sustainable Development and Biodiversity Conservation in Coastal Protection Forests, Bangladesh (SDBC-Sundarbans) Project implemented by GIZ, Dhaka.
5. Reef, R., Feller, I. C., & Lovelock, C. E. (2010). *Nutrition of mangroves*. Tree Physiology, 30(11), 1148–1160.
6. Sarker, S. K., Matthiopoulos, J., Mitchell, S. N., Ahmed, Z. U., Al Mamun, M. B., & Reeve, R. (2019a). 1980s–2010s: *The world's largest mangrove ecosystem is becoming homogenous*. Biological Conservation, 236, 79–91.
7. Sarker, S. K., Reeve, R., &Matthiopoulos, J. (2021). *Solving the fourth-corner problem: Forecasting ecosystem primary production from spatial multispecies trait-based models*. Ecological Monographs, 91(eCM.1454).
8. Sarker, S. K., Reeve, R., Paul, N. K., &Matthiopoulos, J. (2019b). *Modelling spatial biodiversity in the world's largest mangrove ecosystem—the Bangladesh Sundarbans: A baseline for conservation*. Diversity and Distributions, 25, 1–14.
9. Sarker, S. K., Reeve, R., Thompson, J., Paul, N. K., &Matthiopoulos, J. (2016). *Are we failing to protect threatened mangroves in the Sundarbans world heritage ecosystem?*Scientific Reports, 6, 21234.
10. Saenger, P. (2002). *Mangrove ecology, silviculture, and conservation*. Springer Netherlands.
11. Siddiqi, N. A. (2001). *Mangrove forestry in Bangladesh*. Institute of Forestry and Environmental Sciences, University of Chittagong, Chittagong, Bangladesh.
12. Tomlinson, P. B. (1986). *The botany of mangroves*. Cambridge University Press.

Course No: FES 08213222	**Credit: 1.0**	**Year: Third**	**Semester: Second**
Fieldwork in Forests (Mangrove Forest)			**Course Status: Practical**

Course Rationale:
This course will allow a student to gain hands on experience of the mangrove and coastal forest ecosystem and forestry sector situation in Bangladesh.

Course Objectives:
The objectives of the course are

- To make the students understand the mangrove and coastal forest diversity, their trends, applications, roles in environmental stability.

- To facilitate the students develop skills for practicing wood seasoning and preservation, tea, coffee and rubber, protected area and wildlife management, social forestry activities, forest engineering and harvesting;
- To familiarize the students with the professional roles in forest and forestry sectors.
- To help the students apply the acquired knowledge and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Diversity of mangrove and coastal forests in Bangladesh and its components.
2. Wood seasoning and preservation, tea, coffee and rubber, protected area and wildlife management, social forestry activities, forest engineering and harvesting.
3. The overall situation of different forest, forestry and environmental sector related areas of Bangladesh
4. Nursery and Plantation management (Hardening; root pruning; performing tending operation both in nursery and plantation; integrated pest management; Tree planting; nursing; reporting etc.)
5. Practical application of all core courses of the current semester will be demonstrated.

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Evaluate diversity mangrove and coastal forest, its trends, applications, role in environmental stability.

CO2 Generalize knowledge about wood seasoning and preservation, tea, coffee and rubber, protected area and wildlife management, social forestry activities, forest engineering and harvesting

CO3 Relate the professional roles in forest and forestry sectors

CO4 Criticize the relevant field level management issues to meet the needs of the country.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3						3		3	
CO2	3	3								
CO3	3						3		3	
CO4	3						3		3	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL06	CA03, CA04, CA05, SA02
CO2	TL04, TL06	CA03, CA04, CA05, SA02
CO3	TL04, TL06	CA03, CA04, CA05, SA02
CO4	TL04, TL06	CA03, CA04, CA05, SA02
CO5	TL04, TL06	CA03, CA04, CA05, SA02

Recommended References:

1. Documents provided by the related visited areas' official/s.
2. Documents provided by the related course teacher/s.

Note: In case of unavoidable situation, this course will be exchanged with course no. **FES 08213126**

Course No: FES 08213223	**Credit: 3.0**	**Year: Third**	**Semester: Second**

Social Forestry and Forestry Extension	Course Status: Theory

Course Rationale:
This course aims to provide the basic concepts and attributes of community engagement in the forestry sector and the means and approaches of disseminating knowledge and practices using extension approaches at the doorstep of the farmers and forestry practitioners. Social forestry principles and practices of Bangladesh and neighboring countries and the concepts and role of extension in popularizing social forestry is another rationale for offering the course.

Course Objectives:
The objectives of the course are to-

- Make the students understand the basic concepts, definitions and features of community-based forestry.
- Give an overview about the history and development of social forestry, global and Bangladesh perspectives.
- Make the students understand multiple benefits of social forestry by reviewing best practices of social forestry of Bangladesh and neighboring countries with special reference to evaluate the impacts of social forestry on forest conservation, livelihoods, and rural development.
- Help the students accumulate ideas about the concepts, goals and principles of forestry extension.
- Acquaint the students with the basic tools and techniques of community organization for forestry and natural resources extension, evaluation and monitoring.

Course Content:
Social Forestry

1. **Introduction**: Overview of forest resources and land use in Bangladesh; basic features of social forestry; historical development of social forestry globally and in Bangladesh; social forestry as a resource system; social forestry as a development program; distinguishing characteristics of social forestry from other forestry development alternatives; key objectives and goals of social forestry (e.g., poverty alleviation, environmental conservation, community empowerment).
2. **Social Forestry and Community Participation**: The role of local communities in social forestry programs; models of community participation: decision-making, benefit-sharing, and management rights; gender, equity, and social inclusion in social forestry.
3. **Socio-Economic Impacts of Social Forestry in Bangladesh**: Economic benefits—livelihoods, employment, and income generation through forestry activities; social benefits—empowerment, education, and health improvements in rural areas; environmental benefits—conservation of biodiversity, watershed protection, and carbon sequestration.
4. **Tools of Appraising Social Forestry**: Rural appraisal techniques and their basic features: D&D, RRA, PRA, SIA, SWOT and Project Management Framework.
5. **Social Forestry Practices in Bangladesh and Other Countries**: Comparing social forestry practices in various Asian and African countries; comparative analysis of social forestry policies globally.
6. **Contemporary Issues Affecting Social Forestry of Bangladesh**: Sustainability: concept, attributes and importance in development; local institutions: nature, types and their roles; legal framework, land tenure, gender perspective, conflict and its resolution.
7. **Monitoring, Evaluation and Impact Assessment in Social Forestry** : Methods and tools for monitoring and evaluating social forestry projects; impact assessment frameworks: socio-economic, environmental, and cultural dimensions; indicators of success and failure in social forestry programs; community-based monitoring and participatory evaluation approaches.

Forestry Extension

1. **Introduction**: Basic concepts and features of extension and community engagement; forestry extension goals and objectives; forestry extension strategies, programs and methods; forestry extension perspectives, role of forestry extension in forestry development.
2. **Principles and Context of Forestry Extension**: Motivation, learning, diffusion and adoption of innovation, group processes and dynamics, communication; elements/components of forestry extension: forestry extension agencies and individuals, their roles and challenges; stakeholders of forestry extension.
3. **Forestry Extension Program Development**: Various types of extension programs, their strengths and limitations; blueprint vs. the learning approach to program development; program development cycle.
4. **Forest Extension Programs in Bangladesh and Other Asian Countries**: Current status, challenges and future development clues.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand the basic concepts, attributes and philosophy of social forestry, principles and impacts of social forestry on the environment and society.
CO2 Employ the principles and applications of various appraising tools used in social forestry.
CO3 Determine the basic understanding of theoretical concept, approaches and practices of forestry extension and realize the contemporary development in social forestry and forestry extension of Bangladesh as well as other countries.
CO4 Combine the role of social forestry and forestry extension in community development.
CO5 Categorize the tools and techniques of extension to ensure effective community engagement.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3			2						
CO2						2	3			
CO3	3			3	1					
CO4		2						1	3	2
CO5		1				1	3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Rao, Y. S., et al. (1985). *Community forestry: Socio-economic aspects*. FAO Regional Office for Asia and the Pacific.
2. Rebugio, L. L. (Ed.). (1990). *Readings on rural development: Social forestry and forestry extension* (Vol. 3, Social forestry). UNDP/FAO/85/011, Field Document No. 23, IFCU.
3. Khan, N. A., Chowdhury, J. K., Huda, K. S., &Mondal, M. I. (2004). *An overview of social forestry in Bangladesh*. Bangladesh Forest Department & University of Chittagong.
4. Khan, N. A., Millat-e-Mustafa, M., Ahmed, F. U., & Hossain, M. K. (2005). *An annotated bibliographic review of community forestry research in Bangladesh*. University of Chittagong.
5. Khan, N. A. (2008). *Society and social forestry: Patronage and land tenure in Bangladesh and Thailand.* University of Chittagong Press.
6. Lacuna-Richman C. (2012). Growing from Seed: An introduction to Social Forestry, World Forest Series, Vol.11, Springer Nature.
7. Vaarde T. H.(2023). Social Forestry: Tending the Land as People of Place, Synergetic Press.

Course No: FES 08213224	Credit: 2.0	Year: Third	Semester: Second
Forest Development Planning			Course Status: Theory

Course Rationale:
This course will provide an adequate knowledge of the basic concepts of development planning to help the forestry sector to integrate the principles of sustainable development to the country's policies and programs, and to reverse the loss of environmental resources.

Course Objectives:
The objectives of this course are to-

- Enhance the skill of the basic concepts of economic development.
- Apply the knowledge of principles on development planning.
- To understand basic knowledge of MDGs and SDGs.
- Enhance the skill of sector analysis and forecasting.
- Apply the knowledge of project evaluation.

Course Content:

1. **Introduction**: Economic Growth and Economic Development**;** Vicious Cycle of Poverty**;** Drivers of Slow Economic Development in Bangladesh; Role of Forestry in Economic Development.
2. **Principles of Development Planning:** Plan and Planning; characteristics of a Good Plan; Types of Plan; Forest Development Planning; Need for Forest Development Planning in Bangladesh; Steps in Planning; Forest Planning in Bangladesh.
3. **MDGs and SDGs:** Millennium Development Goals; Targets with Indicators for Ensuring Environmental Sustainability in Bangladesh; Sustainable Development and Goals; Forestry Sector Policies in Bangladesh.
4. **Sector Analysis and Forecasting:** Sector Analysis. Purposes of Sector Analysis; Steps in Sector Analysis; Constraints in Sector Analysis; Forestry Sector Analysis. Forecasting; Significance; Econometric Analysis for forecasting of forest product market.
5. **Project and Project Evaluation:** Project; Characteristics of Project; Project Cycle; Aspects of Project; Project Evaluation; Methods of Project Evaluation; Risk and Uncertainty; SWOT Analysis; Logical Framework Approach (LFA); Social Benefit/Cost Analysis; Limitations of Social Benefit/Cost Analysis; Externalities of the forestry sector not covered in the analysis; Preparation of a Project Concept Paper (PCP); Development Project Proforma/ Proposal (DPP) preparation.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Interpret the basic concepts of economic development
CO2 Apply the knowledge of principles of development planning
CO3 Summarize the knowledge of MDGs and SDGs
CO4 Relate the basic concepts of sector analysis and forecasting
CO5 Design the knowledge of project evaluation

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	1	1	1		1	1			
CO2	3	1	1			1	2			
CO3	3	1	1	1		1	2			
CO4	3	1	1			1	2			
CO5	3	1	1			1	3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Alam, M. (2009). *Evolution of forest policies in Bangladesh: A critical analysis*. International Journal of Social Forestry (IJSF), 2(2), 149-166.
2. Bhattacharya, D. (2015). *The agenda of sustainable development goals: Implementation challenges for Bangladesh.* The Centre for Policy Dialogue, Bangladesh.
3. Bangladesh Institute of International Strategic Studies (BIISS). (2015*). Moving from MDGs to SDGs – Bangladesh experience and expectation.* Proceedings of the special seminar, September 17, 2015. Bangladesh Institute of International Strategic Studies, Dhaka, Bangladesh.
4. Buongiorno, J., &Gilless, J. K. (2003). *Decision making methods for forest resource management*. Academic Press, Elsevier Science.
5. Hashim, T., Osman, R., &Badioze-Zaman, F. S. (2016). *Poverty challenges in education context: A case study of transformation of the mindset of a non-governmental organization.*International Journal of Advanced and Applied Sciences, 3(11), 40-46.
6. Pant, M. M. (1984). *Forest economics and valuation*. Medhawi Publishers.
7. Pant, M. M. (1990). *Planning forestry development in Bangladesh* (UNDP/FAO/BGD/85/011, Field Document No. 15).

Course No: AMS 05423202	**Credit: 2.0**	**Year: Third**	**Semester: Second**
Biostatistics Laboratory			**Course Status: Practical**

Course Rationale

Biostatistics in Forestry provides the essential framework to measure, analyze, and manage forest resources sustainably. It enables accurate inventory sampling, ecological modeling, and data-driven decisions on conservation, wood production, and ecosystem health. This laboratory-based course provides the inferential statistical framework needed to convert field observations into testable hypotheses and scientifically valid conclusions. Through hands-on use of **R statistical software**, students will move beyond basic data description to practical experience in data analysis, experimental design, hypothesis testing, and predictive modeling. The course bridges the gap between raw forest and environmental data and statistical interpretation, equipping students with essential computational and analytical skills for modern forest management and environmental research.

Course Objectives

- Develop foundational skills in descriptive and exploratory data analysis for summarizing and visualizing forestry and environmental data.
- Establish inferential statistical competence for parameter estimation, hypothesis testing, and decision-making under uncertainty.
- Design and apply appropriate sampling strategies and experimental layouts for reliable field data collection.
- Build proficiency in R for data management, statistical computation, and interpretation of forestry and environmental datasets.

- Analyze ecological and forest system relationships using regression, correlation, and multivariate statistical techniques.
- Evaluate and interpret predictive models, including linear and logistic regression, for forecasting forest and environmental outcomes.

Course Content

1. Biostatistics basics and data visualization: Scope and foundations of biostatistics, use and misuse of statistics in forestry and environmental research, population vs. sample, types of biological data, explanatory and response variables, standard deviation vs. standard error, biological variation vs. sampling precision, exploratory data analysis (EDA): summary statistics, boxplots, histograms, and basic data visualization for field data understanding.

2. Data Acquisition: Sampling & Experimental Design: Sampling process, probabilistic and non-probabilistic sampling methods, sample size determination, law of large numbers, sampling bias in forest inventories, completely randomized design (CRD), randomized block design (RBD), **stratified sampling, cluster sampling**, field experimental design in forestry and environmental studies.

3. Probability & Distribution Theory: Probability concepts and laws, frequentist and Bayesian interpretation, random variables, probability distributions, normal distribution and properties, 68–95–99.7 rule, Central Limit Theorem, sampling distribution concept, Shapiro–Wilk test for normality, R-based implementation.

4. Estimation, Inference & Comparative Tests: Parameter estimation, point and interval estimation, confidence intervals, p-values, degrees of freedom, hypothesis testing framework, z-test (mean and proportion), t-test (one-sample, independent, paired), F-test (variance comparison), ANOVA (one-way), Chi-square test (goodness of fit, independence), Kruskal–Wallis test, Mann–Whitney U test, Wilcoxon signed-rank test, R-based statistical analysis.

5. Modeling Biological Relationships: Covariance, correlation (Pearson's r), simple linear regression, multiple linear regression, regression coefficients estimation, model diagnostics, goodness of fit (R^2), residual analysis, logistic regression (binary outcomes), predictive modeling in forestry and environmental data.

6. Multivariate Statistics for Forest Ecology: Multivariate data structure, multivariate normal distribution, cluster analysis, similarity measures, principal component analysis (PCA), factor analysis, canonical correspondence analysis (CCA), ordination techniques, ecological gradient analysis, species-environment relationship modeling.

Course Learning Outcomes (COs)

Upon successful completion of the course, students will be able to:

COs	Learning Outcome
CO1	Perform exploratory and descriptive analysis of forestry and environmental data, and identify data quality issues including bias and inconsistency.
CO2	Design appropriate sampling strategies and experimental layouts (CRD, RBD, stratified, cluster sampling) for forestry and environmental studies.
CO3	Apply probability concepts, distributions, and normality tests to validate assumptions of biological data and use multivariate statistical techniques to analyze complex ecological relationships and classify forest ecosystems using R.
CO4	Conduct and interpret estimation and hypothesis testing procedures using parametric and non-parametric methods.
CO5	Develop and evaluate regression-based predictive models for forest and environmental outcomes.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3			3				
CO2	3		3			3				
CO3	2		3			3	2			

CO4	3		3			3				
CO5	2		3			3				

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL03, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL03, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL03, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL03, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL03, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Forthofer, R. N., Lee, E. S., & Hernandez, M. (2007). *Biostatistics: A guide to design, analysis, and discovery* (Biostatistics). Elsevier.
2. Gail, M., Krickeberg, K., Samet, J. M., Tsiatis, A., & Wong, W. (2012). *Statistics for biology and health*. Springer.
3. Macfarland, T. W. (2014). *Introduction to data analysis and graphical presentation in biostatistics with R*. Springer.
4. Shahbaba, B. (2012). *Biostatistics with R* (Use R!). Springer.
5. Sokal, R. R., &Rohlf, F. J. (2009). *Introduction to biostatistics ideas. In Clinical and translational science* (pp. 59–68). Elsevier.
6. Venables, W. N., Smith, D. M., & the R Core Team. (2021). *An introduction to R* (Version 4.1.2). https://www.r-project.org/doc/manuals/

Course No: FES 08214121	**Credit: 3.0**	**Year: Fourth**	**Semester: First**
Forest Management			**Course Status: Theory**

Course Rationale:

Numerous aspects need to be considered in forest management; this complexity means that a solid understanding of these aspects is essential. Due to an expanding global demand for forest products it is also necessary to know how to manage forest in a sustainable way. This course describes fundamentals of forest structure, growth and development and also examines the challenges and opportunities for sustainable forest management in both natural and planted forests.

Course Objectives:

The objectives of this course are

- To facilitate necessary knowledge about the scope and complexity of forest management
- To facilitate knowledge about forest structure, growth and development of forest stands
- To get an idea about what sustainable forest management and give examples of sustainable forest practices around the world and how these examples demonstrate sustainability
- To accumulate basic ideas about forest management planning in Bangladesh.

Course Content:

1. Concepts, Philosophy and Ethics of Forest Management
2. Evolution of Forest Management in Bangladesh and Worldwide (collaborative management, community forestry and co-management).
3. Basic elements and methods of Forest Management
4. Fundamentals of Structure, Growth and Development in Stands and Forests
5. Sustained yield concept, definition; forest organization (managerial silviculture); Rotation, Concept of Normal Forest, Growing Stock and Increment, Yield Regulation, stumpage appraisal.
6. Sustainable Forest Management: Concepts and Techniques, Multiple use forestry
7. Forest Management Planning

8. Forest Certification
9. Ecological Forest Management: Ecosystem Management Approaches, Strategies for reducing habitat loss and fragmentation, Forest Landscape Restoration (FLR), Concept and Implementation techniques.
10. Forest Management Practices in the Hill forests, Sal Forests, and Mangrove Forests in Bangladesh.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Identify the basic elements and methods of forest management
CO2 Summarize the history and development of forest management practices in Bangladesh and worldwide
CO3 Discover the fundamentals of forest stand structure, growth and development
CO4 Illustrate the concepts and practices of Sustainable Forest Management, Ecosystem Management and Forest Landscape Restoration
CO5 Recognize the practical processes of Forest management planning in Bangladesh and worldwide.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3			2						
CO3	3					2				
CO4	3					3				
CO5	2					3	1		2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Davis, L. S., Johnson, K. N., &Bettinger, H. (2001). *Forest management*. McGraw-Hill.
2. Rietbergen-McCracken, J., Stewart, M., &Sarre, A. (2008). *The forest landscape restoration handbook*. Earthscan.
3. Kangas, A., Kangas, J., &Kurttila, M. (2008). *Decision support for forest management* (Vol. 16). Springer.
4. Diez, J. J., &García, J. M. (Eds.). (2012). *Sustainable forest management: Current research*.
5. Prakash, R. (1986). *Forest management*. M/S. International Book Distributors.
6. Young, R. A., & Giese, R. L. (2002). *Introduction to forest ecosystem science and management* (3rd ed.). Wiley.
7. Evans, J. (Ed.). (2008). *The forests handbook, volume 1: An overview of forest science*. John Wiley & Sons.
8. Evans, J. (Ed.). (2008). *The forests handbook, volume 2: Applying forest science for sustainable management* (Vol. 2). John Wiley & Sons.
9. Bettinger, P., Boston, K., Siry, J. P., &Grebner, D. L. (2016). *Forest management and planning*. Academic Press.
10. Ontario Ministry of Natural Resources. (2010). *Forest management guide for conserving biodiversity at the stand and site scales: Background and rationale for direction*. Queen's Printer for Ontario.
11. Franklin, J. F., Johnson, K. N., & Johnson, D. L. (2018). *Ecological forest management*. Waveland Press.

Course No: FES 08214122	**Credit: 3.0**	**Year: Fourth**	**Semester: First**

Forest Management Plan	Course Status: Practical

Course Rationale:
Developing Forest management Plans is a key role of a professional forester to manage forests. Therefore, every forestry graduate needs to have practical experience on how to develop a forest management plan. Knowledge gained by forestry students on forest inventories, social and economic surveys during their preparation of the management plans has practical implications for their job life activities as a forest manager.

Course Objectives:
The key objective of this course is to-

- Help students to conduct intensive Fieldwork to collect forestry data for preparing a forest management plan.

Course Content:
1. Fieldwork in a Forest area in Bangladesh and preparing a Forest management plan.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Compile, process and analyze collected primary and secondary data.
CO2 Design a forest management plan based on the collected data.
CO3 Apply the knowledge of forestry and environmental science in resourse management.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	1	2	2		3	2		3	
CO2	3	1	3			3	2		1	1
CO3	3		3			1	1			1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Davis, L. S., Johnson, K. N., & Bettinger, H. (2001). *Forest management*. McGraw-Hill.
2. Pant, M. M. (1990). *Forest resource management* (Field Document No. 2). UNDP/FAO/RGD/85/011, Institute of Forestry, Chittagong University.
3. Prakash, R. (1986). *Forest management.* M/S. International Book Distributors.
4. Young, R. A., & Giese, R. L. (2002). *Introduction to forest ecosystem science and management* (3rd ed.). Wiley.
5. Kangas, A., Kangas, J., & Kurttila, M. (2008). *Decision support for forest management* (Vol. 16). Springer Science & Business Media.

Course No: FES 05114121	Credit: 3.0	Year: Fourth	Semester: First
Genetics and Tree Improvement			Course Status: Theory

Course Rationale:
Genetics and tree improvement is an advanced course of forestry being offered to 4th year 1st semester students. The course gives students a very basic overview of some principles in fundamental areas of plant genetics and biotechnology to apply this knowledge to improve the traits of forest species, particularly improving the traits of trees.

Course Objectives:
The objectives of the course are to-

- Help the students to develop knowledge in plant breeding and biotechnology
- Enhance the skills of students for using the genetic variation, selection strategies, and transgenic approaches in tree improvement program.
- Lay the foundation for the study of more advanced plant breeding, biotechnology and molecular biology methodology and quantitative genetics for future higher study.

Course Content:
1. Introduction: Concepts, scope, and history of tree/crop improvement; cell, chromosome, central dogma of molecular biology, genetic code, Genetic markers in forest trees, application of genetics in tree breeding and its importance.
2. Population and evolution genetics: species, population, gene pool, speciation, Mandel's law, the Hardy-Weinberg theorem, genotype, phenotype and allele, genetic variation, microevolution.
3. Quantitative genetics: Polygenic traits and heritability and estimating heritability.
4. Tree improvement: Base populations, Candidate trees, Provisional plus tree, plus trees, Seed production area (SPA), Seed orchard, different types of selection, introducing exotic species, genetic gain, Genetic testing – mating designs, breeding values, breeding methods in details (mutation, polyploidy breeding, Hybridization).
5. Clonal forestry and in vitro propagation: Macro propagation – grafting, rooted cuttings**;** Micro propagation – axillary shoot multiplication, adventitious shoot production, somatic embryogenesis, Tissue culture.
6. Genetic engineering: Transformation approaches**,** Development of transgenic plant through Agrobacterium mediated transformation, Societal/ethical considerations of GMOs.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand the concepts of plant genetics, basic terminologies used in genetics and inheritance
CO2 Analyze the genetics and environmental traits, segregation of traits over generation, new species formation and natural selection and genetic drift in natural populations.
CO3 Generate hybridization, mass propagation, Cloning and tissue culture for improving the desired traits.
CO4 Develop the transgenic plant through Agrobacterium mediated transformation

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2	2		2	
CO2	3		3			3	3		2	
CO3	3			2	1	3	3	3	2	1
CO4	3			2	1	3	3	3	2	1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Chandrasekaran, S. N., &Parthasarathy, S. V. (1975). *Cytogenetics and plant breeding* (Revised ed.). Krishnaswamy P. Varadachary& Co.
2. Elliott, J. (1958). *Plant breeding and cytogenetics*. McGraw Hill Publications.
3. Goodenough, U. (1984). *Genetics*. Holt-Saunders International.
4. Jain, K., &Kharkwal, M. C. (2004). *Plant breeding: Mendelian to molecular approaches*. Narosa Publishing House.
5. Fehr, W. R. (1987). *Principles of cultivar development (Vol. 1). Theory and technique*. Macmillan Publishing Company.
6. Allard, R. (1960). *Principles of plant breeding*. John Wiley & Sons, Inc.
7. Snustad, P., Simmons, M., & Jenkins, J. (1997). *Principles of genetics*. John Wiley & Sons, Inc.
8. White, T. L., Adams, W. T., & Neale, D. B. (2007). *Forest genetics*. CABI Publishing.
9. Finkeldey, R., &Hattemer, H. H. (2004). *Tropical forest genetics*. Springer.
10. Kumar, S., &Fladung, M. (2004). *Molecular genetics and breeding of forest trees*. CRC Press.
11. Stewart, C. N., Jr. (2016). *Plant biotechnology and genetics: Principles, techniques, and applications*. John Wiley & Sons, Inc.

Course No: FES 05114122	**Credit: 1.0**	**Year: Fourth**	**Semester: First**
Genetics and Tree Improvement Laboratory			**Course Status: Practical**

Course Rationale:

Genetics and tree improvement Laboratory is the followed-up course of the theory FES 411. After successfully completing the laboratory works, students will be familiar with the active areas of plant breeding as well as will be conversant with biological research and scientific methods of improving the traits of trees.

Course Objectives:

The objective of the course is to-

- Apply the knowledge in selecting superior parental genotypes, hybridization and mass propagation of quality plant materials by simple Cloning and tissue culture methods.

Course Content:

1. Emasculation and hybridization
2. Practice of stem cutting, grafting and budding
3. Practice of tissue culture in the laboratory
4. Visiting and Demonstration of seed orchard, seed production area, plantation and natural forest for the practice of selecting superior parental trees, Plus tree selection

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1. Perform crossing
CO2. Propagate seedlings by stem cutting, developing a new grafted or budded plant
CO3. Select candidate tree, plus tree, how to establish a seed orchard and seed production area

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3			3	3	2	2	1
CO2	3		1			3	3	2	1	1
CO3	3		2			3	1	2	2	1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Reinert, J., & Yeoman, M. M. (1982). *Plant cell and tissue culture: A laboratory manual* (eBook ISBN 978-3-642-81784). Springer-Verlag Berlin Heidelberg.
2. Neumann, K.-H., Kumar, A., & Imani, J. (2009). *Plant cell and tissue culture: A tool in biotechnology, basics and application* (eBook ISBN 978-3-540-93883-5). Springer-Verlag Berlin Heidelberg.
3. R, A., & Joy, P. P. (2014). *Plant biotechnology laboratory manual*. Aromatic and Medicinal Plants Research Station, Kerala, India.

Course No: FES 07314121	**Credit: 3.0**	**Year: Fourth**	**Semester: First**
Urban Forestry and Land Use Planning			**Course Status: Theory**

Course Rationale:

This course will provide adequate knowledge on the basic concepts of planning and managing urban green-spaces and land use planning. This course will be a handout and guide to play a key role by the forestry graduate inUrban forestry and land use planning to mitigate the needs for sustainable land use.

Course Objectives:

The objectives of the course are to-

- Understand the complex characteristics of the urban ecosystem and land use planning for sustainable management policy.
- Develop skills on green space planning and design for ensuring landscape aesthetics, urban recreation, climate change adaptation, and mitigation.
- Provide the knowledge on how to assess the ecosystem services and urban natural resources.
- Enhance the basic knowledge of urban forestry and land use planning.

Course Content:

1 Introduction to urban forestry: Definition; Types of urban forestry; Benefits of urban forestry; Arboriculture; Rooftop gardening.

2 Urban Forestry practices in Bangladesh and other countries: Major cities of Bangladesh and scope of urban forestry in those cities; Urban forestry practices in Thailand, India, China, Malaysia, Brazil.

3 Appraisal and inventory of urban forestry: Assessment of the ecosystem services of urban forest; Urban natural resources accounting and valuation; Street Tree Inventory; Park and Other Urban Natural Resources Inventories; Causes of changes in urban landscape pattern (abiotic, biotic, human land use and disturbance).

4 Urban forestry and landscape management: Urban forest planning, policy and management; Sustainable urban green space management; Management of pests and diseases in urban forestry; Ecological restoration and sustainability of urban forest; Conservation of urban wildlife and wetland; community engagement in urban forest management.

5 Urban forestry and Climate change: Application of GIS for climate change impact assessment on urban areas; Urban resiliency assessment to climate change; Disaster risk reduction strategy; Land use change and zoning of urban area; Climate change mitigation and adaptation through urban forestry.

6 Land use planning: Concept of land, land-use, land cover and land use planning; significance of land use planning; goals of land-use planning; trade-offs between conflicting goals; planning at different levels; land use in relation to sectoral and development planning; steps in the planning process; the need for flexibility; contents of the land-use plan. Current land administration of Bangladesh and its core functions.

7 Land classification: Traditional system of land classification in Bangladesh; land capability classification of Bangladesh; forest land classes of Bangladesh; ecological regions of Bangladesh.

8 Steps in land-use planning: Establishing goals and term of reference; organizing the work; analyzing the problems; identifying the opportunities for change; evaluating land suitability; appraisal of the alternative environmental, economic and social analysis; choosing the best option; preparing the land-use plan; implement the plan; monitoring and revision of the plan.
9 Landscape Planning: Definition, its relationship to other areas of forestry, Principles and elements of landscape planning; Landscape design; Landscape architecture and regional planning; Necessity of landscape planning in the urban areas; Planting and gardening; Site development.
10 Urban recreation and tourism planning: Landscape aesthetics; Urban tourism planning and management; Heritage and cultural tourism; Choice of tree species for plantation in urban areas.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1	Generalize the basic concepts and interpret the importance of urban forestry, land-use planning and management
CO2	Evaluate urban forest planning and management techniques, and develop a sound urban forest management program.
CO3	Apply the knowledge of urban forestry and land use planning
CO4	Interpret current scientific thinking in the planning and management of urban forests in the face of climate change.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3		3							
CO3	3		3							
CO4	3		3							

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Konijnendijk, C. C., Nilsson, K., Randrup, T. B., &Schipperijn, J. (Eds.). (2005). *Urban forests and trees: A reference book*. Springer.
2. Miller, R. W., Hauer, R. J., & Werner, L. P. (2015). *Urban forestry: Planning and managing urban greenspaces*. Waveland Press, Inc.
3. Bradley, G. A. (1995). *Urban forest landscapes: Integrating multidisciplinary perspectives*. University of Washington Press.
4. Salbitano, F., Borelli, S., Conigliaro, M., & Chen, Y. (2016). *Guidelines on urban and peri-urban forestry* (FAO Forestry Paper No. 178). Food and Agriculture Organization of the United Nations.
5. AO. (1993). *Guidelines for land-use planning* (Reprinted 1996). FAO Development Series 1. Food and Agriculture Organization of the United Nations.
6. Haque, S. M. S. (2013*). Introduction to forestry in Bangladesh. Degradation of upland watershed in Bangladesh project,* Institute of Forestry and Environmental Sciences, University of Chittagong, Bangladesh & United States Department of Agriculture (USDA), Grant No. BG-ARS-123.
7. Hassan, M. M. (1999). *Soils of Bangladesh: Their genesis, classification, and use potential.* Published by Murshed Salam.

Course No: AMS 05424101	Credit: 3.0	Year: Fourth	Semester: First
Research Methodology			Course Status: Theory

Course Rationale:
Research methodology is a cornerstone of undergraduate forestry education, providing students with the essential skills to design, conduct, and communicate scientific investigations in forestry and environmental sciences. This course introduces fundamental research concepts, ethical practices, and analytical techniques, empowering students to contribute effectively to both academic and applied research in their field.

Course Objectives:
The objectives of this course are to-

- Help students understand fundamental research concepts and design principles.
- Enable students to distinguish between various sampling and data collection methods.
- Develop students' skills in analyzing and interpreting research data.
- Teach students how to write an effective research proposal.
- Improve students' ability to produce ethical academic/scientific writing.

Course Content:

1. **Introduction:** research concept, philosophy, purpose and types of research, research process.
2. **Research methods:** Phases in forestry Research (pre-fieldwork, fieldwork, post-fieldwork), selection of research methods.
3. **Experimental designs in forestry research:** principles of experimental designs, basic experimental designs, identification of variables in research question, Experimental artifacts and their elimination process.
4. **Field sampling:** population $v_{s.}$ sample, various sampling methods, sampling design, sample size, sampling and non-sampling errors.
5. **Data analysis:** Statistical concepts and their application in forestry, Continuous data and Discrete data, descriptive statistics (data display; data distribution; Bell Curve or Normal Curve; measures of centre and variation in data; probability density function-PDF, test of data normality); inferential statistics (estimation, sampling error, sampling variability, sampling distribution), significance level, degrees of freedom, confidence interval; hypothesis testing (type I and type II errors, Z-test, t-test, χ^2 -test, ANOVA); correlation and regression methods (error and sum of squares-SSE, SSR, SST, MS); post-hoc analysis.
6. **Research proposal writing:** pros and cons of academic/non-academic research proposal writing, elements of an academic research proposal, identification and formulation of research problems, development of research objectives, questions and hypotheses.
7. **Critical reading and abstracting:** strategies in critical reading and abstracting research papers; strategies for reading academic texts/books/journal articles; taking notes from academic texts.
8. **Literature review:** why to write a literature review, what literature to review, what questions the review should answer, structure of the literature review, final checklist in literature review.
9. **Writing research reports:** from proposal to thesis; writing thesis, scientific report and scientific journal article, their structure; argumentation and reasoning in scientific writing; organization of result and discussion; use of language and technical English in writing; scientific ethics and professionalism; intellectual property and plagiarism; use of graphics in the report, methods of citing references; thesis quality.
10. **Plagiarism:** Different forms of plagiarism; demonstration of different forms of plagiarism; scientific ethics and professionalism; intellectual property right.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand and explain the fundamental concepts of research and compare different research methodologies.
CO2 Design a sampling strategy and construct an experimental plot for forestry research.
CO3 Select appropriate statistical tests for data analysis and interpret the results accurately.
CO4 Formulate a research proposal, including conceptual frameworks, objectives, and methodologies

CO5 Execute technical writings following research ethics and professionalism.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3					2	2			
CO3			3			2	2			
CO4						3	2			
CO5		3						2		

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Tan, W. (2023). *Research methods: A practical guide for students and researchers* (2nd ed.). World Scientific.
2. Macfarland, T. W. (2014). *Introduction to data analysis and graphical presentation in biostatistics with R.* Springer.
3. Mishra, &Alok. (2011). *Handbook of Research Methodology*. Educreation Publishing.
4. Kothari, C. R. (2004). *Research methodology: Methods and techniques* (2nd ed.). New Age International (P) Ltd.
5. Schimel, J. (2023). *Writing science: How to write papers that get cited* (2nd ed.). Oxford University Press.
6. Ruxton, G. D., &Colegrave, N. (2022). *Experimental design for the life sciences* (5th ed.). Oxford University Press.
7. Corgill, M., & O'Connor, P. (2013). *Writing scientific research articles: Strategy and steps* (2nd ed.). Wiley-Blackwell.
8. Shankar, G., &Nandal, R. (2023). *Statistical methods for forestry and environmental data*. CRC Press.

Course No: AMS 05424102	**Credit: 1.0**	**Year: Fourth**	**Semester: First**
Research Methodology Laboratory			**Course Status: Practical**

Course Rationale:
In order to conserve biodiversity, expand forest resources/ecosystems and protect the surrounding environment in the face of continuous human pressure and global climate change scenario, forestry graduates should have practical methodological knowledge on forestry research and contemporary environmental issues.

Course Objectives:
The objectives of this course are

- To acquaint students with basic knowledge of research and it's design
- To help them differentiate among different sampling and data collection methods
- To help them to develop skills for data collection for getting results.
- To develop skills to write a manuscript and its ethical issues for journal publication.

Course Content:

1. Data analysis using a computer aided statistical package (e.g. Introduction to R, SPSS, etc.).
2. Identification of a research problem/research gap.
3. Practice of scientific communication.
4. Using reference managing software (e.g. Mendeley).

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Analyze data with software.
CO2 Interpret the results of statistical analyses.
CO3 Identify research gaps on contemporary topics.
CO4 Develop materials for scientific communication.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1			3							
CO2						3				
CO3						3				
CO4		3							2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO2	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO3	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03
CO4	TL03, TL04, TL05, TL06	CA03, CA04, CA05, SA03

Recommended References:

1. Magno, V. C. (1986). *Community forestry handbook*. FAO/UNDP Project BGD/81/028.
2. Stapleton, P., et al. (2004). *Manual on technical writing*. IPGRI, Future Harvest.
3. Paradis, E. (2005). *R for beginners* (1st ed.). F-34095 Montpellier Cedex 05, France.
4. R Development Core Team. (2005). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. ISBN 3-900051-07-0. URL: http://www.R-project.org.
5. Venables, W. N., Smith, D. M., & the R Core Team. (2021). *An introduction to R* (Version 4.1.2). 1-105 pp.
6. Mishra, P., &Alok, S. (2011). *Handbook of research methodology*. Educreation Publishing. ISBN: 978-1-5457-0340-3.
7. Kothari, C. R. (2004). *Research methodology: Methods and techniques* (2nd ed.). New Age International (P) Ltd. ISBN 978-81-224-2488-1.
8. Mendeley. (2010). *Getting started with Mendeley* (1-16 pp.). https://desktop-download.mendeley.com/download/Getting_Started_Guide.pdf

Course No: FES 05214121	**Credit: 3.0**	**Year: Fourth**	**Semester: First**
Environmental Pollution and Climate Change			**Course Status: Theory**

Course Rationale:

Environmental pollution and climate change are among the top issues of public concern and a topic of much media and political discussion. The aim of this course is to identify key environmental problems and discuss scientific aspects of addressing these problems in the face of climate change.

Course Objectives:
The objectives of this course are to-

- Understand the different types of environmental pollution, their causes, effects, and the control measures available at local, national, and global levels.
- Know the basics of climate change, its impacts and potential solutions.
- Explore international and national legal frameworks addressing climate change and environmental protection, including treaties and policies measures.
- Empower students with the knowledge to make responsible decisions for environmental sustainability and climate resilience.

Course Content:

1. **Environmental Pollution:** Definition, Major categories of environmental pollution: causes, effects and control measures of (i) Air pollution, (ii) Water pollution, (iii) Soil and marine pollution, (iv) Noise pollution, (v) Thermal pollution, (vi) Radioactive pollution. Role of individuals in pollution prevention and social aspects of environmental pollution. Case studies on environmental pollution in Bangladesh.
2. **Waste Management and Sustainable Practices:** Types of waste (solid, hazardous, e-waste, medical, plastic) and their sources (industrial, municipal, agricultural, etc.). Basic waste treatment and disposal techniques: landfilling, incineration, composting, recycling and reuse. Concept of sustainable waste management.
3. **Energy and the Environment:** Renewable and non-renewable energy resources (overview), including solar, wind, hydropower and biomass. Role of forests in energy production. Biomass energy sources: wood, forest residues and organic waste. Basic concepts of bioenergy, biofuels and biochar.
4. **Introduction to Climate Change:** Weather, climate and climate variability (basic concepts); Scientific understanding of climate change; Impacts of climate change: agriculture, water and energy systems, health, society, extreme events and sea-level rise; Basic idea of climate models and future projections; Climate change scenario in global and Bangladesh context; Introduction to major international frameworks: UNFCCC, IPCC, Paris Agreement and COP (overview).
5. **Climate Change Adaptation and Mitigation:** Definition and importance of adaptation and mitigation. Basic adaptation strategies (community and ecosystem-based). Overview of mitigation approaches: greenhouse gas emission reduction and carbon sequestration. Introduction to mechanisms: CDM, REDD+, carbon trading and carbon tax (conceptual understanding). Concept of resilience: community and ecosystem resilience.
6. **Disaster Management:** Definition and objectives of disaster management. Common disasters: flood, cyclone, drought, earthquake.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand different kinds of pollution, their causes and effects at the global, regional, and local level, analyze various control measures of pollution, and familiarize students with different case studies on environmental pollution.

CO2 Analyze the key drivers and common features of climate change, and beware of the current and future scenarios of climate change.

CO3 Explain the concepts and physical science basis of climate change, assess the vulnerability to climate change impacts with a special emphasis on agriculture and natural resources, and identify adaptation and mitigation options.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3			2		3		3	1	1
CO2	3			2	2	3	2			
CO3	3			3		3	2	2	2	1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Andrews, S., &Natkin, M. (1983). *World environment handbook: A dictionary of government natural resource management agencies in 144 countries.* World Environment Centre.
2. Anon. (1984). *Air pollutants effects on forest ecosystem.* The Acid Rain Foundation.
3. Anon. (1979). *Fundamentals of air pollution control technologies. Special Project Research on Deflection & Control of Environmental Pollution.*
4. Bashar, S. H. M., &Reazuddin, M. (1990). *Towards sustainable development: Issues of environmental pollution in Bangladesh.* Ministry of Environment & Forest and National Conservation Strategy Secretariat, BARC.
5. Anon. (1990). *UNEP Environmental Data Report*(2nd ed.). UNEP.
6. Letcher, T. M. (2016*). Climate Change: Observed Impacts on Planet Earth* (2nd ed.). Elsevier.
7. Archer, D. (2011). *Global warming: Understanding the forecast.* Wiley.
8. Houghton, J. (2009). *Global warming: The complete briefing* (4th ed.). Cambridge University Press.

Course No: FES 00314122	**Credit: 1.0**	**Year: Fourth**	**Semester: First**
Seminar			**Course Status: Practical**

Course Rationale:

Acquiring speaking and presentation ability is crucial for securing a place in job market as well as in enduring personal aptitude. This course will allow a student to gain hands-on experience of the presentation of forestry and environment related scientific research and popular issues in front of the audience.

Course Objectives:

The objectives of the course are to-

- Enhance the knowledge and skills of presentation on topics related to forest, forestry and environment.
- Help the students with the knowledge and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. **Introduction to Seminar:** Principles and basic features of seminar**.**
2. **Scientific seminar:** What is scientific seminar; key features of scientific seminar, how to extract key message from articles, field documents, books, chapter etc.
3. **Methods of delivering seminar:** Preparation of PPT; Design, timing and message selection for the presentation

Course Learning Outcomes (COs) (3-5):

After the successful completion of the course, students will be able to

CO1 Deliver presentation individually or in a group using multiple approaches.
CO2 Interpret the knowledge of important forestry and environmental issues and research findings.
CO3 Gain the capacity to point out the relevant field level management issues to meet the needs of the country.
CO4 Combine the knowledge and skills of scientific or research article presentation.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	3			2		2		3	
CO2	3		3			1	2		2	
CO3	2						3	2		2
CO4	2			2					2	

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL02, TL05, TL06	CA04, CA05, SA03
CO2	TL02, TL05, TL06	CA04, CA05, SA03
CO3	TL02, TL05, TL06	CA04, CA05, SA03
CO4	TL02, TL05, TL06	CA04, CA05, SA03

Recommended References: As per suggestions of the course teacher

Course No: FES 07224121	**Credit: 2.0**	**Year: Fourth**	**Semester: First**
Dendrochronology			**Course Status: Theory**

Course Rationale:
Dendrochronology, the study of tree rings, offers valuable insights into the historical growth patterns, physiological processes, and anatomical responses of trees to their environment. Tree rings serve as natural archives of ecological information, enabling researchers to determine tree age with precision and to assess past environmental conditions. This information is essential for forest management practices, as it can quickly provide data on tree age, growth rates, regeneration patterns, and survival dynamics. This course aims to introduce students to the fundamental principles and applications of dendrochronology, with a focus on understanding long-term tree responses to environmental variability and change.

Course Objectives:
This course is designed to:
1. Introduce students to the concept and formation of growth rings in trees.
2. Provide knowledge about tree species that develop distinct growth ring boundaries.
3. Examine the environmental and physiological factors that limit tree growth and influence xylem development.
4. Explore how extreme climatic events—such as droughts and floods—impact tree growth and wood anatomy.
5. Develop a comprehensive understanding of how environmental conditions shape tree growth over time.

Course Content:
1. **Introduction to Dendrochronology:** Fundamentals of tree-ring analysis; historical development of the field; key applications; major subfields including dendroecology, dendrohydrology, dendromorphology, and dendroclimatology; overview of wood anatomy; challenges and opportunities in tropical dendrochronology.
2. **Tree Ring Width Analysis:** Principles of limiting factors in tree growth; identification of growth ring boundaries; detection of anomalies such as false and wegging rings; methods for measuring ring widths; techniques in cross-dating, trend analysis, standardization, and indexing; development of tree-ring chronologies and master chronologies.

3. **Dendroclimatology:** Environmental drivers of tree growth; analysis of key climate variables (e.g., precipitation, temperature, humidity, drought indices); interpreting long-term climate trends and their influence on growth; assessing tree responses to extreme climatic events.
4. **Wood Anatomical Time-Series Analysis:** Structure and function of xylem tissue; processes of xylogenesis and cambial activity; anatomical measurements related to growth; estimating hydraulic conductivity and its ecological significance.
5. **Dendrochemistry:** Techniques in chemical analysis of tree rings, including cellulose extraction, stable isotope analysis, and elemental composition studies; applications of dendrochemistry in understanding environmental influences on tree physiology.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Generalize the concept of growth ring boundary based on different macro and microscopic features of wood.
CO2 Understand the tree ring width and the factors affecting ring width or tree growth.
CO3 Critically analyse the climate seasonality which are related with tree ring formation.
CO4 Illustrate the xylogenesis or wood formation process.
CO5 Evaluate the impact of climate on tree growth, wood anatomical processes and finally carbon sequestration.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3					2				
CO2	2					2				
CO3	3					2				
CO4	1					1				
CO5	2					2				

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Schweingruber, F. H. (1988). *Tree rings: Basics and applications of dendrochronology*. D. Reidel.
2. Cook, E. R., &Kairiukstis, L. A. (1990). *Methods of dendrochronology: Applications in the environmental sciences.* Kluwer Academic Publishers.
3. Speer, J. S. (2012). *Fundamentals of tree-ring research*. University of Arizona Press.
4. Fritts, H. C. (2001). *Tree rings and climate*. The Blackburn Press.
5. Gartner, H., &Schweingruber, F. H. (2014). *Microscopic preparation techniques for plant stem analysis*. Kessel Publishing House.

Course No: FES 08214221	**Credit: 3.0**	**Year: Fourth**	**Semester: Second**
Forest and Environment Policy, Law and Administration			**Course Status: Theory**

Course Rationale:
Legal instruments and their effective implementation are essential for sustainable natural resource management. This course provides fundamental knowledge and understanding of legal and policy issues relevant to the forestry and environmental sectors. The course focuses on raising stakeholder awareness to combat illegal practices detrimental to our forests and the environment. It explores how national and global policy instruments shape this legal framework, and how policies are formulated, including the factors that directly or indirectly influence the process. The course is designed to ensure the effective management of forest resources and support the creation of a pollution-free environment, contributing to the broader goal of sustainable development.

Course Objectives:
The objectives of the course are to-

- Familiarize students with the significance of legal and policy instruments related to forests and the environment.
- Understand current terminologies, facts, and issues directly or indirectly linked to planning, management, and decision-making in forestry and environmental fields.
- Provide insights into the development of Bangladesh's forest and environmental policies, acts, and rules, including their limitations and suggestions for improvement.
- Emphasize the effective use of legal frameworks to promote sustainable development.

Course Content:

1. **Introduction:** Concepts, goals and tools of Forest and Environmental policy. Legal terminology in forestry and environment. Definitions, types, and functions of policies, acts, and rules Historical background of Bangladesh's forest and environmental legal instruments.
2. **Institutions in forestry and environment:** Key organizations in Bangladesh's forestry and environmental sectors, their roles and objectives; challenges faced by these institutions and potential strategies for improvement.
3 **Tools and approaches of Forest and Environmental Policy: Key e**lements, principles and tools used in policy formulation**,** implementation and evaluation. Factors influencing the development of forest and environment policies.
4 **Forest and Environmental Policies of Bangladesh:** Historical development and analysis of Bangladesh's forest and environmental policies. Objectives, key statements, implementation challenges, and strategies for policy enhancement.
5 **Forest and environmental governance in Bangladesh:** Current status, challenges and clues for development. Selected best practices from other countries.
6. **Global forest and environmental policy instruments (ICTPs):** Introduction toenvironmental diplomacy: concept, nature and importance.Various international policy instruments on forestry and environment. Influence of global conventions, treaties and protocols (ICTPs) on Bangladesh's policy development. Overview of major instruments including CBD, UNFCCC, UNCCD, CITES, SDGs, Ramsar Convention, Paris Agreement and Forest Principles

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand key concepts, legal terminology, and policy features related to forestry and environmental management.
CO2 Identify and assess institutions involved in the legal and policy frameworks related to forest and environmental governance.
CO3 Analyze historical trends in the development of legal and policy frameworks and their implications for conservation and community development.
CO4 Interpret existing acts, rules, policies, and national documents that influence biodiversity conservation and community development for sustainable resource management
CO5 Examine global ICTPs and evaluate their relevance and impact on national legal and policy frameworks.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
C01	1		-							
CO2		1								
CO3				2	3	1	1			
CO4	1		2							
CO5	1				2			1	1	2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Srivastav, A. (2024). *Forest policies, laws and governance in India: Conservation challenges in the face of climate change* (422 pp.). Springer.
2. Birkland, T. A. (2019). *An introduction to the policy process: Theories, concepts, and models of public policy making*. Routledge.
3. McDermott, C., Cashore, B., &Kanowski, P. (2010). *Global environmental forest policies: An international comparison*. Routledge.
4. Government of Bangladesh. (2016). *Bangladesh national conservation strategy (2016–2031), Part I: Executive summary* (103 pp.). IUCN and Ministry of Environment and Forest, Dhaka, Bangladesh.
5. Government of Bangladesh. (2018). *Bangladesh Tiger Action Plan (2018–2027)*. Ministry of Environment, Forest and Climate Change, Dhaka, Bangladesh.
6. Government of Bangladesh. (2018). *Bangladesh Elephant Conservation Plan (2018–2027)*. Ministry of Environment, Forest and Climate Change, Dhaka, Bangladesh.
7. Rahman, L. M. (2023). *Forest Act guide* (in Bengali). Ain Publisher.
8. Eisma-Osorio, R.-L. (Ed.). (2020). *The impact of environmental law: The stories of the world we want*. IUCN Academy of Environmental Law Series.
9. Government of Bangladesh. (n.d.). *The Constitution of the People's Republic of Bangladesh*. [Note: If a specific year of publication or last amendment is available, replace (n.d.) with that year.]
10. Sample, V. A., & Cheng, A. S. (2003). *Forest conservation policy: A reference handbook* (344 pp.). ABC-CLIO.
11. Yashmi, P., Enters, T., & Leslie, R. N. (2010). *Forest policies, legislation and institutions in Asia and the Pacific* (Working Paper No. APFSOS II/WP/2010/34). FAO.

Course No: FES 08214222	**Credit: 4.0**	**Year: Fourth**	**Semester: Second**
Research Project			**Course Status: Practical**

Course Rationale:

This course will allow a student to gain hands-on experience of the conduct of scientific research related to forests, forestry and environment sectors.

Course Objectives:

The objectives of the course are to-

- Help the students to develop the ability to conduct research works on any forests, forestry and environment sectors' related issues.
- Develop skills to write a scientific dissertation

- Understand the ways of presentation of any forest, forestry and environment sectors' related scientific research article presentation in front of the audience.
- Facilitate the acquired knowledge and skills to solve the relevant field level management issues to meet the needs of the country.

Course Content:

1. Conduct research work on any forests, forestry and environmental sectors related issues.
2. Write a scientific dissertation.
3. Presentation of the selected scientific research articles related to forests, forestry and environmental sectors.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Conduct scientific research work on any forests, forestry and environmental sectors related issues.

CO2 Prepare a scientific dissertation.

CO3 Present a scientific research article in front of audiences.

CO4 Realize the significance of research, dissertation and presentation in professional life and Present themselves professionally in front of the audiences..

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3		3			3	3	2		
CO2		3		2		3	3	2		
CO3		3			2	2	2		2	
CO4		3	3		3	3	3	3	3	2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL02, TL05, TL06	CA05, SA03
CO2	TL02, TL05, TL06	CA05, SA03
CO3	TL02, TL05, TL06	CA05, SA03
CO4	TL02, TL05, TL06	CA05, SA03

Recommended References:

1. Any documents supplied by the related course teacher (s)/supervisor (s).
2. Collection of relevant documents from different libraries, internet, and others sources including personal contact with relevant person (s).

Course No: FES 05214221	Credit: 2.0	Year: Fourth	Semester: Second
Environmental Impact Assessment			Course Status: Theory

Course Rationale:
Environmental impact assessment has become an important tool for development decision-making. This course will facilitate students to play an important role in decision-making on the environmental consequences of proposed actions; and promote environmentally sound and sustainable development through the identification of appropriate enhancement and mitigation measures.

Course Objectives:
The objectives of this course are to

- To facilitate necessary knowledge to improve the environmental design and the social aspects of the proposal
- To enhance the skill on identifying appropriate measures for mitigating the potential impacts of the proposal
- To provide the knowledge of guiding principles of EIA good practice, generalized EIA Process and different methods of impact identification and management
- To apply the knowledge to assess the strength and weakness of EIA reports
- To facilitate students in decision making, including setting the environmental terms and conditions for implementing the proposal
- To foster the analytical and critical learning to deliver theoretical background on how to perform environmental auditing.

Course Content:

1. **Introduction to Environmental Impact Assessment (EIA):** Definition, historical development of EIA, Principles and characteristics of EIA, Importance of EIA authorities/ organizations involved in the EIA processes, EIA guidelines prescribed by FAO.
2. **EIA Stages:** The EIA process before starting the EIA, screening, preliminary assessment, organization, scoping, the EIA study, identification, prediction, evaluation, mitigation, documentation, EIA follow up;, EIA reporting and review.
3. **EIA Methods:** Methods for impact identification-Ad hoc, Checklists, Matrices, Networks and overlays.
4. The role and experience of donor/multi-national organization with EIA
5. **Social and Health impact assessment**: Meaning, importance, difference from other forms of assessment, practices
6. **Practical learning from Case studies:** Environmental Impact Assessment (EIA) and Social Impact Assessment (SIA) in Different development projects of Bangladesh: Forestry and Environmental projects such as Commercial logging, Afforestation, Community forestry; Tourism development, Transport project, Power Plant project, Flood management and livelihood improvement project.
7. **Introducing Strategic Environmental Assessment (SEA):** Definition, SEA in plan making, SEA processes, SEA effectiveness, Limitations
8. **Environmental auditing**: Definition, characteristics and types of environmental audit, objectives, scope of environmental audit, function and benefits, criteria to implement a successful environmental audit, relationship between environmental audit and EIA, audit process, requirements for an effective audit, types of information reviewed during audit, quality system audit, audit report.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Know basic knowledge on the aims, objectives and importance of EIA and it stages
CO2 Apply different methods of impact identification, mitigation, and management
CO3 Evaluate the strength and weakness of some EIA case studies of different development projects of Bangladesh specially related to forestry activities
CO4 Understand the role, responsibilities, and functions of donor/multi-national organization with EIA
CO5 Synthesize the knowledge of EIA auditing

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3									
CO2	3			2		2	3			
CO3	1			3		3	3		2	
CO4	3			1						
CO5	3					3	3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Glasson, J., &Therivel, R. (2019). *Introduction to environmental impact assessment* (5th ed.). Routledge.
2. Noble, F. B. (2015*). Introduction to environmental impact assessment: A guide to principles and practice* (3rd ed.).
3. Eccleston, C. H. (2011). *Environmental impact assessment: A guide to best professional practices.* CRC Press.
4. Therivel, R. (2010). *Strategic environmental assessment in action* (2nd ed.). Routledge.
5. HMSO. (1989). *Environmental assessment: A guide to the procedures*. Department of Environment, UK.
6. Brotto, L., &Pettenella, D. (Eds.). (2018). *Forest management auditing: Certification of forest products and services.*Routledge.

Course No: FES 08214224	**Credit: 1.0**	**Year: Fourth**	**Semester: Second**
Internship			**Course Status: Practical**

Course Rationale:
This course will allow a student to gain hands-on experiences on various management activities of government, non-government, private, multinational organizations working on forestry and environment sectors.

Course Objectives:
The objectives of the course are

- To build technical and professional capacity of the intern
- To gain new skills and opportunities that may not be possible to achieve in the classroom activities
- To gain technical skills within the sector of the choice of intern for working independently and in a team in the professional life
- To apply the acquired academic knowledge during internship and to learn soft skills such as time management, ability to work in team, developing adaptability in new working environment.
- To develop skills on planning future works of the organizations and formulating strategies to implement the plans.

Course Content:
Course content will be designed by the organization of choice of the intern focusing on the developing soft skills such as management, communication, team work, leadership, problem solving strategy and report writing.

Course Learning Outcomes (COs):
After the successful completion of the course, students will be able to

CO1 Gain experiences on office services, field level management works on any forests, forestry and environmental sectors related issues.
CO2 Develop their technical knowledge and communication skills to solve a problem.
CO3 Learn to share their experiences through presentation and report writing.

Mapping of COs with POs

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1O
CO1	3		3			1		3		3
CO2		3		3	2		2	3	2	
CO3		3		2		2		2		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL04, TL05, TL06	CA04, SA03
CO2	TL04, TL05, TL06	CA04, SA03
CO3	TL04, TL05, TL06	CA04, SA03

Recommended References: As per the suggestions of the course teacher

Course No: FES 05224221	**Credit: 2.0**	**Year: Fourth**	**Semester: Second**
Biodiversity Conservation			**Course Status: Theory**

Course Rationale:
This course will provide an introduction to the role of science in the conservation of animals and plants. Course lectures will provide a rigorous introduction to the motivation for, methods of, and major challenges facing contemporary conservation biology. Moreover it will focus on an introduction to the scientific study and analysis of biodiversity; an understanding of the major threats to biodiversity; an overview of contemporary conservation strategies that adequately address the difficult social, economic, political and ethical dimensions' attendant to contemporary conservation problems and an analysis of recent and historic success and failures in conservation efforts.

Course objectives:
The objectives of the course are to-

- Provide the knowledge about conservation of biology and tools used by conservation biologists**.**
- Make the students understand the scientific evidence for biodiversity change in the modern era and the details of contemporary causes of diversity loss.
- Get an idea about the ecological, social, and economic impacts of diversity loss.
- Apply the knowledge of management principles and tools that are used to conserve diversity at levels from genes to landscapes.

Course Content:

1. **Conservation Biology:** Basic concepts of conservation and conservation biology. History of conservation. Conservation ethics.
2. **Biodiversity and its importance:** Defining biodiversity: Biodiversity Pattern (Genetic, species and ecosystem diversity), Spatial Scales of Biodiversity, Instrumental and instinct values of Biodiversity, the uniqueness value of species.

3. **Role of Ecosystem Management:** The values of ecosystems; Objectives of Ecosystem management; Functional diversity, Role of Ecosystems and landscapes management in conserving biodiversity.
4. **Threats to Biodiversity:** Mass extinctions and global change, Extinction processes; Processes that diminish genetic diversity and species diversity, Ecosystem degradation and loss; Overexploitation; Invasive exotics.
5. **Measuring and Monitoring Biodiversity:** Comparison of structural diversity among ecosystems/ forest communities, Purpose of Monitoring, Monitoring approach, Monitoring biodiversity using indicators, Challenges to monitoring.
6. **Conservation Strategies:** Types of conservation strategies, Evidence-Based Conservation, reserve selection, design and management; Managing ecosystems and populations; Zoos and gardens. State of forest biodiversity conservation in Bangladesh.
7. **The Human Factors:** Social Factors, economics, politics and action, Conventions and Organization related to biodiversity conservation (CBD, IUCN, WWF, CITES, WCPA).

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Interpret the basic concepts of conservation and conservation ethics and explain the term biodiversity in species, ecosystem and evolutionary-centric approach.

CO2 Point out the current threats to biodiversity to relate them to human feedback on conservation efforts.

CO3 Distinguish between fine and coarse filter approach of biodiversity conservation.

CO4 Evaluate the knowledge of conservation management to address conservation problems and priorities.

CO5 Relate how interdisciplinary work (particularly involving the domains of policy, economics, cultural orientation and ethics) is essential to addressing problems in nature conservation. Research in assessing species diversity and species-specific conservation approach.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2				3	2			
CO2	3					3	2			2
CO3	3				2	3	2			2
CO4	2				2	3				
CO5	3					3	3			

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04,CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01

CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Berkes, F. (2004). *Rethinking community-based conservation.* Conservation Biology, 18(4), 621–630.
2. Brooks, T. M., Mittermeier, R. A., da Fonseca, G. A. B., Gerlach, J., Hoffmann, M., Lamoreux, J., Chapin, F. S., Zavaleta, E. S., Eviner, V. T., Naylor, R. L., Vitousek, P. M., Reynolds, H. L., Hooper, D. U., Lavorel, S., Sala, O. E., Hobbie, S. E., Mack, M. C., &Díaz, S. (2000). *Consequences of changing biodiversity.* Nature, 405, 234–242.
3. Gardner, T. (2010). *Monitoring forest biodiversity: Improving conservation through ecologically responsible management.*Earthscan LLC.
4. Hunter, M. L., & Gibbs, J. P. (2007). *Fundamentals of conservation biology* (3rd ed.). Blackwell Publishing.

Course No: FES 05224223	Credit: 2.0	Year: Fourth	Semester: Second
Protected Area Management			Course Status: Theory

Course Rationale:

This course will provide knowledge on the basic concepts of the IUCN protected area categories, their management focusing on wildlife and ecotourism. The main emphases of this course are to increase benefits of the local community around the forest protected areas, improve the recreational opportunities and restore the environmental sustainability of the forest by conserving biodiversity specially the wildlife. Moreover this course will create sentiments on remnant biodiversity conservation and ecotourism management.

Course objectives:

The objectives of the course are to-

- Facilitate the students achieve necessary skills to solve the relevant field level issues regarding community based forest management in the forest protected area.
- Help them conceptualize basic theories of ecotourism in the protected area and their importance in Bangladesh.
- Provide basic ideas about protected areas related policy and legislation.

Course Content:

1. **Introduction:** History, concepts and objectives, importance and legal status of protected areas, IUCN protected area management categories.
2. **Protected areas of Bangladesh:** Problems and prospects, objectives, existing management systems, concepts and principles of co-management, co-management in Bangladesh; history, types and status.
3. **Protected area management techniques:** Planning and design, administration and maintenance, facility improvement and development, management capacity and Buffer zone
4. **Measuring protected area management effectiveness:** IUCN Framework to Measure the Effectiveness of Protected Area Management, Numeric Methodology to Evaluate Protected Area Systems, Procedure to Rate Management Effectiveness in Protected Wilderness Areas, Monitoring System.
5. **Eco-tourism management:** Concepts and principles of ecotourism, importance of ecotourism, role of different organizations in promoting eco-tourism in Bangladesh. Concept and types of tourist carrying capacity; relationship between impact and use level, methods of tourist carrying capacity assessment, Numerical-based carrying capacity assessment- definition and assessment procedure of Physical carrying capacity (PCC), Real carrying capacity (RCC) and Effective or permissible carrying capacity (ECC), sustainable eco-tourism management in the protected areas.
6. **Policy and legislation:** protected areas related policy and legislation.

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Interpret the basic concepts of IUCN protected area categories management principles.
CO2 Learn how to engage community in protected area management.
CO3 Distinguish between conservation and ecotourism development concepts.
CO4 Manage protected areas in conserving biodiversity and creating sustainable ecotourism.
CO5 Evaluate the protected areas related policy and legislation.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3				2					
CO2	1	2		2			3		3	
CO3	3			2	1					
CO4			3	2			3			
CO5	3							3	1	1

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Mukul, S. A., & Rashid, A. Z. M. M. (Eds.). (2017). *Protected areas: Policies, management, and future directions.* Nova Science Publishers.
2. Chowdhury, M. S. H. (2014). *Forest conservation in protected areas of Bangladesh: Policy and community development perspectives.* Springer.
3. Sinclair, A. R. E., Fryxell, J. M., &Caughley, G. (2006). *Wildlife ecology, conservation, and management.* Blackwell Publishing.
4. Krausman, P. R., & Cain, J. W. III. (2013). *Wildlife management & conservation.* The John Hopkins University Press.
5. Hussain, K. Z. (1974). *An introduction to wildlife of Bangladesh.* Government of Bangladesh.
6. Sarker, S. U. (1992). *Parks and wildlife management.* UNDP/FAO Project BGD/85/011, IFCU.
7. Watson, J. E. M., Dudley, N., Segan, D. B., & Hockings, M. (2014). *The performance and potential of protected areas.* Nature, 515, 67–73.
8. Oldekop, J. A., Holmes, G., Harris, W. E., & Evans, K. L. (2016). *A global assessment of the social and conservation outcomes of protected areas.* Conservation Biology, 30(1), 133-141.
9. Cifuentes, M., Izurieta, A., & De Faría, H. (2000). *Measuring protected area management effectiveness.* WWF, GTZ, IUCN.

Course No: FES 05214225	Credit: 3.0	Year: Fourth	Semester: Second
Environmental Governance and Diplomacy			Course Status: Theory

Course Rationale: This course will provide an adequate knowledge of the basic concepts of environmental governance, its implication to natural resource management (NRM) particularly forest governance. How the knowledge of governance and its various attributes can be applied in the policy formulation and negotiation process is also an important purview of the course.

Course Objectives: The salient objectives of the course are:

- To understand the basic concepts of governance
- To provide the knowledge of the current status of environmental governance giving emphasis to NRM
- To apply the knowledge of integrating national and international policy and legal instruments in environmental diplomacy to attain sustainable development goals
- To get an idea about the national and international policy negotiation process.

Course Content:

1. **Introduction to Environmental Governance:** Definition, scope, objectives and basic attributes of governance.
2. **Importance and various types of Governance:** Why governance is important? Various types of governance and their implications in policy management.
3. **Governance Principles and Sustainable Development:** Historical development of governance mechanism. International Principles of sustainable development and their relation to governance.
4. **Environmental Governance in Bangladesh:** History of environmental governance in Bangladesh. Problems and prospects of environmental governance. Forest governance and its chronological development. Factors responsible for the poor governance. National strategy to ensure sustainable forest governance.
5. **National Policy Formulation and Governance Outlook:** Formulation process of various national policy documents. The status and influence of environment and forest governance in the policy formulation. Role of International Conventions, Treaties and Protocols (ICTPs) in ensuring effective environmental governance.
6. **Introduction to Environmental Diplomacy and Negotiation process:** Basic concepts, definitions and scopes of environmental diplomacy. Recent development in environmental diplomacy and negotiation process.
7. **Various Process of Negotiation:** Stakeholders of negotiation process. Role play in negotiation process. Various approaches of negotiation. Quality of an environment diplomat. Problems and prospects of environmental diplomacy in Bangladesh

Course Learning Outcomes (COs) (3-5):
After the successful completion of the course, students will be able to

CO1 Understand the basic concept, features and attributes of governance and environmental governance.
CO2 Connect between governance and development and their implications in policy formulation.
CO3 Determine the historical trends of forest and environmental governance in Bangladesh
CO4 Analyse the current forest and environment related policy and plans in Bangladesh and negotiation process.
CO5 Evaluate the current international and national environmental governance issues and processes for better negotiation and representation.

Mapping Course Learning Outcomes (COs) with the POs

COs vs. POs	Fundamental Skill			Social Skill		Thinking Skill		Personal Skill		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3			3		2				
CO2	3			3		2			2	
CO3	3									1

CO4		2					1			
CO5	3			2			3			2

3: Strong 2: Moderate 1: Weak

Mapping Course Learning Outcomes (COs) with the Teaching-Learning & Assessment Strategy

COs	Teaching-Learning Strategy	Assessment Strategy
CO1	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO2	TL01, TL05	CA01, CA03, CA04, CA05, SA01
CO3	TL01, TL05	CA01, CA02, CA03, CA04, CA05, SA01
CO4	TL01, TL05	CA02, CA03, CA04, CA05, SA01
CO5	TL01, TL05	CA02, CA03, CA04, CA05, SA01

Recommended References:

1. Robledo, C., et al. (2008). *Climate change and governance in the forestry sector*. Intercooperation.
2. Dahal, G. R., & Capistrano, D. (2006). *Forest governance and institutional structure: An ignored dimension of community-based forest management in the Philippines.* International Forestry Review, 8(4), 377–394.

PART D

20. **Grading/Evaluation**:

A student will be evaluated continuously in the semester-course system, for theoretical classes s/he will be assessed by class participation, assignments, quizzes, term-test examinations and final examination. For laboratory work s/he will be assessed by observation of the student at work, viva-voce during laboratory works, from his/her written reports and grades of examinations designed by the respective course teacher and the examination committee. For Research Project, the supervisor will give an overall assessment for the student and on her/his research project, which will count 30% of the total marks. Evaluation of the research project by the 2 (two) external examiners who are not involved in supervision/co-supervision will count 40% of the total marks, and from the final presentation in the presence of the examination committee, 30% of marks will be counted. The examination committee will enter the aggregated Marks of the examinees using the software and process the result.

Examination Committee

Formation:

A committee consisting of all the teachers headed by the Head of the discipline will propose examination committees for all existing semesters to the respective Dean within 4 (four) weeks of the semester start. It is not necessary to form separate committees for students having incomplete courses at a particular level; the committee of the nearest level of the same year will be responsible to process the result of the students. But, for the clearing graduates, 4th year 2nd semester committee of immediate past will process the -results of left out students of 4th year 2nd semester, while 4th year 2nd semester is not running.

Members:

The examination committees for different semesters will be comprised of the following members:

Chairman: A teacher not below the rank of Professor of the discipline. In absence of a Professor of the discipline, an Associate Professor/Head of the discipline. Head of the Discipline will be the Chairman of the terminal semesters.

Internal Members: 4 (four) teachers of the discipline.

External members: One teacher/expert (not below the rank of professor or equivalent) of the Major Field from other university/organization and one teacher from each of the disciplines offering the general education courses.

The respective Deans will ratify the list of the examination committees and send to the registrar for approval from the AC.

Change of Committee Members:

If for some genuine reason either the chairman or a member of the examination committee is needed to be changed, then the committee consisting of all the teachers headed by the Head of the discipline will send the nomination/proposal/recommendation to the respective Dean. The Dean will authenticate/approve the change and send it to the Controller of Examinations for book-keeping and to take the approval of the Vice-Chancellor.

Grading Scale:

Distribution of Marks:

The marks of a given course will be as follows:

Class Attendance	10%
Term-Test Examinations	20%
Class Performance (Quizzes, MCQs, fill in the gaps, report writing/presentation/assignments)	10%
Final Examination (25% is the pass mark for the final examination)	60%

Class Participation:

The marks for class participation will be as follows:

Attendance (Percentage)	Marks	Attendance (Percentage)	Marks	Attendance (Percentage)	Marks
95 and above	10	80 to 84	7	65 to 69	4
90 to 94	9	75 to 79	6	60 to 64	3
85 to 89	8	70 to 74	5	Less than 60	0

A student will not be allowed to appear in the examination of a course if his/her class attendance in that course is less than 50%.

Letter Grade and Grade Point:

Letter Grade and corresponding Grade-Point for a course will be awarded from the roundup marks of individual courses as follows:

Numerical Grade	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	C-	2.00
Less than 40%	F	0.00

Candidates for four-year Bachelor's degree will be awarded the degree with Distinction if his/her overall CGPA is 3.75 or above. However a student will not be considered for Distinction if (a) s/he is not a regular student (has

semester drop, incomplete courses in any semester or break of study) (b) has 'F' grade in one or more courses (c) has upgraded her/his GPA through improvement, (d) addicted to drugs, (e) disciplinary action (s) is taken against her/him.

Grades:

GPA:

Grade Point Average (GPA) is the weighted average of the grade points obtained in all the courses completed by a student in a semester.

Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA):

CGPA:

Cumulative Grade Point Average (CGPA) of only major and both major and second major degree will be calculated by the weighted average of every course of previous semesters along with the present semester. For clearing graduates if the roundup value of the third digit after decimal is nonzero the second digit will be incremented by one. A student will also receive a separate CGPA for his second major courses.

Incomplete Courses:

If a student has an incomplete course(s), s/he has to register such an incomplete course(s) from preceding semesters before registering courses from current or successive semesters. If an incomplete course is not available or offered in the running semesters, the student shall take such course(s) when it is available or offered.

Course Withdrawal:

A student can withdraw a course by a written application to the Controller of Examinations through the Head/Director of the discipline/institute two weeks before the examination start. The Controller of Examinations will send the revised registration list(s) to the disciplines before the commencement of semester final examination. There will be no record in transcript if the course is withdrawn.

Course Repetition:

If a student has to repeat a failed or incomplete course and that course is not available/offered any more, the discipline may allow him/her to take an equivalent course from the current curriculum. For clearing graduates, if any incomplete course is not available/offered in the running semester, the discipline may suggest a suitable/equivalent course to complete the credit requirement so required for the degree.

Course Improvement:

A student will be allowed only once to improve maximum of 2 (two) theory courses for which s/he has obtained a B- grade or less in the previous level by registering in the semesters of the immediate next level. Such course grade improvement opportunity shall be given only for 100-300 level courses. If the course grade does not improve then the previous course grade will sustain in grade count. In the case of the course grade improvement, this will be cited/noted in the concerned transcripts beside the grade count as "Improvement."

F Grades:

A student will be given an "F" grade if s/he fails or remains absent in the final examination of a registered course. If a student obtains an "F" grade, her/his grade will not be counted for GPA and s/he will have to repeat the course. An "F" grade will be in her/his record, and s/he will not be eligible for distinction, award, and scholarship of the university.

Annexures

Teaching-Learning & Assessment Strategy

Code	Teaching-learning (TL) strategy	Code	Assessment Strategy
TL 01	Lecture using board/LCD projectors/OHP projectors		Continuous assessment (CA)
TL 02	Assignment/project/seminar/workshop/tutorial	CA 01	Midterm Examination 1
TL 03	Laboratory/Other teaching aids (Audio-visual: film and documentaries, virtual classroom, etc.)	CA 02	Midterm Examination 2
TL 04	Guest lectures/industrial visit/field visit	CA 03	Quiz
TL 05	Self-learning using reference books/research articles/case study/other online materials	CA 04	Assignment
TL 06	Simulation/field demonstration	CA 05	Presentation (Individual/group) /Viva voce
			Summative Assessment (SA)
		SA 01	Semester-end examination (Theory)
		SA 02	Semester-end oral examination
		SA 03	Seminar paper evaluation (Project + Seminar)

Bloom's Taxonomy of Learning Outcomes: Cognitive Domain

Bloom's Level Create	**Key Verbs (keywords)** design, formulate, build, invent, create, compose, generate, derive, modify, develop. Example Learning Outcome: By the end of this lesson, the student will be able to design an original homework problem dealing with the principle of conservation of energy
Bloom's Level Evaluate	**Key Verbs (keywords)** choose, support, relate, determine, defend, judge, grade, compare, contrast, argue, justify, support, convince, select, evaluate. Example Learning Outcome: By the end of this lesson, the student will be able to determine whether using conservation of energy or conservation of momentum would be more appropriate for solving a dynamics problem.
Bloom's Level Analyze	**Key Verbs (keywords)** classify, break down, categorize, analyze, diagram, illustrate, criticize, simplify, associate, differentiate. Example Learning Outcome: By the end of this lesson, the student will be able to differentiate between potential and kinetic energy.
Bloom's Level Apply	**Key Verbs (keywords)** calculate, predict, apply, solve, illustrate, use, demonstrate, determine, model, perform, present. Example Learning Outcome: By the end of this lesson, the student will be able to calculate the kinetic energy of a projectile.
Bloom's Level Understand	**Key Verbs (keywords)** describe, explain, paraphrase, restate, give original examples of, summarize, contrast, interpret, discuss. Example Learning Outcome: By the end of this lesson, the student will be able to describe Newton's three laws of motion to in her/his own words.
Bloom's Level Remember	**Key Verbs (keywords)** list, recite, outline, define, name, match, quote, recall, identify, label, recognize. Example Learning Outcome: By the end of this lesson, the student will be able to recite Newton's three laws of motion.

Table of Specification (TOS) for the Summative and Continuous Assessment (Semester Final Exam) – 2 credit courses
Total Marks: 60 (15 marks in each question)

Questions			Bloom's levels of Cognition					
Q. No.	**Marks**	**CLO No**	**Remember**	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Create**
PART – A								
1(a)								
1(b)								
1(c)								
OR								
1(d)								
1(e)								
1(f)								
2(a)								
2(b)								
2(c)								
PART - B								
3(a)								
3(b)								
3(c)								
OR								
3(d)								
3(e)								
3(f)								
4 (a)								
4 (b)								
4 (c)								
Total Marks								
Total % in each level								
% Marks in Major levels								

Table of Specification (TOS) for the Summative and Continuous Assessment (Semester Final Exam) – 3 & 4 credit courses
Total Marks: 60 (10 marks in each question)

Questions			Bloom's levels of Cognition					
Q. No.	**Marks**	**CLO No**	**Remember**	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Create**
PART – A								
1(a)								
1(b)								
1(c)								
OR								
1(d)								
1(e)								
1(f)								
2(a)								
2(b)								
2(c)								
3(a)								
3(b)								
3(c)								
PART - B								
4(a)								
4(b)								
4(c)								
OR								
4(d)								
4(e)								
4(f)								
5 (a)								
5(b)								
5 (c)								
6(a)								
6(b)								
6(c)								
Total Marks								
Total % in each level								
% Marks in Major levels								